UNIVERSITÀ DEGLI STUDI DI PADOVA

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Mathematical Modelling for Engineering and Science

MASTER'S THESIS

Investigations on some models in gradient elastoplasticity

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Premise

CHAPTER 1

Introductory notions regarding mechanical modeling and plasticity theory

1.1. A general framework for developing mechanical models

In this section we describe a general framework that allows to systematically develop mechanical models. We will use it repeatedly through this thesis. The framework we are going to present, is based on the principle of virtual powers, which is a powerful and elegant tool to describe the equilibrium of the "forces" acting on a mechanical system. The "forces" must be intended in a generalized sense here. In fact, the configurations of the mechanical system are described as elements of a space (possibly infinite dimensional) of configurations, and therefore the generalized forces must be understood as actions that "push" the configuration of the system within this space. Moreover the configuration of the mechanical system can include also the description of internal or microscopic conditions of the system, so that a generalized forces does not necessarily correspond to a mechanical macroscopic distribution of force in the usual sense.

Some historical information about the birth of the principle of virtual powers can be found in [39] and in [45]. Modern works that use the principle of virtual powers to develop systematically mechanical model are [28, 34, 32], where second gradient theories are considered, and [30, 31, 27] that treat materials with internal variables.

1.1.1. Kinematics. Kinematics is concerned with the description of the possible *configurations* of a mechanical system. We assume that they can be described as the elements of a separable and reflexive Banach space V, the *space of configurations*.

Example 1.1. Consider N material points in $\mathbb{R}^d$. Their configuration can be described by the matrices $x = (x_1, \dots x_N) \in \mathbb{R}^{d\times N}$, where $x_\alpha \in \mathbb{R}^d$ with $\alpha \in \{1, \dots, N\}$ is the position of the αth material point.

Example 1.2. The configurations of a deformable body at large strain in $\mathbb{R}^d$, can be identified with the deformations $\chi \in W^{1,p}(\Omega)^d$ with $p \in (1, \infty)$, where $\Omega \subset \mathbb{R}^d$, open and bounded subset of $\mathbb{R}^d$ is some reference configuration of the body. Every $x \in \Omega$ is the reference position of a material point and $\chi(x)$ is the position occupied by that material point in the deformed configuration. The condition of incompenetrability, $\det \operatorname{grad}\chi > 0$ a.e. in Ω, was purposely excluded from the definition of the configuration space in order to preserve the linear structure. Even thought we do not consider the incompenetrability constrain at the level of kinematical description, it can be imposed in a later moment, as we will show.

Example 1.3. Consider a body that can be subject to damage. This can be modeled considering an internal damage variable $\alpha \in W^{1,p}(\Omega)$ with $p \in (1, \infty)$. Where $\alpha = 0$ the material is sound, while where $\alpha = 1$ the material is completely damaged. The material is partially damaged where $\alpha \in (0, 1)$. One should add the constrain $\alpha \in [0, 1]$, but this is not imposes at the level of definition of the space of configurations, again in order to preserve the linear structure.

Examples 1.2 and 1.3 suggest that the assumption that space of configurations has a linear structure is not restrictive in most mechanical models, even when the truly attainable configurations actually should lie in a space M with the structure of a manifold. Indeed in most cases, M can be seen as a subset of a larger linear space of configurations V. Then we can formulate the model on V and impose that the evolution of the configuration of the system is restricted to M by including suitable reactions, as we will see. Thus the geometric complexity of M can be avoided and reduced to the analytical complexity of managing the reactions within V required to enforce the constraint

Remark 1.4. As Example 1.3 show, the configuration of the mechanical system can include also the description of an internal state of the system that does not necessarily correspond with the macroscopic observable placement of the system in the space. This is particularly true also in the case of plasticity theory, that will be discussed in the next section.

1.1.2. Statics. With statics we mean the treatment of equilibrium of forces, in a generalized sense, acting on a mechanical system. Any *generalized force* acting on the mechanical system, can be described as a linear functional $\mathcal{F} \in V^*$. Given any $\tilde{v} \in V$, the duality pairing $\langle \mathcal{F}, \tilde{v} \rangle$ can be interpreted as the projection of $\mathcal{F}$ along the direction $\tilde{v}$. If we interpret $\tilde{v}$ as a virtual rate, that is $\tilde{v} = \dot{\tilde{u}}(0)$ for an arbitrary evolution $\tilde{u} \in C^1([0,1], V)$ of the configuration of the system, then $\langle \mathcal{F}, \tilde{v} \rangle$ is called the virtual power of $\mathcal{F}$ along the virtual rate $\tilde{v}$.

Example 1.5. (Example 1.1 continued) A generalized force can be seen as an element $F = (F_1, \ldots, F_N) \in (\mathbb{R}^{d\times N})^* = \mathbb{R}^{d\times N}$. The virtual power acts on $\tilde{v} = (\tilde{v}_1, \ldots, \tilde{v}_N)$ as

$$\langle F, \tilde{v} \rangle = \sum_{\alpha=1}^{N} F_\alpha \cdot \tilde{v}_\alpha.$$

Here F_α is the force acting on the αth material point.

Example 1.6. (Example 1.2 continued) Consider the functional $\mathcal{F} \in W^{-1,p^*}(\Omega)^d$ of the form

$$\langle \mathcal{F}, \tilde{v} \rangle = \int_\Omega P : \mathrm{Grad}\, \tilde{v}\, dX + \int_\Omega f \cdot \tilde{v}\, dX + \int_{\partial\Omega} h \cdot \tilde{v}\, dX,$$

where $P \in L^{P^*}(\Omega, \mathbb{R}^{d\times d})$, $f \in L^{p^*}(\Omega)^d$ and $h \in L^{p^*}(\partial\Omega)^d$. The first integral represents a generalized force that acts in the direction of the local bending of the body, the second integral represents a volume force distributed over Ω, the last integral a surface force distributed over $\partial\Omega$.

Consider now a time interval $[0, T]$ and an actual evolution of the configuration of the system, $u \colon [0, T] \to V$. For simplicity, in this discussion we assume that $u \in \mathrm{AC}([0, T], V)$ even thought less regular evolutions, for example exhibiting jumps, can be considered as we will see subsequently in this thesis. Along its evolution, the mechanical system is subject to various generalized forces and we call $\mathcal{F} \colon [0, T] \to V^*$ their sum. We require that equilibrium holds, that is

$$\mathcal{F}(t) = 0 \quad \text{for a.e. } t \in (0, T).$$

This can be written also as

$$\langle \mathcal{F}(t), \tilde{v} \rangle = 0 \quad \text{for all } \tilde{v} \in V \text{ and a.e. } t \in (0, T). \tag{1.1}$$

Equation (1.1) is the statement of the principle of virtual powers, that is, the virtual power of the generalized force acting on the system along all virtual rates must vanish.

1.1.3. Constitutive relations. To get an evolution equation, able to predict the evolution of the mechanical system, it is necessary to link $\mathcal{F}$ to the evolution u through constitutive relations. We focus on constitutive relations of the form

$$\mathcal{F}(t) \in \hat{\mathcal{F}}(u(t), \dot{u}(t), t) \quad \text{for a.e. } t \in (0, T),$$

where $\hat{\mathcal{F}} \colon V \times V \times [0, T] \rightsquigarrow V^*$. We postulate that

$$\hat{\mathcal{F}}(u, v, t) = -\partial_u \mathcal{E}(u, t) - \partial_v \mathcal{R}(u, v, t), \tag{1.2}$$

where $\mathcal{E} \colon V \times [0, T] \to (-\infty, \infty]$ and $\mathcal{R} \colon V \times V \times [0, T] \to [0, \infty]$ are called respectively the *energy* and the *dissipation potential*. The ∂ symbol can have different definitions according to the regularity of $\mathcal{E}$ and $\mathcal{R}$. If Fréchet differentiability is assumed then ∂ denotes the Fréchet differential, while if only convexity is assumed it denotes the subdifferential. Less regularity requires to use extended notions of subdifferentials such as the Fréchet subdifferential. In view of (1.2), we can split $\mathcal{F} = \mathcal{F}_c + \mathcal{F}_d$, where $\mathcal{F}_c(t) \in -\partial_u \mathcal{E}(u(t), t)$ is the conservative generalized force and $\mathcal{F}_d(t) \in -\partial_v \mathcal{R}(u(t), \dot{u}(t), t)$ is the dissipative generalized force. The requirement that $\mathcal{F}_d$ has a dissipative behavior can be formulated as

$$\langle \mathcal{F}_d(t), \dot{u}(t) \rangle \leq 0 \quad \text{for a.e. } t \in (0, T).$$

This means that the projection of $\mathcal{F}_d$ along the direction of the rate is negative, or equivalently that the power of $\mathcal{F}_d$ is nonpositive. This is automatically satisfied if

$$\langle \xi, v \rangle \geq 0 \text{ for all } (u, v, t) \in V \times V \times [0, T] \text{ and } \xi \in \partial_v \mathcal{R}(u, v, t).$$

This last condition is called *dissipation inequality*.

In most situations one can split $\mathcal{F} = \mathcal{X} - \mathcal{I}$, where $\mathcal{X}$ and $\mathcal{I}$ from $[0, T]$ to V^* are the external generalized force and the internal generalized force respectively. The internal generalized force is due to the internal response of the mechanical system, while the external generalized force represents the external actions imposed on it. In this case the constitutive relations are usually specialized to

(i) $\mathcal{E}\colon V \times [0,T] \to (-\infty,\infty]$ of the form $\mathcal{E}(u,t) = \Psi(u) - \langle \mathcal{X}(t), u\rangle$, where $\Psi\colon V \to (-\infty,\infty]$ is the free energy, describing the conservative internal response, and $\mathcal{X}\colon [0,T] \to V^*$ is the external generalized force,
(ii) $\mathcal{R}(u,v,t) = \Phi(u,v)$, where $\Phi\colon V\times V \to [0,\infty]$ is the dissipation potential describing the dissipative internal response.

Then the constitutive relations can be written as $\mathcal{I}(t) \in \hat{\mathcal{I}}(u(t),\dot{u}(t))$ for a.e. $t\in(0,T)$, where $\hat{\mathcal{I}}\colon V\times V \rightsquigarrow V^*$ is given by

$$\hat{\mathcal{I}}(u,v) = \partial\Psi(u) + \partial_v\Phi(u,v). \tag{1.3}$$

In this case the equilibrium condition reads

$$\partial\Psi(u(t)) + \partial_v\Phi(u(t),\dot{u}(t)) \ni \mathcal{I}(t) = \mathcal{X}(t) \quad \text{for a.e. } t\in(0,T)$$

or

$$\partial_u\mathcal{E}(u(t),t) + \partial_v\Phi(u(t),\dot{u}(t)) \ni 0 \quad \text{for a.e. } t\in(0,T). \tag{1.4}$$

This is known as the *doubly nonlinear equation*. Now the evolution problem can be formulates as follows.

Problem 1.7. Let $u_0 \in \operatorname{dom}\mathcal{E}(\cdot,0)$ and $\mathcal{X}\colon[0,T]\to V^*$. Find $u\in \mathrm{AC}([0,T],V)$ such that (1.4) holds.

Remark 1.8. Before imposing particular constitutive relations, $\mathcal{I}$ and $\mathcal{X}$ are completely undefined. However we notice that, before making specific constitutive assumptions, when one develop a new mechanical model it is useful to understand which is the particular structure that $\mathcal{I}$ and $\mathcal{X}$ must assume in order to describe the forces involved in the mechanical system of interest. This is the approach that will be followed in this thesis.

Remark 1.9. In terms of Φ, the dissipative inequality reads

$$\langle \xi, v\rangle \geq 0 \text{ for all } (u,v)\in V\times V \text{ and } \xi\in\partial_v\Phi(u,v). \tag{1.5}$$

A possible condition ensuring (1.5) is that $\Phi(u,\cdot)$ is convex and $\Phi(u,0)=0$. Indeed if $(u,v)\in V\times V$ and $\xi\in\partial_v\Phi(u,v)$, then by definition of subdifferential, $0 = \Phi(u,0) \geq \Phi(u,v) + \langle\xi, 0-v\rangle$, so that $\langle\xi,v\rangle \geq \Phi(u,v)\geq 0$.

1.1.4. Adding constraints with the method of Lagrange multipliers. To include a kinematical constrains on the process, say $u(t)\in K$ for all $t\in[0,T]$, one can set $\mathcal{E}(u,t)=\infty$ if $u\notin K$. This can be done by including ι_K in $\mathcal{E}$. In this way $\mathcal{F}_c(t)$ will include the reaction $\mathcal{P}(t)\in -N_K(u(t))\subset V^*$ required to enforce the constrain. Here $N_K = \partial\iota_K$.

Example 1.10. (Example 1.2 continued) Let $\Phi\colon W^{1,p^*}(\Omega)^d \to (-\infty,\infty]$. To describe an elastic response, we choose Ψ of the form

$$\Psi(\chi) = \int_\Omega W(X, \operatorname{Grad}\chi(X))dX,$$

where $W\colon \Omega\times\mathbb{R}^{d\times d} \to (-\infty,\infty]$ is a continuous free energy density. Incompenetrability impose that $W(X,F)=\infty$ if $\det F\leq 0$.

Example 1.11. (Example 1.3 continued) To ensure that $z(x)\in[0,1]$ for a.e. $x\in\Omega$, one choose $\Phi\colon W^{1,p}(\Omega)\to(-\infty,\infty]$ such that $\Phi(z)=\infty$ if $z\notin[0,1]$ on a set of positive measure.

However there are situations in which it would be desirable to treat the constraint reaction $\mathcal{P}$ as an independent variable instead. In this sense $\mathcal{P}$ can be considered as an additional descriptor of the configuration of the system. This is particularly useful if one wants to consider constitutive relations depending on the reaction $\mathcal{P}$ itself, as it happens for the model that is discussed in Part 2. Then we split the generalized force as $\mathcal{F} = -\mathcal{I}+\mathcal{X}+\mathcal{P}$. The constitutive relations are imposed on $\mathcal{I}$ only as

$$\mathcal{I}(t) \in -\partial\Psi(\mathcal{P}(t),u(t)) - \partial\Phi(\mathcal{P}(t),u(t),\dot{u}(t)) \quad \text{for a.e. } t\in(0,T),$$

where now $\Psi\colon V^*\times V\to(-\infty,\infty]$ and $\Phi\colon V^*\times V\times V\to[0,\infty]$. Instead $\mathcal{X}$ is assigned while $\mathcal{P}$ is included among the unknowns of the evolution together with u. Then the evolution equation can be written as

$$\begin{matrix} u(t)\in K \\ \mathcal{P}(t)\in -N_K(u(t)) \end{matrix} \quad \text{and} \quad \partial_u\mathcal{E}(\mathcal{P}(t),u(t),t) + \partial_v\Phi(\mathcal{P}(t),u(t),\dot{u}(t)) \ni \mathcal{P}(t) \quad \text{for a.e. } t\in(0,T), \tag{1.6}$$

where now $\mathcal{E}\colon V^*\times V\times[0,T]\to(-\infty,\infty]$ is defined by $\mathcal{E}(\mathcal{P},u,t) = \Psi(\mathcal{P},u) - \langle\mathcal{X}(t),u\rangle$. Then the evolution problem becomes as follows.

Problem 1.12. Let $u_0\in\operatorname{dom}\mathcal{E}(\cdot,0)$ and $\mathcal{X}\colon[0,T]\to V^*$. Find $u\in\mathrm{AC}([0,T],V)$ and $\mathcal{P}\colon[0,T]\to V^*$ such that (1.6) holds.

1.2. A review of plasticity theory

The origins of the theory of plasticity can be traced back to the experimental work of Tresca [66] in the late nineteenth century on the deformation of metals. He recognized that the behavior of a metal is elastic as long as the stresses do not exceed a certain threshold: beyond the threshold, called the yield point, they "flow" in a plastic way, accumulating permanent deformation. Tresca, probably inspired on Coulomb's work on the stability of soils [17], also proposed a first criterion to describe the yield threshold, based on the maximum shear stress. Shortly afterwards De Saint Venant [63] and Lévy [41] produced the first theory to explain the temporal evolution of a rigid-plastic solid subjected to external loads. At the beginning of the twentieth century the theory was extended by Prandl [58], Von Mises [49] and Reuss [61] to the elastoplastic case. Until the seventies, the theory of plasticity was formulated in incremental terms, i.e. linking the temporal increase in stress to the rate of strain. A variational formulation, mathematically more elegant and robust, was introduced by Moreau [51] in the sixties. With the aim of giving a coherent formulation of the problems of plasticity (and friction), Moreau also invented important notions of convex analysis (such as subdifferentials) necessary to deal with the non-smooth problems of mechanics.

To describe the experimentally observed increase of the yield strength of metals at small scales, since the work of Aifantis [1] in 1984, has been developed plasticity models (see [24] for the early developments) that include a dependence on the gradient of the kinematical field describing the plastic distortion. This allows to include in the model an internal characteristic length scale that is not considered by classical plasticity. One refer to this kind of models as the gradient plasticity theory. Modern formulations are given in [30], [31] and [5]. The importance of the gradient plasticity theory in this thesis is mainly due to a mathematical issue rather then a modeling concern. Indeed we will see that the inclusion of the dependence the plastic distortion gradient will be necessary to ensure compactness properties needed to handle successfully the nonconvex character of plasticity at large strain in Part 1 and the complex nonlinear couplings of the model discussed in Part 2.

The extension of the theory to the case of large deformations has been addressed since the sixties, and a unanimous consensus has never been reached between the different schools [52, 20], especially with regard to the choice and interpretation of the variable describing the permanent plastic deformation. In this thesis we will adopt the approach of Gurtin and Anand [33, 31] and of Mielke et al. [43], the only one for which a theory of existence was successfully obtained.

In this section we give a concise description of (gradient) plasticity theory, both at regime of small and large deformations, inserting it in the general framework proposed in Section 1.1. We will confine ourself to the plasticity of metals; in Part2 we will discuss how to theory can be modified to describe other types of materials. The focus of this discussion will be on the formulation of the mechanical models, in the following chapter we will address the mathematical questions regarding the existence of solutions.

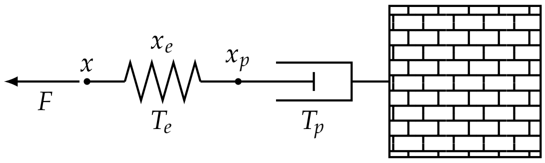
1.2.1. Plasticity theory at small deformations. Let $\Omega \subset \mathbb{R}^d$, a bounded, open and Lipschitz domain, be the reference configuration of a body. Each $x \in \Omega$ is the reference position of a material point belonging to the body. The macroscopic configuration of the body is described by the displacement field $u\colon \Omega \to \mathbb{R}^d$ that assign to each material point $x \in \Omega$ the displacement $u(x)$ from its reference position. The displacement gradient is defined as $H = \operatorname{grad} u$ and is additively decomposed as $H = W + E$, where $W = \operatorname{skew} H$ represents a rigid spin of the material while $E = \operatorname{sym} H$ is the strain. The distortion of the microscopic structure of the body is described by the field $H_p\colon \Omega \to \mathrm{Lin}$, called plastic distortion. We decompose additively $H_p = W_p + E_p$, where $W_p = \operatorname{skew} H_p$ is the plastic spin and $E_p = \operatorname{sym} H_p$ is the plastic strain. In what follows we assume[1] for simplicity that that $W_p = 0$. Moreover we also add the hypotheses of plastic incompressibility $\operatorname{tr} E_p = 0$ so that $E_p\colon \Omega \to \mathrm{DevSym}$: this hypotheses is suitable for the description of metals but it is not good for other kind of materials such as geomaterials, as we will discuss in Part 2. Now the configuration of the system is described by the couples $z = (u, E_p) \in Z$ where the space Z is chosen case by case depending on the particular constitutive model.

The plastic strain E_p can be considered as the elastically relaxed strain of the material, so we define the elastic strain as $E_e = E - E_p$.

Remark 1.13. To clearly understand the significance of the definition of elastic strain E_e, it is helpful to have this picture in mind. We view the strain E as "resting upon" the plastic strain E_p, with the elastic strain E_e acting as the coupling mechanism that transfers force between them. This concept can be illustrated with the following simple mechanical model. Consider a damper with one end fixed to a wall and the other at horizontal position $x_p \in \mathbb{R}$, and a spring in series with the damper whose end is at

[1]Nonzero plastic spin are considered by the micropolar continuum theories, or Cosserat theories ,introduced by the Cosserat brothers [16] at the beginning of the twentieth century. They are suitable for describing, for example, granular materials when is relevant the change of the orientations of the grains.

horizontal position $x \in \mathbb{R}$. The configuration of the system is described by the couples $z = (x, x_p) \in \mathbb{R}^2$, and we view x as the "macroscopic" configuration and x_p as the internal or "microscopic" configuration. We define $x_e = x - x_p \in \mathbb{R}$ the length of the spring.



On x acts the external force $F \in \mathbb{R}$, the wall exert a force $T_p \in \mathbb{R}$ on the damper and the "macroscopic" and "microscopic" points x and x_p exchange a force $T_e \in \mathbb{R}$. If we call $\tilde{w} = (\tilde{v}, \tilde{v}_p) \in \mathbb{R}^2$ a generic virtual rate of the system, and we define accordingly $\tilde{v}_e = \tilde{v} - \tilde{v}_e \in \mathbb{R}$ the elastic virtual rate, this configuration of forces is described by the internal and external virtual powers

$$\langle \mathcal{I}, \tilde{w} \rangle = T_e \tilde{v}_e + T_p = T_e(\tilde{v} - \tilde{v}_p) + T_p \tilde{v}_p \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = F\tilde{v}$$

respectively. Thus the equilibrium condition $\mathcal{I} = \mathcal{X}$ can be written as the "macroscopic" balance of forces $T_e = F$ together with the "microscopic" balance of forces $T_p = T_e$.

Motivated by the consideration in Remark 1.13, we consider the virtual rates $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$ of z, we define the elastic virtual rate as $\tilde{V}_e = E\tilde{v} - V_p$ and we consider and internal and an external virtual powers functionals of the form

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_\Omega T_e : \tilde{V}_e dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \vdots \operatorname{grad} \tilde{V}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} \tag{1.7}$$

respectively. In the expression of the internal virtual power, $T_e \colon \Omega \to \mathrm{Sym}$ is the elastic stress, $T_p \colon \Omega \to \mathrm{DevSym}$ is the plastic microforce and $K_p \colon \Omega \to \mathrm{DevSym} \otimes \mathbb{R}$ is the plastic microstress. The fact that K_p appears qualifies the plasticity theory we are developing as a gradient plasticity theory. In contrast to classical plasticity, where $K_p = 0$, gradient plasticity allows a "contact" interaction at the level of the microscopic structure. The external virtual power includes a volumetric force $f \colon \Omega \to \mathbb{R}^d$ and a surface force $h \colon \partial\Omega \to \mathbb{R}^d$. Also external microforces acting on the microscopic configuration, i.e. terms conjugate to the plastic virtual rate $\tilde{V}_p$, could be added, but in order to keep the discussion simpler we neglect them.

By a simple application of the Green theorem and the fundamental lemma of the calculus of variations, one easily get the strong form of the equilibrium condition $\mathcal{I} = \mathcal{X}$,

$$\begin{aligned} \operatorname{div} T_e + f = 0 \quad & \text{in } \Omega \\ T_e n = h \quad & \text{in } \partial\Omega \\ \operatorname{dev} T_e - T_p + \operatorname{div} K_p = 0 \quad & \text{in } \Omega \\ K_p n = 0 \quad & \text{in } \partial\Omega. \end{aligned}$$

The first two conditions express the balance of macroscopic forces, while the last two represents the balance of microscopic forces.

The choice of constitutive relations is driven by the idea that the elastic response is conservative while the plastic response is dissipative. Thus one choose a free energy $\Psi \colon Z \to \mathbb{R}$ and a dissipation potential $\Phi \colon Z \to [0, \infty)$ of the form

$$\Psi(z) = \int_\Omega W_e(E_e) dx + \int_\Omega W_p(E_p, \operatorname{grad} E_p) dx \quad \text{and} \quad \Phi(\dot{z}) = \int_\Omega R(\dot{E}_p) dx,$$

where

(i) the elastic energy density $W_e \colon \mathrm{Sym} \to \mathbb{R}$ and the plastic energy density $W_p \colon \mathrm{DevPla} \to \mathbb{R}$ are differentiable and
(ii) the plastic dissipation potential density $R \colon \mathrm{DevSym} \to [0, \infty)$ is convex and satisfies $R(0) = 0$.

The condition $\mathcal{I} \in \partial\Psi(z) + \partial\Phi(\dot{z})$ imposes that

$$T_e = \frac{\partial W_e}{\partial E_e}(E_e), \qquad T_p \in \frac{\partial W_p}{\partial E_p}(E_p, \operatorname{grad} E_p) + \partial R(\dot{E}_p) \quad \text{and} \quad K_p = \frac{\partial W_p}{\partial \operatorname{grad} E_p}(E_p, \operatorname{grad} E_p).$$

Usually one assumes a linear elastic response, corresponding to

$$W_e(E_e) = \frac{1}{2}\mathfrak{C}E_e : E_e = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2,$$

where $\kappa > 0$ is the bulk modulus and μ is the shear modulus. A possible choice of the plastic free energy is

$$W_p(E_p, \operatorname{grad} E_p) = w_p(E_p) + \frac{1}{2} l^2 \mathfrak{B}_2 |\operatorname{grad} E_p|^2$$

where $\mathcal{B}_2 > 0$ is a constant with the dimensions of a stress and $l > 0$ is a constant with the dimensions of length. The term containing the gradient of E_p act as a regularization of the plastic strain field, since it penalize sharp variations of E_p. Moreover it includes an internal length scale on the model, that is otherwise absent in the classical non-gradient plasticity theory. The quantity $B_p = w_p'(E_p)$ is called back-stress and represents the conservative part of the plastic microforce. The classical Von Mises rate rate-independent constitutive relation for the dissipative part of the plastic microforce is obtained choosing $R(\dot{E}_p) = Y|\dot{E}_p|$, where $Y > 0$ is the yield strength. In this case

$$\partial R(\dot{E}_p) = \begin{cases} B(0, Y) & \text{if } \dot{E}_p = 0 \\ \left\{ Y \dfrac{\dot{E}_p}{|\dot{E}_p|} \right\} & \text{if } \dot{E}_p \neq 0. \end{cases}$$

Now, the microforce balance can be written as

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in \partial R(\dot{E}_p).$$

The mechanical meaning of Von Mises costitutive relation is that, when $|\operatorname{dev} T_e - B_p + \operatorname{div} K_p| < Y$, then $\dot{E}_p = 0$ and no plastic strain accumulates. If one choose $w_p(E_p) = \frac{1}{2}\mathfrak{B}_1|E_p|^2$, then $B_p = \mathfrak{B}_1 E_p$ so that the back-stress model an increasing resistance of the material to plastic flow. This behavior is common in metals and is called hardening. Refined hardening models can be obtained by choosing w_p appropriately.

Remark 1.14. More general expression of R can be obtained as follows. One choose a bounded, closed and convex set $C \subset \operatorname{DevSym}$ and impose that

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in C \quad \text{and} \quad \dot{E}_p \in N_C(\operatorname{dev} T_e - B_p + \operatorname{div} K_p)$$

where N_C is the outer normal cone to C. The first condition is the requirement that the dissipative plastic force must belong to C, the second is a flow rule of the plastic strain. Using a well known result of convex analysis [23, Corollary 5.2], this is equivalent to

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in \partial \sigma_C(\dot{E}_p)$$

where $\sigma_C \colon \operatorname{DevSym} \to \mathbb{R}$ is the support function of C, defined by

$$\sigma_C(A) = \sup_{S \in C} S \cdot A.$$

This behavior is obtained by choosing the dissipation potential density $R = \sigma_C$.

1.2.2. Plasticity theory at large deformations. Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz, be the reference configuration of the body. The macroscopic configuration of the body is described by the placement map $\chi \colon \Omega \to \mathbb{R}^d$ that assign to each material point $x \in \Omega$ its position $\chi(x)$. The local state of deformation is described by the deformation gradient $F = \operatorname{grad} \chi \colon \Omega \to \operatorname{Lin}$. The Cauchy deformation tensor is defined as $C = F^T F$ and have the important property of being invariant under superimposed rigid placement: if $\chi^*(X) = Q\chi(X) + a$, with $Q \in \operatorname{Orth}^+$ and $a \in \mathbb{R}^d$, then $F^* = QF$ and $C^* = C$. The distortion of the microscopic structure is described by the plastic distortion $F_p \colon \Omega \to \operatorname{Lin}$. Then the configuration of the system is described by the couple $z = (\chi, F_p) \in Z$ where the particular definition of the function space Z depends on the choice of constitutive relations and it is not important now. The virtual rates are denoted as $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$.

Remark 1.15. In accordance with the discussion made in Section 1.1, we do not include the constrain $\det F > 0$ (the impenetrability condition) and $\det F_p = 1$ (the plastic incompressibility condition) in the definition of Z in order to preserve the linear structure of the space of configurations.

The plastic distortion can be considered as the locally elastically relaxed state of the material, so we define the elastic distortion as $F_e = FF_p^{-1}$ for all $z = (\chi, F_p)$ with $\det F_p > 0$. A measure of the elastic distortion that is not affected by superimposed rigid placements[2] is the Cauchy elastic distortion tensor $C_e = F_e^T F_e = F_p^{-T} C F_p^{-1}$. Given a time evolution $z = (\chi, F_p) \colon \Omega \times [0, T] \to \mathbb{R}^d \times \operatorname{Lin}$ with $\det F_p > 0$, we have that

$$\dot{F}_e = \dot{F} F_p^{-1} - F F_p^{-1} \dot{F}_p F_p^{-1} = \dot{F} F_p^{-1} - F_e \dot{F}_p F_p^{-1}.$$

[2]Given $Q \in \operatorname{Orth}^+$ and $a \in \mathbb{R}^d$, if one superimpose the rigid placement $x \mapsto x^* = Qx + a$, then the placement transforms as $\chi^* = Q\chi + a$ while $F_p^* = F_p$ does not change since it is referred to the reference configuration.

Consistently with that, for all $z, \tilde{w} \in Z$, we define $\tilde{V}_e = (\operatorname{grad} \tilde{v})F_p^{-1} - F_e \tilde{V}_p F_p^{-1}$.

Motivated by Remark 1.13, we decide that for $z = (\chi, F_p) \in Z$ with $\det F_p > 0$, the internal and and external virtual power functionals are of the form

$$(1.9) \qquad \langle \mathcal{I}, \tilde{w} \rangle = \int_\Omega P_e : \tilde{V}_e dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \vdots \operatorname{grad} \tilde{V}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v}$$

Here $P_e \colon \Omega \to \mathrm{Lin}$ is the elastic first Piola stress, $T_p \colon \Omega \to \mathrm{Lin}$ is the plastic microforce, $K_p \colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$ is the plastic microstress, $f \colon \Omega \to \mathbb{R}^d$ is the external volumetric force and $h \colon \partial\Omega \to \mathbb{R}^d$ is the external surface force. The strong form of the equilibrium equation $\mathcal{I} = \mathcal{X}$ can be easily obtained using the Green lemma and the fundamental lemma of the calculus of variations; we find

$$\begin{gathered} \operatorname{div}(P_e F_p^{-T}) + f = 0 \quad \text{in } \Omega \\ P_e F_p^{-T} n = h \quad \text{in } \partial\Omega \\ F_e^T P_e F_p^{-T} - T_p + \operatorname{div} K_p = 0 \quad \text{in } \Omega \\ K_p n = 0 \quad \text{in } \partial\Omega. \end{gathered}$$

This corresponds to the balance of macroscopic and microscopic forces.

The choice of the constitutive relations follows the same criteria used with the small deformation model. We consider $\Psi \colon Z \to \mathbb{R} \cup \{\infty\}$ and $\Phi \colon Z \times Z \to [0, \infty]$ of the form

$$\Psi(z) = \int_\Omega W_e(F_e) dx + \int_\Omega W_p(F_p, \operatorname{grad} F_p) dx \quad \text{and} \quad \Phi(z, \dot{z}) = \int_\Omega R(F_p, \dot{F}_p) dx.$$

This gives the constitutive relations

$$P_e \in \frac{\partial W_e}{\partial F_e}(F_e), \quad T_p \in \frac{\partial W_p}{\partial F_p}(F_p, \operatorname{grad} F_p) + \frac{\partial R}{\partial \dot{F}_p}(F_p, \dot{F}_p) \quad \text{and} \quad K_p \in \frac{\partial W_p}{\partial \operatorname{grad} F_p}(F_p, \operatorname{grad} F_p).$$

The elastic free energy density $W_e \colon \mathrm{Lin} \to \mathbb{R} \cup \{\infty\}$ is assumed to satisfy $W(F_e) = \infty$ if and only if $\det F_e \leq 0$ (in order to enforce the impenetrability constrain) and to be differentiable on its domain. The requirement that the energy is indifferent with respect to superimposed rigid motions impose W_e to be of the form $W_e(F_e) = \hat{W}_e(F_e^T F_e)$ for some $W_e \colon \mathrm{Sym} \to \mathbb{R} \cup \{\infty\}$. Similarly the plastic free energy density $W_p \colon \mathrm{Lin} \times (\mathrm{Lin} \otimes \mathbb{R}) \to \mathbb{R} \cup \{\infty\}$ is supposed to satisfy $W_p(F_p, \operatorname{grad} F_p) = \infty$ if $\det F_p \neq 1$ so that the plastic incompressibility is implied. The analogue of the Von Mises constitutive relation (known as J_2 flow theory) is obtained choosing

$$R(F_p, \dot{F}_p) = \begin{cases} Y|\dot{F}_p F_p^{-1}| & \text{if } \det F_p = 1 \\ \infty & \text{otherwise.} \end{cases}$$

Notice that in this definition appears the deviatoric plastic rate $L_p = \dot{F}_p F_p^{-1}$ (see also Remark 1.16).

Remark 1.16. A more geometric approach to plasticity theory at large strain (that we do not adopt in the spirit of the discussion we made in Section 1.1) is the following. Let M be the manifold of the couples $z = (\chi, F_p) \in Z$ such that $\det F > 0$ and $\det F_p = 1$. It is a Lie group (at least formally) with respect to the group operation

$$\begin{aligned} \bullet \colon & M \times M \to M, \\ & (z^1, z^2) \mapsto z^1 \bullet z^2 = (\chi^1 \circ \chi^2, (F_p^1 \circ \chi^2) F_p^2). \end{aligned}$$

Given a time evolution $z = (\chi, F_p) \colon [0, T] \to M$, we define the velocity gradient as $L = \dot{F} F^{-1}$ and the plastic rate as $L_p = \dot{F}_p F_p^{-1} \colon \Omega \times [0, T] \to \mathrm{Lin}$. Notice that the plastic strain rate is deviatoric, i.e., $L_p(x, t) \in \mathrm{Dev}$ for all $(x, t) \in \Omega \times [0, T]$. Indeed since $\det F_p = 1$, then

$$0 = \frac{d}{dt} \det F_p = \operatorname{cof} F_p : \dot{F}_p = \det F_p F_p^{-T} : \dot{F}_p = I : \dot{F}_p F_p^{-1} = \operatorname{tr} L_p.$$

If we define the elastic rate as $L_e = \dot{F}_e F_e^{-1}$ we easily see that $L = L_e + L_p$. This means that the Lie algebra is the linear space $\mathfrak{m}$ of the virtual rates $\tilde{w} = (\tilde{v}, \tilde{L}_p)$ with $\tilde{v} \colon \Omega \to \mathbb{R}^d$ and $\tilde{L}_p = \tilde{V}_p F_p^{-1} \colon \Omega \to \mathrm{DevSym}$. Consistently we let $\tilde{L} = (\operatorname{grad} \tilde{v}) F^{-1}$ and $\tilde{L}_e = \tilde{L} - \tilde{L}_p$. Then we can consider the internal and external virtual powers functional $\mathcal{I}$ and $\mathcal{X}$ as elements of $\mathfrak{m}^*$ of the form

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_\Omega T_e : \tilde{L}_e dx + \int_\Omega \left(T_p : \tilde{L}_p + K_p \vdots \operatorname{grad} \tilde{L}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v},$$

where T_p and K_p does not coincides with the ones appearing in (1.9).

CHAPTER 2

Solution concepts and general existence results

2.1. An existence result for evolutionary variational inequalities

Let V be a separable and reflexive Banach space. Consider a mechanical system with space of configurations V and assume that

(E) the free energy $\Phi\colon V \to \mathbb{R}$ is quadratic in the sense that $\Psi(u) = \frac{1}{2}\langle \mathcal{A}u, u\rangle$ for some $\mathcal{A} \in \mathscr{L}(V, V^*)$, and Ψ has superlinear growth, i.e.,
$$\lim_{\|u\|\to\infty} \frac{\Psi(u)}{\|u\|} = \infty,$$

(D) the dissipation potential $\Phi\colon V \to [0, \infty]$ is convex, proper, l.s.c., positively 1-homogeneous and Lipschitz on its domain.

Assumption (E) immediately implies that Ψ is Fréchet differentiable with Fréchet differential $\partial\Psi(u) = \mathcal{A}$ for all $u \in V$. Since Φ is convex and Lipschitz continuous on its domain then it admits a non empty subdifferential $\partial\Phi(v)$ for all $v \in \operatorname{dom}\Phi$. **Per il teorema di Rockafeller sappiamo che è subdifferenziabile nell'interno del suo dominio, poi però data la Lipschitzianità su tutto il bordo immagino che si possa dimostrare che è subdifferenziabile su tutto il dominio in realtà.** In this case Problem 1.7 can be formulated explicitely as follows.

Problem 2.1. Given $\mathscr{X} \in H^1((0,T), V^*)$ with $\mathcal{X}(0) = 0$, find $u \in \mathrm{AC}([0,T], V)$ such that $u(0) = 0$ and

$$\langle \mathcal{A}u(t), v - \dot{u}(t)\rangle + \Phi(v) - \Phi(\dot{u}(t)) \geq \langle \mathcal{X}(t), v - \dot{u}(t)\rangle \quad \text{for all } v \in V \tag{2.1}$$

for a.e. $t \in (0,T)$.

Inequalities of the form (2.1) are called *evolutionary variational inequalities*. The following theorem ensure the existence and uniqueness of solutions to Problem 2.1.

Theorem 2.2. Let V a separable and reflexive Banach space and let assumption (E) and (D) hold. Then for every $\mathcal{X} \in H^1((0,T), V^*)$ with $\mathcal{X}(0) = 0$, there exists a unique $u \in \mathrm{AC}([0,T], V)$ solution to Problem 2.1.

Proof. See [35]. **Riportarla**. □

2.2. Energetic solutions

2.2.1. Motivation. Let Q be a separable and reflexive Banach space. Let $\mathcal{E}: Q\times[0,T] \to \mathbb{R}$ Fréchet differentiable such that $\mathcal{E}(\cdot, t)$ is convex for all $t \in [0,T]$ and let $\Phi\colon Q\times[0,\infty]$ be convex, lower semicontinuous and positively 1-homogeneous. Consider the following doubly nonlinear equation

$$\partial_q\mathcal{E}(q(t), t) + \partial\Phi(\dot{q}(t)) \ni 0. \tag{2.2}$$

We say that $q \in \mathrm{AC}([0,T], V)$ is a differentiable solution if (2.2) holds for a.e. $t \in (0,T)$. From Proposition B.33 this is equivalent to

$$-\partial_q\mathcal{E}(q(t), t) \in \partial\Phi(0) \quad \text{and} \quad \Phi(\dot{q}(t)) = -\langle \partial_q\mathcal{E}(q(t), t), \dot{q}(t)\rangle = -\frac{d}{dt}\mathcal{E}(q(t), t) + \partial_t\mathcal{E}(q(t), t) \tag{2.3}$$

for a.e. $t \in (0,T)$. The first condition is the requirement that the conservative forces belong to the admissibility set $\partial\Phi(0)$. From the definition of subdifferential, this is equivalent to
$$\Phi(\hat{q} - q(t)) \geq \Phi(0) - \langle \partial_q\mathcal{E}(q(t), t), \hat{q} - q(t)\rangle \quad \text{for all } \hat{q} \in Q.$$

Exploiting the convexity of $\mathcal{E}(\cdot, t)$ it follows that
$$\mathcal{E}(\hat{q}, t) - \mathcal{E}(q(t), t) \geq \lim_{\epsilon\to 0} \frac{\mathcal{E}(q(t) + \epsilon(\hat{q} - q(t)), t) - \mathcal{E}(q(t), t)}{\epsilon} = \langle \partial_q\mathcal{E}(q(t), t), \hat{q} - q(t)\rangle.$$

Then the first condition in (2.3), implies

$$\mathcal{E}(q(t), t) \leq \mathcal{E}(\hat{q}, t) + \Phi(\hat{q} - q(t)) \quad \text{for all } \hat{q} \in Q, \tag{2.4}$$

Since Φ is 1-homogeneous, also the inverse implication holds, indeed

$$\langle \partial_q \mathcal{E}(q(t),t), \hat{q} - q(t)\rangle = \lim_{\epsilon\to 0} \frac{\mathcal{E}(q(t)+\epsilon(\hat{q}-q(t)),t) - \mathcal{E}(q(t),t)}{\epsilon} \geq \frac{\Phi(\epsilon(\hat{q}-q(t)))}{\epsilon} = \Phi(\hat{q}-q(t)).$$

The mechanical meaning of condition (2.4) is the following: the energy $\mathcal{E}(q(t),t) - \mathcal{E}(\hat{q},t)$ released by the system while passing from $q(t)$ to $\hat{q}$ cannot exceed the dissipation $\Phi(\hat{q}-q(t))$. This is a stability condition. The second condition in (2.3) is the power balance. Integrating it on the interval $(0,t)$ for $t \in (0,T]$ we get the energy balance

$$\mathcal{E}(q(t),t) = \mathcal{E}(q(0),0) + \int_0^t \partial_t \mathcal{E}(q(s),s)ds + \int_0^t \Phi(\dot{q}(s))ds.$$

The first integral is the work of external forces, while the second integral is the dissipation. The following lemma gives an alternative way to express the dissipation.

Lemma 2.3. For all $0 \leq s < t \leq T$ let

$$\mathrm{Diss}_\Phi(q,[s,t]) = \inf\left\{ \sum_{k=1}^N \Phi(q(t_k) - q(t_{k-1})) \,\middle|\, s = t_0 < \cdots < t_N = t \right\}.$$

Then

$$\mathrm{Diss}_\Phi(q,[s,t]) = \int_s^t \Phi(\dot{q}(r))dr.$$

Proof. Consider any subdivision $\{t_k\}_{k=0}^N$ of $[s,t]$. By Jensen inequality and in view of the 1-homogeneity of Φ, it holds that

$$\Phi(q(t_k) - q(t_{k-1})) = \Phi\left((t_k - t_{k-1})\frac{1}{t_k - t_{k-1}} \int_{t_{k-1}}^{t^k} \dot{q}(r)dr\right) \leq \int_{t_{k-1}}^{t_k} \Phi(\dot{q}(r))dr \quad \text{for all } k.$$

Then the inequality

$$\mathrm{Diss}_\Phi(q,[s,t]) \leq \int_s^t \Phi(\dot{q}(r))dr$$

holds. To prove the other inequality we split the interval $[s,t]$ into intervals

$$I_k^n = \left[s + \frac{k-1}{n}(t-s), s + \frac{k}{n}(t-s)\right]$$

for $k \in \{1,\ldots,n\}$ with $n \in \mathbb{N}$. Then, by lower semicontinuity of Φ and by the Fatou's lemma,

$$\begin{aligned}\int_s^t \Phi(\dot{q}(r))dr &\leq \int_s^t \liminf_{n\to\infty} \frac{n}{t-s}\Phi\left(q\left(r+\frac{t-s}{n}\right) - q(r)\right)dr \leq \\ &\leq \liminf_{n\to\infty} \frac{n}{t-s} \int_s^{t-\frac{t-s}{n}} \Phi\left(q\left(r+\frac{t-s}{n}\right) - q(r)\right)dr.\end{aligned}$$

Now

$$\begin{aligned}\int_s^{t-\frac{t-s}{n}} \Phi\left(q\left(r+\frac{t-s}{n}\right) - q(r)\right)dr &= \sum_{k=1}^{n-1} \int_0^{\frac{t-s}{n}} \Phi\left(q\left(\frac{k}{n}(t-s)+r\right) - q\left(\frac{k-1}{n}(t-s)+r\right)\right)dr \leq \\ &\leq \int_0^{\frac{t-s}{n}} \sum_{k=1}^{n-1} \Phi\left(q\left(\frac{k}{n}(t-s)+r\right) - q\left(\frac{k-1}{n}(t-s)+r\right)\right)dr \leq \\ &\leq \frac{t-s}{n}\mathrm{Diss}_\Phi(q,[s,t]).\end{aligned}$$

We have proved that

$$\int_s^t \Phi(\dot{q}(r))dr \leq \mathrm{Diss}_\Phi(q,[s,t])$$

as desired. □

Summarizing, we have seen that (2.2) is equivalent

$$\mathcal{E}(q(t),t) \leq \mathcal{E}(\hat{q},t) + \Phi(\hat{q}-q(t)) \quad \text{for all } \hat{q} \in Q \text{ and all } t \in [0,T],$$

$$\mathcal{E}(q(t),t) = \mathcal{E}(q(0),0) + \int_0^t \partial_t \mathcal{E}(q(s),s)ds + \mathrm{Diss}_\Phi(q,[0,t]) \quad \text{for all } t \in [0,T].$$

This conditions makes sense also if $\mathcal{E}$ is not Fréchet differentiable and q is not absolutely continuous.

This results suggest to define energetic solutions directly as evolutions $q\colon [0,T] \to Q$ satisfying (2.5) also when less regularity is assumed. We now explore further this possibility. Let Q be a separable and reflexive Banach space and $\mathcal{E}\colon Q \times [0,T] \to [0,\infty]$. Rather than using a dissipation potential, we describe the system's dissipation in terms of a dissipation distance. This is defined as a map $\mathcal{D}\colon Q \times Q \to [0,\infty]$, where $\mathcal{D}(q_0, q_1)$ denotes the minimum energy dissipated by the system when it jumps from q_0 to q_1. The dissipation along an evolution $q\colon [0,T] \to Q$ in the time interval $[s,t]$ is defined as

$$\mathrm{Diss}_{\mathcal{D}}(q,[s,t]) = \sup\left\{ \sum_{k=1}^{N} \mathcal{D}(q(t_{k-1}), q(t_k)) \,\middle|\, s = t_0 < \cdots < t_N = t \right\}.$$

Then we say that q is an energetic solution if

$$\mathcal{E}(q(t),t) \le \mathcal{E}(\hat{q},t) + \mathcal{D}(q(t),\hat{q}) \quad \text{for all } \hat{q} \in Q \text{ and all } t \in [0,T],$$

$$\mathcal{E}(q(t),t) = \mathcal{E}(q(0),0) + \int_0^t \partial_t \mathcal{E}(q(s),s)ds + \mathrm{Diss}_{\mathcal{D}}(q,[0,t]) \quad \text{for all } t \in [0,T].$$

In the following, the definition will be made more precise.

Remark 2.4. Given a dissipation potential $\Phi\colon Q \times Q \to [0,\infty]$, $(q,\dot{q}) \mapsto \Phi(q,\dot{q})$, such that $\Phi(q,\cdot)$ is positively 1-homogeneous for all $q \in Q$, we can define the dissipation distance as

$$\mathcal{D}(q_0,q_1) = \inf\left\{ \int_0^1 \Phi(q(t),\dot{q}(t))dt \,\middle|\, q \in C^1([0,T],Q),\ q(0) = q_0 \text{ and } q(1) = q_1 \right\}. \tag{2.7}$$

Notice that the range of the integral can be choosen arbitrary due to the positive 1-homogeneity of Φ. The particular choice (2.7) of $\mathcal{D}$ is motivated by the following *formal* consideration. Let $q\colon [0,T] \to Q$ be an energetic solution. Fix $t \in [0,T]$, $w \in Q$, $\epsilon > 0$ and let $\tilde{q}(r) = q(t) + r\epsilon w$. Then

$$\mathcal{E}(q(t),t) \le \mathcal{E}(q(t)+\epsilon w,t) + \mathcal{D}(q(t),q(t)+\epsilon w) \le \mathcal{E}(q(t)+\epsilon w,t) + \int_0^1 \Phi(\tilde{q}(r),\dot{\tilde{q}}(r))dr$$

so that

$$-\frac{\mathcal{E}(q(t)+\epsilon w,t) - \mathcal{E}(q(t),t)}{\epsilon} \le \frac{1}{\epsilon}\int_0^1 \Phi(q(t)+r\epsilon w,\epsilon w)dr = \int_0^1 \Phi(q(t)+r\epsilon w,w)dr.$$

Assuming enough regularity, we let $\epsilon \to 0$ and we get

$$-\langle \partial_q \mathcal{E}(q(t),t), w\rangle \le \Phi(q(t),w).$$

This means that $-\partial_q \mathcal{E}(q(t),t) \in \partial\Phi(q(t),\cdot)(0)$. Differentiating the energy balance

$$-\partial_q \mathcal{E}(q(t),t) = \Phi(q(t),\dot{q}(t)).$$

This means that $-\partial_q \mathcal{E}(q(t),t) \in \partial\Phi(q(t),\cdot)(q(t))$. Thus (2.7) formally implies that energetic solutions are also solutions to the Biot inclusion.

2.2.2. Assumptions and preliminary results.

Assume that

(S1) Y and Z separable and reflexive Banach spaces,

(S2) $\mathscr{Y}$ and $\mathscr{Z}$ weakly closed subsets of Y and Z respectively.

Let $Q = Y \times Z$ and $\mathcal{Q} = \mathscr{Y} \times \mathscr{Z}$. The assumptions on the energy functional are that

(E1) $\mathcal{E}\colon [0,T] \times \mathcal{Q} \to (-\infty,\infty]$,

(E2) for all $t \in [0,T]$ and $E \in \mathbb{R}$ the set $\mathscr{D}_E := \{q \in \mathcal{Q} \mid \mathcal{E}(q,t) \le E\}$ is bounded and weakly closed

(E3) there exists $\mathscr{D} \subset \mathcal{Q}$ such that $\operatorname{dom}\mathcal{E}(\cdot,t) = \mathscr{D}$ for all $t \in [0,T]$,

(E4) for all $q \in \mathscr{D}$, the function $\mathcal{E}(q,\cdot)$ is $C^1([0,T])$ and
  (i) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q,t)| \le C_1^E(C_0^E + \mathcal{E}(q,t))$ for all $q \in \mathcal{Q}$ and $t \in [0,T]$,
  (ii) for each $E \in \mathbb{E}$ there exists a modulus of continuity ω^E such that for all $q \in \mathscr{D}_E$ and $t,s \in [0,T]$

$$|\partial_t \mathcal{E}(q,t) - \partial_t \mathcal{E}(q,s)| \le \omega^E(|t-s|).$$

Notice that (E2) is equivalent to say that $\mathcal{E}$ is weakly lower semicontinuous and has bounded sublevels. Moreover (E4, i) implies

$$\mathcal{E}(q,t_1) + C_0^E \le e^{C_1^E|t_1-t_0|}\left(\mathcal{E}(q,t_1) + C_0^E\right) \text{ for all } q \in \mathscr{D} \text{ and } t_0,t_1 \in [0,T].$$

This follows from the following lemma.

Lemma 2.5. Let $f \in C^1([0,T])$ and there exists $a \in \mathbb{R}$ and $b > 0$ such that $|\dot{f}(t)| \leq b(f(t)+b)$ for all $t \in [0,T]$. Then for all $t_0, t_1 \in [0,T]$ it holds $f(t_0) + a \leq e^{b|t_1-t_0|}(f(t_1)+a)$.

Proof. Let $z(t) = f(t) + a$. Then $|\dot{z}| \leq bz$. It follows that $\dot{z} \leq bz$ and

$$z(t) \leq z(t_0) + \int_{t_0}^{t} bz(s)ds \quad \text{for all } 0 \leq t_0 \leq t_1 \leq T.$$

By (B.1) it follows that $z(t_1) \leq e^{b(t_1-t_0)}z(t_0)$ all for all $0 \leq t_0 \leq t_1 \leq T$. But it holds also $\dot{z} \leq -bz$. If we set $\zeta(t) = z(T-t)$ then $\dot{\zeta} \leq b\zeta$. Then reasoning as before we get $z(t_1) \leq \zeta(T-t_1) \leq e^{b(t_0-t_1)}\zeta(T-t_0) \leq e^{b(t_0-t_1)}z(t_0)$ for all $0 \leq t_1 \leq t_0 \leq T$. This concludes the proof. □

The assumptions on the dissipation distance are

(D1) $\mathcal{D}\colon \mathcal{Z} \times \mathcal{Z} \to [0,\infty]$ is a quasidistance, that is $\mathcal{D}(z_0,z_1) = 0$ if and only if $z_0 = z_1$ and $\mathcal{D}(z_0,z_2) \leq \mathcal{D}(z_0,z_1) + \mathcal{D}(z_1,z_2)$ for all $z_1, z_2 \in \mathcal{Z}$,

(D2) $\mathcal{D}$ is weakly lower semicontinuous, meaning that $z_n \rightharpoonup z$ and $\hat{z}_n \rightharpoonup \hat{z}$ implies $\mathcal{D}(z,\hat{z}) \leq \liminf_{n\to\infty} \mathcal{D}(z_n,\hat{z}_n)$.

For a given $q = (y,z) \in \mathcal{Q}$, sometimes we write $\mathcal{D}(q)$ to mean $\mathcal{D}(z)$.

Definition 2.6. A sequence $(q_n,t_n) \in \mathcal{Q} \times [0,T]$ is said to be stable if

$$\sup_n \mathcal{E}(q_n,t_n) \leq \infty \quad \text{and} \quad q_n \in \mathcal{S}(t_n)$$

where the stable set is defined by

$$\mathcal{S}(t) = \{q \in \mathcal{Q} \mid \mathcal{E}(q,t) < \infty \text{ and } \mathcal{E}(q,t) \leq \mathcal{E}(\hat{q},t) + \mathcal{D}(q,\hat{q}) \text{ for all } \hat{q} \in \mathcal{Q}\}$$

We add the following assumptions regarding the stable set

(C1) for all stable sequences $(q_n,t_n) \in \mathcal{Q}\times[0,T]$ such that $q_n \rightharpoonup q$ and $t_n \to t$, it holds that $\partial_t\mathcal{E}(q_n,t) \to \partial_t\mathcal{E}(q,t)$,

(C2) the stable sets are closed in the sense that, if $(q_n,t_n) \in \mathcal{Q} \times [0,T]$ is a stable sequence such that $q_n \rightharpoonup q$ and $t_n \to t$, then $q \in \mathcal{S}(t)$.

2.2.3. Definition of energetic solutions.

We define energetic solutions as follows.

Definition 2.7. A map $q\colon [0,T] \to \mathcal{Q}$ is called energetic solution if it is stable at every instant, that is

$$q(t) \in \mathcal{S}(t) \text{ for all } t \in [0,T],$$

if $t \mapsto \partial_t\mathcal{E}(q(t),t)$ belongs to $L^1((0,T))$ and the energy identity

$$\mathcal{E}(q(t),t) + \mathrm{Diss}_{\mathcal{D}}(z,[s,t]) = \mathcal{E}(q(0),0) + \int_0^t \partial_t\mathcal{E}(q(s),s)ds$$

holds for all $t \in [0,T]$. Here the dissipation is defined as

$$\mathrm{Diss}_{\mathcal{D}}(z,[s,t]) = \sup\left\{\sum_{k=1}^{N} \mathcal{D}(z(t_{k-1}),z(t_k)) \,\middle|\, s = t_0 < \cdots < t_N = t\right\}$$

for all $0 \leq s < t \leq T$.

2.2.4. Existence theorem.

Theorem 2.8. (existence of energetic solution) If the hypotheses (S), (E), (D) and (C) holds, then for all $q_0 \in \mathcal{S}(0)$ there exists a bounded energetic solutions q such that $q(0) = q_0$.

2.2.5. Tools to verify the compatibility conditions.

Proposition 2.9. Assume that the following property holds

(PC1) for all stable sequence (q_k,t_k) such that $(q_k,t_k) \rightharpoonup (q,t)$ and for all $\hat{q} \in \mathcal{Q}$, there exist a sequence $\hat{q}_k$ such that

$$\limsup_{k\to\infty} \left(\mathcal{E}(\hat{q}_k,t_k) + \mathcal{D}(q_k,\hat{q}_k)\right) \leq \mathcal{E}(\hat{q},t) + \mathcal{D}(q,\hat{q}).$$

Then for all stable sequence (q_k,t_k) such that $(q_k,t_k) \rightharpoonup (q,t)$ it holds that $\lim_{k\to\infty}\mathcal{E}(q_k,t_k) = \mathcal{E}(q,t)$. Moreover (C2) holds.

Assume that in addiction to (PC1) also the following property holds

(PC2) for all $\epsilon > 0$ and $E \in \mathbb{R}$, there exists $\delta > 0$ such that for all $t, s \in [0, T]$ and $q \in \mathcal{Q}$ the implication

$$\mathcal{E}(q, t) \le E \text{ and } |t - s| < \delta \text{ implies } |\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| < \epsilon$$

holds.

Then (C1) holds.

2.3. Balanced-viscosity solutions

2.3.1. Introduction. Let

$$V \text{ a separable and reflexive Banach space.}$$

Consider an evolution $u\colon [0, T] \to V$ and the doubly nonlinear equation for the evolution

$$\partial_u \mathcal{E}(u(t), t) + \partial\Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0, T).$$

Here $\mathcal{E}$ is the energy and Φ is the dissipation potential. In application usually $\mathcal{E}(u, t) = \Psi(u) - \langle \mathcal{F}(t), u\rangle$ where Ψ is the free energy and $\mathcal{F}$ is the external action. For a given $\eta > 0$ consider the time-rescaled evolution $u\colon (0, \eta^{-1}T) \to V$ and the time-rescaled equation

$$\partial_u \mathcal{E}(u(t), \eta t) + \partial\Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0, \eta^{-1}T). \tag{2.8}$$

For $\eta \to 0$ this describe the situation in which the external actions are mush slower with respect to the characteristic relaxation time of the system. Equation (2.8) can be written as

$$\partial_u \mathcal{E}(u_\eta(t), t) + \partial\Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0, T) \tag{2.9}$$

where $u_\eta(t) = u(\eta^{-1}t)$ and $\Phi_\eta(v) = \eta^{-1}\Phi(\eta v)$.

2.3.2. Assumptions. The energy $\mathcal{E}$ is supposed to satisfy the following assumptions.

$\mathcal{E} \in C^1(V \times [0, T])$

for all $E > 0$ the sublevel $D_E = \{u \in V \mid \sup_{t\in[0,T]} \mathcal{E}(u, t) \le D_E\}$ is weakly compact

($\mathcal{E}$.3) there exists $C > 0$ such that $|\partial_t \mathcal{E}(u, t)| \le C(1 + \mathcal{E}(u, t))$ for all $u \in V$ and $t \in [0, T]$

The dissipation potential is defined by

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$ its rescaled version is

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

Here $\mathcal{R}$ is a rate-independent dissipation potential and $\mathcal{V}$ is a rate-dependent dissipation potential that models viscosity. It is assumed that $\mathcal{R}$ and $\mathcal{V}$ satisfy the following assumptions.

(Φ.1) $\mathcal{R}$ and $\mathcal{V}$ from V into $[0, \infty)$ are continuous, convex and strictly positive in $V \setminus \{0\}$

(Φ.2) $\mathcal{R}$ is positively 1-homogeneous

$\mathcal{V}(v) = \phi(\|v\|)$ with $\phi \in C^1([0, \infty))$ convex such that $\phi(0) = 0$, $\phi'(0) = 0$, $\phi(r) > 0$ for $r > 0$ and $\lim_{r\to\infty} \phi'(r) = \infty$

Notice that from Proposition B.33 it holds $\mathcal{R}(0) = 0$. Moreover notice that ϕ can be extended to an even $C^1(\mathbb{R})$ function.

2.3.3. Preliminary results regarding the dissipation potential. Let $K = \partial\mathcal{R}(0)$. By Proposition B.33 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$. Notice that there exists a constant $M > 0$ such that $\mathcal{R}(v) \le M\|v\|$ for all $v \in V$. Indeed by contradiction assume that there exist $v_n \in V \setminus \{0\}$ such that $\mathcal{R}(v_n) \ge n\|v_n\|$. Since $\mathcal{R}$ is positively homogeneous the inequality implies $\mathcal{R}(n^{-1}v_n/\|v_n\|) \ge 1$ so $\liminf_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) \ge 1$. But $n^{-1}v_n/\|v_n\| \to 0$ so by continuity of $\mathcal{R}$ it follows $\lim_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) = 0$. This is the desired contradiction.

Since Φ_1 is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta\to 0^+} \Phi_\eta(v) = \inf_{\eta>0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

It simple to check that $\Phi^*_\eta(\xi) = \eta^{-1}\Phi^*(\xi)$. It is possible to obtain a more explicit representation formula for Φ^*. By the inf-convolution duality formula

$$\Phi^*_\eta(\xi) = \left(\mathcal{R} + \mathcal{V}_\eta\right)^*(\xi) = \inf_{\zeta\in V}\left(\mathcal{R}^*(\zeta) + \mathcal{V}^*_\eta(\xi - \zeta)\right) = \eta^{-1}\inf_{\zeta\in K} \mathcal{V}^*(\xi - \zeta) = \eta^{-1}\min_{\zeta\in K} \mathcal{V}^*(\xi - \zeta). \tag{2.10}$$

Last equality holds true because, since K is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

Lemma 2.10. The function ϕ^* satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.

Proof. Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. □

Using that lemma and (2.10)

$$\Phi_\eta^*(\xi) = \eta^{-1} \min_{\zeta \in K} \phi^*(\|\xi - \zeta\|). \tag{2.11}$$

2.3.4. Absolutely continuous and BV evolutions. Let $\mathcal{R}\colon V \to [0,\infty)$ satisfying (Φ.1) and (Φ.2). We say that a map $u\colon [0,T] \to V$ is $\mathrm{AC}_{\mathcal{R}}([0,T],V)$ if there exists $m \in L^1((0,T),[0,\infty))$ such that

$$\mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \leq t_0 < t_1 \leq T. \tag{2.12}$$

Lemma 2.11. Let $u \in \mathrm{AC}_{\mathcal{R}}([0,T],V)$. Then

$$\mathcal{R}[\dot{u}](t) \doteq \lim_{h\to 0} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right)$$

exists for a.e. $t \in (0,T)$ and $\mathcal{R}[\dot{u}] \in L^1((0,T))$. Moreover for all m as in (2.12)

$$\mathcal{R}[\dot{u}](t) \leq m(t) \quad \text{for a.e. } t \in (0,T).$$

Proof. For any $s \in (0,T)$ and $t \in (0,T)$ let $\delta_s(t) = \mathcal{R}(u(t) - u(s))$. Then

$$\delta_s(t_1) - \delta_s(t_0) \leq \mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for all } 0 \leq t_0 < t_1 \leq T.$$

The second inequality holds since $\mathcal{R}$ is sub additive by Proposition B.33 so

$$\delta_s(t_1) = \mathcal{R}(u(t_1) - u(s)) \leq \mathcal{R}(u(t_1) - u(t_0)) + \mathcal{R}(u(t_0) - u(s)) = \mathcal{R}(u(t_1) - u(t_0)) + \delta_s(t_0).$$

Then the map $t \mapsto \delta_s(t) - \int_0^t m(r)dr = \beta_s(t)$ is non increasing on $(0,T)$ and in particular it is differentiable a.e. on $(0,T)$. This in turns implies that δ_s is differentiable a.e. on $(0,T)$ and

$$\dot{\delta}_s(t) \leq \delta(t) \doteq \liminf_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right) \quad \text{for a.e. } t \in (0,T).$$

Note that

$$0 \leq \delta(t) \leq \liminf_{h\to 0^+} \frac{1}{h} \int_t^{t+h} m(s)ds = m(t) \quad \text{for a.e. } t \in (0,T)$$

and this implies $\delta \in L^1((0,T))$. Since $\delta_s(t) = \beta_s(t) + \int_0^t m(r)dr$ and β_s is non increasing then the singular part of the distributional derivative of δ_s is non positive and

$$\mathcal{R}(u(t) - u(s)) = \delta_s(t) \leq \int_s^t \dot{\delta}_s(r)dr \leq \int_s^t \delta(r)dr. \tag{2.13}$$

Let

$$\gamma(t) \doteq \limsup_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right).$$

By (2.13) then $\delta(t) \leq \gamma(t) \leq \delta(t)$. This proves that

$$\delta^+(t) \doteq \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right) = \delta(t) \quad \text{exists for a.e. } t \in (0,T).$$

Applying the previous argument to $t \mapsto u(T-t)$ and $v \mapsto \mathcal{R}(-v)$ in place of u and $\mathcal{R}$ respectively, follows that

$$\delta^-(t) = \lim_{h\to 0^-} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right) = \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t) - u(t-h)}{h}\right) \quad \text{exists for a.e. } t \in (0,T)$$

and satisfy the conditions

$$\mathcal{R}(u(t) - u(s)) \leq \int_s^t \delta^-(r)dr.$$

and $\delta^-(t) \leq m(t)$ for a.e. $t \in (0,T)$. This means that δ^+ and δ^- coincides both with the minimal $m \in L^1((0,T),(0,\infty))$ for which (2.12) holds. □

Given a map $u\colon [0,T] \to V$ we define the total variation on $[a,b] \subset [0,T]$ as

$$\mathrm{Var}_{\mathcal{R}}(u,[a,b]) = \sup \left\{ \sum_{k=1}^{N} \mathcal{R}(u(t_k)) - \mathcal{R}(u(t_{k-1})) \,\middle|\, a = t_0 < \cdots < t_N = b \right\}.$$

We say that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$ if $\mathrm{Var}_{\mathcal{R}}(u,[0,T]) < \infty$.

Lemma 2.12. It holds that $\mathrm{AC}([0,T],V) \subset \mathrm{AC}_{\mathcal{R}}([0,T],V)$ and $\mathrm{BV}([0,T],V) \subset \mathrm{BV}_{\mathcal{R}}([0,T],V)$.

Proof. Recall that there exists $M > 0$ such that $\mathcal{R}(v) \le M\|v\|$. Let $u \in AC([0,T],V)$. Then

$$\mathcal{R}(u(t_1) - u(t_0)) \le M\|u(t_1) - u(t_0)\| \le \int_{t_0}^{t_1} Mm(s)ds.$$

This shows that $u \in \mathrm{AC}_{\mathcal{R}}([0,T],V)$. Next let $u \in \mathrm{BV}([0,T],V)$. Then

$$\mathrm{Var}_{\mathcal{R}}(u,[0,T]) \le M\mathrm{Var}(u,[0,T]).$$

This implies that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$. □

2.3.5. A chain rule formula for the energy functional.

Proposition 2.13. Let $u \in AC([0,T],V)$. Then $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$ and

$$\frac{d}{dt}\mathcal{E}(u(t),t) = \langle \partial_u \mathcal{E}(u(t),t), \dot{u}(t)\rangle + \partial_t \mathcal{E}(u(t),t) \quad \text{for a.e. } t \in (0,T). \tag{2.14}$$

Proof. *Step 1.* Since u is continuous then $K = u([0,T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in K. Then up to a subsequence $t_n \to t$ and by continuity $u_n = u(t_n) \to u(t)$.
Step 2. Since $\mathcal{E} \in C^1(V \times [0,T])$ then by Proposition B.14 it is Lipschitz in $K \times [0,T]$. Let $L > 0$ be the Lipschitz constant.
Step 3. Let $0 \le s < t \le T$ and $m \in L^1((0,T),(0,\infty))$ as in Definition B.4. Then

$$|\mathcal{E}(u(t),t) - \mathcal{E}(u(s),s)| \le L\left(\|u(t) - u(s)\| + |t-s|\right) \le L\int_s^t (m(r) + r)\, dr.$$

This proves that $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$.
Step 4. By Proposition B.5 it follows that u is a.e. differentiable in $(0,T)$. This implies formula (2.14). □

2.3.6. Existence results in presence of positive viscosity.

Proposition 2.14. Let $u_0 \in V$. Then there exists $u_\eta \in \mathrm{AC}([0,T],V)$ such that $u_\eta(0) = u_0$ and

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for a.e. } t \in (0,T).$$

Moreover the identity

$$\mathcal{E}(u_\eta(t),t) + \int_s^t \left(\Phi_\eta(\dot{u}_\eta(r)) + \Phi_\eta^*(-\partial_u\mathcal{E}(u_\eta(r),r)\right) dr = \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(r),r)dr \tag{2.15}$$

holds for every $0 \le s < t \le T$.

Proof. I still have to report the proof of existence. We prove the identity (2.15). By the properties of subdifferentials

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\langle \partial_u \mathcal{E}(u_\eta(t),t), \dot{u}_\eta(t)\rangle \quad \text{for a.e. } t \in (0,T).$$

By (2.14) we have

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\frac{d}{dt}\mathcal{E}(u_\eta(t),t) + \partial_t\mathcal{E}(u_\eta(t),t) \quad \text{for a.e. } t \in (0,T).$$

Integrating this identity in (s,t) we get the thesis. □

2.3.7. Contact dissipation potential. The dissipation contact potential $\mathfrak{p}\colon V \times V^* \to [0,\infty)$ is

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\Phi_\eta(v) + \Phi^*_\eta(\xi)\right) = \inf_{\eta>0}\eta^{-1}\left(\Phi(\eta v) + \Phi^*(\xi)\right).$$

By (2.11)

$$\begin{aligned}\mathfrak{p}(v,\xi) &= \inf_{\eta>0}\left(\mathcal{R}(v) + \eta^{-1}\phi(\eta\|v\|) + \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|)\right) = \\ &= \mathcal{R}(v) + \inf_{\eta>0}\left[\eta^{-1}\min_{\zeta\in K}\left(\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)\right)\right] = \\ &= \mathcal{R}(v) + \min_{\zeta\in K}\inf_{\eta>0}\frac{\phi(\eta\|v\|)+\phi^*(\|\xi-\zeta\|)}{\eta}\end{aligned}$$

where last equality holds because the minimum point does not depends on η. But for all r and γ in $\mathbb{R}$

$$\inf_{\eta>0}\frac{\phi(\eta r)+\phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r)+\phi^*(\gamma)}{\eta} \geq \gamma r$$

and the identity holds for η such that $\gamma = \phi'(\eta r)$. Then

$$\mathfrak{p}(v,\xi) = \mathcal{R}(v) + \|v\|\min_{\zeta\in K}\|\xi-\zeta\|. \tag{2.16}$$

Let $\mathfrak{f}\colon V\times V\times[0,T]\to[0,\infty)$ defined by

$$\mathfrak{f}(u,v,t) = \mathfrak{p}(v,-\partial_u\mathcal{E}(u,t)) = \mathcal{R}(v) + \mathfrak{e}(u,t)\|v\|$$

where $\mathfrak{e}\colon V\times[0,T]\to[0,\infty)$ is defined by

$$\mathfrak{e}(u,t) = \min_{\zeta\in K}\|\partial_u\mathcal{E}(u,t)+\zeta\|.$$

Notice that $\mathfrak{e}(u,t)=0$ if and only if $-\partial_u\mathcal{E}(u,t)\in K$.

2.3.8. Parametrized balanced-viscosity solutions. Let $u_\eta \in \mathrm{AC}([0,T],V)$ be the solution to the doubly nonlinear equation

$$\partial_u\mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot u_\eta(t)) \ni 0 \quad \text{for } t\in(0,T)$$

The identity (2.15) implies

$$\mathcal{E}(u_\eta(t),t) + \int_s^t \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dr \leq \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(t),t)dr \tag{2.17}$$

for all $0\leq s<t\leq T$.

Lemma 2.15. There exists a constant $C>0$ such that

$$\int_0^T \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dt \leq C \quad \text{for all } \eta>0.$$

Proof. Since $\mathfrak{f}\geq 0$ by ($\mathcal{E}$.3) then inequality (2.17) implies

$$\mathcal{E}(u_\eta(t),t) \leq \mathcal{E}(u_0,0) + \int_0^t \partial_t\mathcal{E}(u(s),s)ds \leq \mathcal{E}(u_0,0) + C\int_0^t(1+\mathcal{E}(u_\eta(s),s))ds.$$

By ($\mathcal{E}$.3) and Gronwall inequality $-1\leq\mathcal{E}(u_\eta(t),t)\leq C_1$ for all $t\in[0,T]$ and $\eta>0$. Then (2.17) implies

$$\int_0^T \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dt \leq 1 + \mathcal{E}(u_0,0) + CT(1+C_1).$$

This concludes the proof. □

Let $\boldsymbol{s}_\eta\colon[0,T]\to[0,S_\eta]$ defined by

$$\boldsymbol{s}_\eta(t) = t + \int_0^t \mathfrak{f}(u_\eta(s),\dot u_\eta(s),s)ds \quad \text{and} \quad S_\eta = \boldsymbol{s}_\eta(T). \tag{2.18}$$

Let also

$$\boldsymbol{t}_\eta = \boldsymbol{s}_\eta^{-1}\colon[0,S_\eta]\to[0,T] \quad \text{and} \quad \boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta)\colon[0,S_\eta]\to V.$$

To write the time-rescaled energy identity we introduce the map $\mathfrak{F}_\eta\colon V\times[0,T]\times V\times(0,\infty)\to[0,\infty)$ defined by

$$\mathfrak{F}_\eta(u,t,v,a)=a\Phi_\eta(v/a)+\Phi^*_\eta(-\partial_u\mathcal{E}(u,t))-a\partial_t\mathcal{E}(u,t)=\mathcal{R}(v)+\mathfrak{E}_\eta(u,t,v,a)-a\partial_t\mathcal{E}(u,t),$$

where $\mathfrak{E}_\eta\colon V\times[0,T]\times V\times(0,\infty)\to[0,\infty)$ is given by

$$\begin{aligned}\mathfrak{E}_\eta(u,t,v,a)&=a\mathcal{V}_\eta(v/a)+a\Phi^*_\eta(-\partial_u\mathcal{E}(u,t))=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\min_{\zeta\in V^*}\phi^*(\|\partial_u\mathcal{E}(u,t)+\zeta\|)=\\&=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\phi^*\left(\min_{\zeta\in V^*}\|\partial_u\mathcal{E}(u,t)+\zeta\|\right)=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\phi^*(\mathfrak{e}(u,t)).\end{aligned}$$

Then (2.15) becomes

$$\mathcal{E}(\boldsymbol{u}_\eta(t),\boldsymbol{t}_\eta(t))+\int_s^t\mathfrak{F}_\eta(\boldsymbol{u}_\eta(r),\boldsymbol{t}_\eta(r),\dot{\boldsymbol{u}}_\eta(r),\dot{\boldsymbol{t}}_\eta(r))dr=\mathcal{E}(\boldsymbol{u}_\eta(s),\boldsymbol{t}_\eta(s))\quad\text{for all }0\le s<t\le S_\eta.$$

Identity (2.18) implies

$$1=\dot{\boldsymbol{t}}_\eta(s)+\mathfrak{f}(\boldsymbol{u}_\eta(s),\dot{\boldsymbol{u}}_\eta(s),\boldsymbol{t}_\eta(s))\quad\text{for a.e. }s\in(0,S_\eta).$$

Now let $\mathfrak{F}\colon V\times[0,T]\times V\times[0,\infty)\to[0,\infty]$ defined by

$$\mathfrak{F}(u,t,v,a)=\mathcal{R}(v)+\mathfrak{E}(u,t,v,a)-a\partial_t\mathcal{E}(u,t)$$

where $\mathfrak{E}\colon V\times[0,T]\times V\times[0,\infty)\to[0,\infty]$

$$\mathfrak{E}(u,t,v,a)=\begin{cases}\iota_K(-\partial_u\mathcal{E}(u,t)) & \text{if } a>0\\ \mathfrak{e}(u,t)\|v\| & \text{if } a=0.\end{cases}$$

Definition 2.16. We say that a pair $(\boldsymbol{u},\boldsymbol{t})\colon[0,S]\to V\times[0,T]$ is an admissible parametrized curve if

(i) $\boldsymbol{t}$ is non decreasing and absolutely continuous, $\boldsymbol{u}\in \mathrm{AC}_{\mathcal{R}}([0,S],D_E)$ for some $E>0$

(ii) $\boldsymbol{u}$ locally absolutely continuous (with respect to $\|\cdot\|$) in the open set

$$G=\{s\in[0,S]\mid\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))>0\}=\{s\in[0,S]\mid-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))\notin K\}$$

(iii) the estimate

$$\int_0^S\mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s)ds=\int_0^S\mathcal{R}[\dot{\boldsymbol{u}}](s)ds+\int_G\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|ds<\infty$$

holds.

For every admissible parametrized curve and for a.e. $s\in[0,S]$ is well defined

$$\mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](s)=\begin{cases}\mathcal{R}[\dot{\boldsymbol{u}}](s)+\iota_K(-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)))-\dot{\boldsymbol{t}}(s)\partial_t\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(s)) & \text{if }\dot{\boldsymbol{t}}(s)>0\\ \mathcal{R}[\dot{\boldsymbol{u}}](s)+\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| & \text{if }\dot{\boldsymbol{t}}(s)=0.\end{cases}$$

where is implicit that if $s\notin G$ then $\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|=0$.

We denote with $\mathscr{A}([0,S],V\times[0,T])$ the set of all admissible parametrized curves and we say that $(\boldsymbol{u},\boldsymbol{t})$ is

(i) non degenerate if $\dot{\boldsymbol{t}}(s)+\mathcal{R}[\dot{\boldsymbol{u}}](s)>0$ for a.e. $s\in[0,S]$

(ii) surjective if $\boldsymbol{t}(0)=0$ and $\boldsymbol{t}(S)=T$

(iii) $\boldsymbol{m}$-normalized with $\boldsymbol{m}\in L^\infty((0,S),(0,\infty))$ if $(\boldsymbol{u},\boldsymbol{t})$ fulfills

$$\dot{\boldsymbol{t}}(s)+\mathcal{R}[\dot{\boldsymbol{u}}](s)+\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|=\boldsymbol{m}(s)\quad\text{for a.e. }s\in(0,S)$$

Definition 2.17. A parametrized balanced-viscosity solution is a non degenerate and surjective curve $(\boldsymbol{u},\boldsymbol{t})\in\mathscr{A}([0,S],V\times[0,T])$ satisfying

$$\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(t))+\int_s^t\mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](r)dr=\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))\quad\text{for all }0\le s<t\le S.$$

Lemma 2.18. Let $\boldsymbol{t}_n\in\mathrm{AC}([0,S],[0,T])$ non decreasing and surjective. Let $E>0$ and $\boldsymbol{u}_n\in\mathrm{AC}([0,S],D_E)$. Assume that there exists $L>0$ such that

$$\dot{\boldsymbol{t}}_n(s)+\mathfrak{f}(\boldsymbol{u}_n(s),\dot{\boldsymbol{u}}_n(s),\boldsymbol{t}_n(s))\le L\quad\text{for a.e. }s\in(0,S)\tag{2.19}$$

Then up to a extracting a subsequence, $(\boldsymbol{u}_n,\boldsymbol{t}_n)$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u},\boldsymbol{t})\in\mathscr{A}([0,S],D_E\times[0,T])$. Moreover

$$\dot{\boldsymbol{t}}(s)+\mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s)\le L\quad\text{for a.e. }s\in(0,S)$$

and the following lower semicontinuity properties holds

$$\liminf_{n\to\infty}\int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))dr \geq \int_s^t \mathcal{R}[\dot{\boldsymbol{u}}](r)dr$$

$$\liminf_{n\to\infty}\int_s^t \mathfrak{e}(\boldsymbol{u}_n(r),\boldsymbol{t}_n(r))\|\dot{\boldsymbol{u}}_n(r)\|dr \geq \int_s^t \mathfrak{e}(\boldsymbol{u}(r),\boldsymbol{t}(r))\|\dot{\boldsymbol{u}}(r)\|dr$$

$$\liminf_{n\to\infty}\int_s^t \mathfrak{E}_{\eta_n}(\boldsymbol{u}_n(r),\boldsymbol{t}_n(r),\dot{\boldsymbol{u}}_n(r),\dot{\boldsymbol{t}}_n(r))dr \geq \int_s^t \mathfrak{E}(\boldsymbol{u}(r),\boldsymbol{t}(r),\dot{\boldsymbol{u}}(r),\dot{\boldsymbol{t}}(r))dr$$

for every $0 \leq s < t \leq S$ and $\eta_n \in (0,T)$ with $\eta_n \to 0$.

Proof. *Step 1.* From the fact that $\boldsymbol{t}_n(0) = 0$ and that $\dot{\boldsymbol{t}}_n$ are uniformly bounded (by (2.19)), it follows that $\boldsymbol{t}_n$ are uniformly Lipschitz continuous. Then by Ascoli–Arzelà up to extracting a subsequence, $\boldsymbol{t}_n$ converges uniformly to some $\boldsymbol{t} \in C^{0,1}([0,T])$. VERIFICARE CHE $\boldsymbol{t}$ SODDISFA LE PROPRIETÀ RICHIESTE.

Step 2. Since $\mathcal{R}$ is continuous and D_E is weakly compact we can prove that for all $\delta > 0$ there exists $M_\delta > 0$ such that

$$|v-w| \leq \delta + M_\delta \mathcal{R}(v-w).$$

Indeed by contradiction assume that there is some $\delta > 0$ such that for all n there exist v_n and w_n with

$$|v_n - w_n| \geq \delta + n\mathcal{R}(v-w).$$

Since D_E is weakly compact, we can assume that $v_n \rightharpoonup v$ and $w_n \rightharpoonup w$. Up to extracting a subsequence we can also assume that $v_n \to v$ and $w_n \to w$ in H. QUI VA SISTEMATA LA SITUAZIONE. IL MOTIVO È CHE SERVIREBBE CONVERGENZA FORTE NELLA (SEMI)NORMA IN CUI $\mathcal{R}$ È CONTINUA. Then

$$|v-w| \geq \delta + \limsup_{n\to\infty} n\mathcal{R}(v_n - w_n).$$

This is a contradiction because necessarily $v \neq w$ so that $\mathcal{R}(v-w) > 0$ and in this case

$$\limsup_{n\to\infty} n\mathcal{R}(v_n - w_n) = \infty.$$

Let $\omega\colon [0,\infty) \to [0,\infty)$ defined by $\omega(r) = \inf_{\delta>0}(\delta + M_\delta r)$. Notice that ω is non decreasing and $\lim_{r\to 0}\omega(r) = 0$.

Step 3. By Jensen inequality and assumption (2.19)

$$\mathcal{R}(\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s)) = |t-s|\mathcal{R}\left(\frac{1}{|t-r|}\int_s^t \dot{\boldsymbol{u}}_n(r)dr\right) \leq \int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))drl \leq L|t-s|.$$

Then

$$|\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s)| \leq \omega(\mathcal{R}(\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s))) \leq \omega(L|t-s|).$$

This implies that $\boldsymbol{u}_n$ are equicontinuous with respect to $|\cdot|$. Then up to extracting a subsequence

$$\lim_{n\to\infty}\sup_{s\in[0,S]}|\boldsymbol{u}(s)-\boldsymbol{u}_n(s)| = 0$$

for some $\boldsymbol{u} \in \mathrm{AC}_{|\cdot|}([0,T],V)$. It follows thatB □

Theorem 2.19 (existence of parametrized solutions). Let $u_\eta \in \mathrm{AC}([0,T],V)$ be solutions to the doubly nonlinear equation (2.9) with fixed initial condition $u_0 \in V$. Consider non decreasing and surjective absolutely continuous time rescalings $\boldsymbol{t}_\eta\colon [0,S] \to [0,T]$ and let $\boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta)$. Suppose that as $\eta \to 0$

$$\boldsymbol{m}_\eta = \dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta) \overset{*}{\rightharpoonup} \boldsymbol{m} \quad \text{in } L^\infty((0,S)) \tag{2.20}$$

for some $\boldsymbol{m} \in L^\infty((0,S),(0,\infty))$. Then there exists a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_\eta, \boldsymbol{t}_\eta)$ converges uniformly on $[0,S]$ to a parametrized balanced viscosity solution $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V\times[0,T])$. Moreover $(\boldsymbol{u},\boldsymbol{t})$ is $\boldsymbol{m}$-normalized.

Proof. Arguing as in the proof of Lemma 2.15 one finds that there exists $E > 0$ such that $\boldsymbol{u}_\eta(s) \in D_E$ for all $\eta > 0$ and $s \in [0,S]$. Since (2.20) holds, there exists $L > 0$ such that $\|\dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta)\|_\infty \leq L$. Then the hypotheses of Lemma 2.18 are satisfied and we can find a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_{\eta_n}, \boldsymbol{t}_{\eta_n})$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V\times[0,T])$. Since $\mathcal{E} \in C^1(V\times[0,T])$ then

$$\mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \to \mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)) \quad \text{for all } s \in [0,S]. \tag{2.21}$$

Now, since $\dot{\boldsymbol{t}}_\eta$ are bounded in $L^\infty((0,S))$, up to extracting a subsequence it holds that $\dot{\boldsymbol{t}}_{\eta_n} \xrightarrow{*} \dot{\boldsymbol{t}}$ in $L^\infty((0,T))$. In particular $\dot{\boldsymbol{t}}_{\eta_n} \rightharpoonup \dot{\boldsymbol{t}}$ in $L^1((0,T))$. Then by Lemma 2.20

$$\liminf_{n\to\infty} \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r))dr \geq \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}(r), \boldsymbol{t}(r))dr. \tag{2.22}$$

Now taking the inferior limit as $n \to \infty$ of the energy identity

$$\mathcal{E}(\boldsymbol{u}_{\eta_n}(t), \boldsymbol{t}_{\eta_n}(t)) + \int_s^t \mathfrak{F}_{\eta_n}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r), \dot{\boldsymbol{u}}_{\eta_n}(r), \dot{\boldsymbol{t}}_{\eta_n}(r))dr \leq \mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \quad \text{for all } 0 \leq s < t \leq T$$

taking into account (2.21), (2.22) and the lower semicontinuity estimates of Lemma 2.18

$$\mathcal{E}(\boldsymbol{u}(t), \boldsymbol{t}(t)) + \int_s^t \mathfrak{F}[\boldsymbol{u}, \boldsymbol{t}, \dot{\boldsymbol{u}}, \dot{\boldsymbol{t}}](r)dr \leq \mathcal{E}(\boldsymbol{u}(s), \boldsymbol{t}(s)) + \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}(r), \boldsymbol{t}(r))dr$$

for all $0 \leq s < t \leq T$. ADESSO MANCA DA OTTENERE L'ALTRA STIMA (E VOLENDO MOSTRARE CHE SODDISFA LA GIUSTA NORMALIZZAZIONE). □

Lemma 2.20. Let $f_n, f, m_n, m\colon (0,T) \to [0,\infty]$ be measurable functions that satisfy

$$\liminf_{n\to\infty} f_n(s) \geq f(s) \quad \text{for a.e. } s \in (0,T)$$

$$m_n \rightharpoonup m \quad \text{in } L^1((0,T)).$$

Then

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \geq \int_0^T f(s)m(s)ds.$$

Proof. Let $g_k(s) = \inf_{n\geq k} \min\{f_n(s), n\}$. Notice that $g_k \in L^\infty((0,T))$. Fix k and notice that for all $n \geq k$

$$\int_0^T f_n(s)m_n(s)ds \geq \int_0^T g_k(s)m_n(s)ds.$$

Taking the inferior limit as $n \to \infty$

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \geq \int_0^T g_k(s)m(s)ds.$$

Now g_k is an increasing sequence and

$$\lim_{k\to\infty} g_k(s) = \lim_{k\to\infty} \inf_{n\geq k} \min\{f_n(s), n\} = \liminf_{n\to\infty} f_n(s) \geq f(s) \quad \text{for a.e. } s \in (0,S).$$

The passing to the limit $k \to \infty$ and using monotone convergence the proof is concluded. □

2.3.9. An example. Let $V = L^2(\Omega)$ with $\Omega \subset \mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v|dx \quad \text{and} \quad \mathcal{V}(v) = \frac{1}{2}\int_\Omega |v|^2 dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v) - \mathcal{R}(w)| \leq \int_\Omega \big||v| - |w|\big|\, dx \leq \int_\Omega |v - w|dx \leq C\|v - w\|_2.$$

It holds that K is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi \in V^* = L^2(\Omega)$ satisfies $\|\xi\|_\infty \leq 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v\in V} \int_\Omega (\xi v - |v|)\, dx \leq \sup_{v\in V} \int_\Omega (\xi v - |v|)\, dx \leq \sup_{v\in V} \int_\Omega (|\xi| - 1)|v|dx \leq 0$$

and $\mathcal{R}^*(\xi) \geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon > 0$ and $w \in L^1(\Omega)$ such that $\langle \xi, w\rangle \geq 1 + \epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta > 0$ there exist $v \in L^2(\Omega)$ such that $\|w - v\| \leq \delta$. Then for all $\lambda > 0$

$$\mathcal{R}^*(\xi) \geq \langle \xi, \lambda v\rangle - \|\lambda v\|_1 \geq \lambda\left(\langle \xi, w\rangle - \|w\|_1 - \delta(1 + \|\xi\|)\right) \geq \lambda\left(\epsilon - \delta(1 + \|\xi\|)\right).$$

Choosing δ such that $\epsilon - \delta(1+\|\xi\|) > 0$ and letting $\lambda \to \infty$ we get $\mathcal{R}(\xi) = \infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (2.10)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta} \min_{\zeta\in K} \|\xi - \zeta\|_2^2 = \frac{1}{2\eta} \int_\Omega \min_{|\zeta|\leq 1} |\xi - \zeta|^2 dx.$$

By (2.16) it follows

$$\mathfrak{p}(v, \xi) = \|v\|_1 + \|v\|_2 \min_{\zeta\in K} \|\xi - \zeta\|_2.$$

Part 1

One dimensional model of a plastic cylindrical bar

CHAPTER 3

Introduction

In a Mémoire of 1885 [15], the French engineer M. Considère collected his studies on the experimental tests he carried out on the mechanical behavior of structural metals, whose use as building materials had become widespread. Part of his study is dedicated to the tensile test of cylindrical samples: a cylindrical bar is placed under tension by controlling its elongation and the axial force to which it is subjected is measured. The experimental data reported by Considère are shown in Figure 3.1. In the

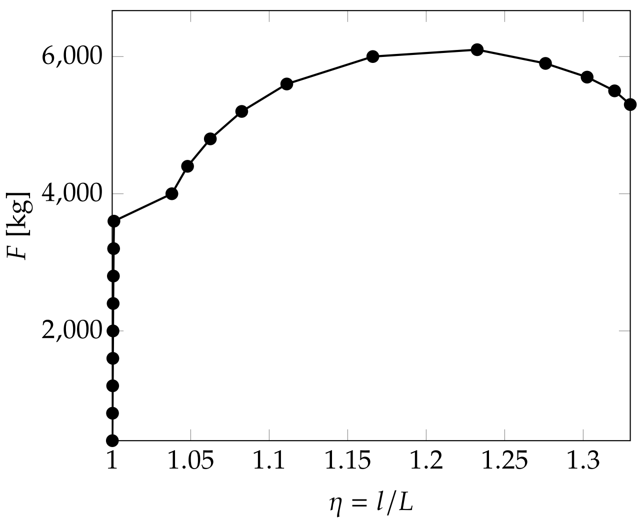


FIGURE 3.1. Trend of the tensile force F as the elongation $\eta = l/L$ varies (where L is the initial length and l the length in the deformed configuration) taken from the experimental data in [15]. The sample is made of soft iron and has a length $L = 201$ mm and a diameter of 16 mm in the initial configuration.

first phase of the test, the behavior of the specimen is elastic (the steep part of the graph in the figure). Subsequently the yield point is reached and at first the axial force increases as the deformation increases. This behavior is known as *hardening* and is due the increase in the internal resistance of the material to yield as the plastic deformation accumulates. The hardening effect competes with the shrinkage of the specimen section which tends to weaken the material, since the axial force is distributed over an increasingly smaller section. This leads to a decrease in the slope in the curve in the figure, until the force begins to decrease when the geometric effect of the decrease in section prevails over hardening. Considère observed that reaching the maximum tensile force corresponds approximately to the moment in which the bar, which had initially deformed uniformly along its entire length, begins to shrink more at a localized point, which experimentally is a priori undetermined, depending on the imperfections of the specimen. This phenomenon, which was already known before Considère studies although it had not yet been characterized quantitatively, is known as *necking*. The criterion that links the start of the necking to the achievement of the maximum traction force is now known as *Considère criterion*. Considère experimentally observed that the length of the region affected by the neck is independent of the length of the bar but increases with the diameter.

The first contributions towards the theoretical understanding of the necking phenomenon in the context of the theory of plasticity date back to the second half of the twentieth century, when the phenomenon was studied as a manifestation of the loss of stability of the *triavial solution*, i.e. the one in which the bar remains in a state of uniform deformation. In a 1958 work [36], R. Hill developed a general theory to deal with the phenomena of bifurcation and loss of stability within the theory of incremental plasticity. The theory of incremental plasticity was the framework used before the modern theory of plasticity based on solid variational foundations, the one we discussed in Section 1.2 of Chapter 1, was introduced in the 1960s. In the theory of incremental plasticity the constitutive relation links the stress rate to the strain rate, so that the *incremental problem* of determining the velocity field of the elastoplastic solid at any instant emerges naturally. Hill's bifurcation theory concerns the uniqueness of the solution to such an incremental problem. Using Hill's bifurcation theory, in the 1970s [37, 38] it was demonstrated that the loss of stability of the uniform solution occurs shortly after the tensile force reaches its maximum, in agreement with the Considère criterion. In the same period, the first numerical simulation using finite elements of the occurrence of the necking phenomenon was also carried out, allowing for the first time to accurately predict the shape of the neck [53].

In this part of the thesis we aim to study the necking phenomenon in the context of the modern mathematical theory of plasticity. The final objective would be to obtain results similar to those obtained in the context of incremental plasticity, but inserted into the more elegant and mathematically robust conceptual framework of modern plasticity theory. In this context we would like to study necking, not as a loss of uniqueness of the incremental problem, but rather as an energetic instability of the uniform solution that leads to energetically favored localized configuration.

With the hope of being able to obtain the most explicit and analytically tractable results, we have developed a simplified model to describe the metal bar, obtained through a dimensional reduction of the complete three-dimensional plasticity theory. The idea is to retain the essential ingredients of the modern variational formulation, while eliminating the geometric complexity of the complete problem. The approach used is inspired by Antman's rod theory [2] and the one-dimensional approach to necking discussed in [3], which however are limited to elasticity theory alone. We therefore assume that each configuration of the bar can be described through only three scalar fields defined along the length of the bar: the average axial position of the sections, which represents the longitudinal kinematics; a dilation parameter that quantifies the widening or narrowing of the cross section; and finally a third scalar parameter representing the average axial plastic distortion.

In Chapter 4 we formulate a version of the model valid in the regime of small deformations. The existence and uniqueness of solutions to the initial boundary value problem produced by this model is established by resorting to the general theory of existence for evolutionary variational inequalities presented in Section 2.1 of Chapter 2. We then move on to study the initial boundary value problem associated with the tensile test. As one might expect, this model is not able to capture the onset of necking, which as we have discussed is an instability phenomenon intrinsically due to the nonlinearity induced by large deformations, and we show this by explicitly determining the solution. In Chapter 5 we refine the model so that it can describe the behavior of the bar at large deformations. The existence of solutions to the resulting evolutionary problem is studied through the notion of energy solutions described in Section 2.2 of Chapter 2. We then approach the study of the stability of the trivial solution using the notion of global stability provided by the definition of energy solution itself. If we ignore the effect of plasticity by reducing ourselves to the purely elastic case we are able to obtain an explicit result which indicates that the solution in which the bar deforms uniformly loses stability after the tension force reaches its maximum. In the plastic case we are unable to obtain an equally explicit result. We will propose some heuristic considerations which suggest that the instability of the trivial solution also occurs in this case after the force reaches its maximum, in accordance with the Considére criterion.

Our analysis cannot be considered conclusive since it is not yet able to show the occurrence of necking in a complete and rigorous way, nor is it capable of qualitatively describing the salient properties of the behavior of the bar following the triggering of the instability. We hope that a deeper analysis of this simplified one-dimensional model will provide clearer insight into the mechanisms that govern the onset and evolution of necking.

CHAPTER 4

One-dimensional model at small deformations

In this chapter we develop and analyze a version of the one dimensional model aiming to describe the behavior of the metallic bar under tension under the assumption of small deformations. We split our discussion into three sections. The first section is devoted to the development of the model, the second to an existence and uniqueness results. The last section treats the explicit solution of two boundary value problems describing the tensile testing of the bar. In both the cases we show that the small deformation model is not able to predict the necking phenomenon.

4.1. Formulation

Before starting to develop the true one dimensional model, we look at the full multidimensional model of the bar but considering a constrained kinematics that does not allows for bending and shear. This will provide the basic intuition upon which we will be able to build the reduced dimensional model.

4.1.1. Motivation from the multidimensional kinematics. Let $\Omega = (0, L)\times\Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix} \tag{4.1}$$

where u_1 and u_2 are functions from $(0, L)$ to R. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix} \tag{4.2}$$

where $p\colon (0, L) \longrightarrow \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$.

We now look how the elastic free energy looks like when the displacement is of the form (4.1) and the plastic distortion is of the form (4.2). Assuming an isotropic elastic response the elastic free energy density is

$$W_e(E_e) = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. But in our case

$$\operatorname{tr} E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix},$$

so that

$$W_e(E_e) = \frac{1}{2}\kappa\left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2\mu|u_2'|^2.$$

We now compute the elastic free energy per unit axial length. This will suggest us how to appropriately choose the free energy in the one dimensional model. To this aim we integrate the elastic free energy density over the section. We find

$$\int_\Sigma W_e(E_e)dy = \frac{1}{2}A\kappa\,(e_1 + (d-1)e_2)^2 + \frac{d-1}{d}A\mu(e_1 - e_2)^2 + \frac{1}{2}J\mu|u_2'|^2, \tag{4.3}$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

4.1.2. Kinematics of the one dimensional model. We are now ready to build a purely one-dimensional model of the bar, drawing inspiration from the multidimensional kinematics discussed above. We stress that this formulation should be viewed as an independent model with its own intrinsic validity, regardless of the multidimensional framework, which was presented in the previous subsection solely as a motivation.

Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is now described by the couples $z = (u, p) \in Z$ where $u_i\colon (0, L) \to \mathbb{R}$ for $i \in \{1, 2\}$ and $p\colon (0, L) \to \mathbb{R}$. Here u_1 is the longitudinal displacement of the sections in the axis direction and u_2 is the quantity describing the transversal expansion of the sections. The configuration of the microscopic structure of the bar is described by the plastic distortion $p\colon (0, L) \to \mathbb{R}$. As suggested by the relations we found in previous subsection, we define $e_1 = u_1' - p$ as the quantities quantifying the elastic distortion.

The space Z of configurations $z = (u, p)$ will be clearly defined subsequently when the existence result will be proved. The generic virtual velocity will be denoted as $\tilde{w} = (\tilde{v}, \tilde{q}) \in Z$.

4.1.3. Statics. With the same reasoning that lead to choose the internal power](1.7) in the context of the multidimensional plasticity theory, now we assume that the generalized internal force in our one-dimensional model is of the form

$$\langle \mathcal{I}, \tilde{w}\rangle = \int_0^L \left(\zeta\tilde{v}_2' + \sigma_1(\tilde{v}_1' - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d-1)^{-1}\tilde{q})\right) dx + \int_0^L \tau\tilde{q}dx.$$

The first integral describes the elastic response that couples the macroscopic kinematic variables u with the microscopic variable p. Here $\zeta\colon (0, L) \to \mathbb{R}$, $\sigma_1\colon (0, L) \to \mathbb{R}$ is the longitudinal elastic stress and $\sigma_2\colon (0, L) \to \mathbb{R}$ is the longitudinal elastic force. The last integral describes the microscopic response and $\tau\colon (0, L) \to \mathbb{R}$ is the microscopic force.

Similarly, the external generalized force is required to be of the form

$$\langle \mathcal{X}, \tilde{w}\rangle = \int_0^L f \cdot \tilde{v}dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Here $f\colon (0, L) \to \mathbb{R}^2$ is an external force distributed along the bar and $h\colon \{0, L\} \to \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

The application of he principle of virtual powers, $\langle \mathcal{I}, \tilde{w}\rangle = \langle \mathcal{X}, \tilde{w}\rangle$ for all $\tilde{w} \in Z$, leads to equilibrium equations whose strong form is given by the following proposition.

Proposition 4.1. The local form of the equilibrium condition is

$$\sigma_1' + f_1 = 0 \quad \text{in } (0, L) \tag{4.4a}$$

$$h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\} \tag{4.4b}$$

$$-\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L) \tag{4.4c}$$

$$h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\} \tag{4.4d}$$

$$\sigma_1 - (d-1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L) \tag{4.4e}$$

Proof. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1'\tilde{v}_1 dx + \sigma_1(L)\tilde{v}_1(L) - \sigma_1(0)\tilde{v}_1(0) = \int_0^L f_1\tilde{v}_1 dx + h_1(L)\tilde{v}_1(L) - h_1(0)\tilde{v}_1(0).$$

We get (4.4a) and (4.4b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta'\tilde{v}_2 dx + \zeta(L)\tilde{v}_2(L) - \zeta(0)\tilde{v}_2(0) + \int_0^L \sigma_2\tilde{v}_2 dx = \int_0^L f_2\tilde{v}_2 dx + h_2(L)\tilde{v}_2(L) - h_2(0)\tilde{v}_2(0).$$

We get (4.4c) and (4.4d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d-1)^{-1}\sigma_2 - \tau)\tilde{q}dx = 0.$$

This implies (4.4e). □

4.1.4. Constitutive relations. To find appropriate constitutive relations for our one-dimensional model, we first choose the free energy and dissipation potential functionals. The free energy functional $\Psi \colon Z \to \mathbb{R}$ is chosen as

$$\Psi(z) = \int_0^L \big(W_e(e_1, e_2) + W_p(p)\big)\, dx$$

where $W_e \colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and $W_p \colon \mathbb{R} \to \mathbb{R}$ are now the elastic and plastic free energies per unit axial lenght. We specialize to the case

$$W_e(e_1, e_2) = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 \quad \text{and} \quad W_p(p) = \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2. \tag{4.5}$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. The above formula for W_e is motivated by what we found in (4.3) when discussing the multidimensional model of the bar. The dissipation potential functional $\Phi \colon Z \to \mathbb{R}$ is supposed to be

$$\Phi(\dot{z}) = \int_0^L R(\dot{p})dx \tag{4.6}$$

where $R \colon \mathbb{R} \to [0, \infty]$ is the plastic dissipation potential per unit axial length, which is required to be convex and satisfying $R(0) = 0$. A psiibile particular R could be the one dimensional analog of the Von-Mises yield criterion, i.e. $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength.

Consistently with that choices of Ψ and Φ, applying (1.3) we find the following constitutive relations

$$\begin{aligned}
\zeta &= \mu l^2 u_2' \\
\sigma_1 &= \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\
\sigma_2 &= (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\
\tau &\in Bp + \partial R(\dot{p}).
\end{aligned}$$

Notice that the general discussion of Section 1.1 of Chapter 1 implies that these constitutive relations are compatible with the dissipation inequality.

Remark 4.2. (the Mandel stress) Introducing the constitutive relations (4.7) into (4.4e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We will refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is interpreted as the elastic force felt by the microscopic configuration and we notice that it depends on the shear modulus only.

4.2. Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions of evolutions . For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

We define the spaces $U = H^1((0, L), \mathbb{R} \times \mathbb{R})$ and $P = L^2((0, L))$. Let also $Z = U \times P$ the space of configurations of the system. We will write the generic configuration as $z = (u, p) \in Z$ with $u \in U$ and $p \in P$ and the generic virtual rate as $w = (v, q) \in Z$ with $v \in U$ and $q \in P$.

The evolution problem associated with the model described in this chapter is of the form of an evolutionary variational inequality that has been studied in Section 2.1 of Chapter 2. According to (4.5), the free energy $\Psi \colon Z \to \mathbb{R}$ is defined by

$$\Psi(z) = \int_0^L \left[\frac{1}{2}\kappa(u_1' + (d-1)u_2)^2 + \frac{1}{2}\mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)^2 + \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2\right] dx$$

Its Fréchet differential is $\mathcal{A} = \partial\Psi\colon Z \to Z^*$ given by

$$\langle \mathcal{A}z, w\rangle = \int_0^L l^2\mu u_2' v_2' dx + \int_0^L \left[\kappa(u_1' + (d-1)u_2) + \mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)\right](v_1' - q)dx +$$
$$+ \int_0^L \left[(d-1)\kappa(u_1' + (d-1)u_2) - \mu\left(u_1' - u_2 - \frac{d}{d-1}q\right)\right]\left(v_2 + (d-1)^{-1}p\right)dx +$$
$$+ \int_0^L Bpq dx.$$

According to (4.6) the dissipation potential $\mathcal{R}\colon Z \to \mathbb{R}\cup\{\infty\}$ is defined by

$$\mathcal{P}(w) = \int_0^L R(q)dx.$$

The external generalized force is $\mathcal{X}\colon [0,T] \to Z^*$,

$$\langle \mathcal{X}(t), w\rangle = \int_0^T f(t)\cdot v dx.$$

Now to prove the existence of solution we need only to satisfy the assumptions of Theorem 2.2. This will be done under the following assumptions:

(G) $L > 0$ and $d = 2$ or $d = 3$,
(ψ) κ, μ, l and B are positive constants,
(R) $R\colon \mathbb{R} \to \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$,
(f) $f \in H^1((0,T), U^*)$ and $f(0) = 0$.

Theorem 4.3. Assume hypotheses (G), (ψ), (R) and (f). Then there exists a unique solution to the evolutionary problem associated with .

Proof. We verify the hypothesis of Theorem 2.2.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z\rangle \geq \alpha\|z\|_Z^2$. If we call $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned}
\langle \mathcal{A}z, z\rangle = 2\int_0^L \psi dx &= \kappa\|e_1 + (d-1)e_2\|_2^2 + \mu\|e_1 - e_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 = \\
&= \kappa\|u_1' + (d-1)u_2\|_2^2 + \mu\left\|u_1' - u_2 - \frac{d}{d-1}p\right\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 \geq \\
&\geq (1-\theta)\kappa\|u_1'\|_2^2 + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa\|u_2\|_2^2 + (1-\tau)\mu\|u_1' - u_2\|_2^2 + \\
&\qquad l^2\mu\|u_2'\|_2^2 + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2 \\
&\geq \left[(1-\theta)\kappa + (1-\eta)(1-\tau)\mu\right]\|u_1'\|_2^2 + \\
&\quad + \left[\left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa + \left(1 - \frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + \\
&\qquad + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2
\end{aligned}$$

for all θ, τ and η in $(0,\infty)$. We restrict to θ, τ and η in $(0,1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned}
\langle \mathcal{A}z, z\rangle \geq C_1&\left[(1-\theta)\kappa + (1-\eta)(1-\tau)\mu\right]\|u_1\|_{1,2}^2 + \\
&+ \left[C_2 l^2\mu + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa + \left(1 - \frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_{1,2}^2 + \\
&\qquad + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2
\end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|_{1,2}^2$, $\|u_2\|_{1,2}^2$ and $\|p\|_2^2$ are positive and we get the desired coercivity.

Step 2. Since $\Phi(w) = Y\|q\|_1$ we see immediately that Φ is convex, proper, continuous and 1-homogeneous.
Step 3. Since $f \in H^1((0,T), U^*)$ and $f(0) = 0$ then $\mathcal{X} \in H^1((0,T), Z^*)$ and $\mathcal{X}(0) = 0$. □

4.3. Analytical solution to the tensile test problem

In this section we obtain explicit solutions to the boundary value problem describing the tensile test. We will consider two different variants of the problem: in the first variant the ends are left free to slide in the transversal direction, while in the second variant the ends are clamped. We will see that the behavior of the solutions in this two situations is qualitatively similar, and in particular necking does not occur.

4.3.1. Tensile test with free-sliding ends. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $h_2(L,t) = h_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (4.4a) and (4.4b) we have that $\sigma_1 = F$ on all $(0,L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned}
\sigma_2 &= (d-1)\left(F - \mu(e_1 - e_2)\right) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) = \\
&= (d-1)F - d\frac{\mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \\
&= \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.
\end{aligned}$$

Now from equation (4.4c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L,t) = h_2(0,t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = F.$$

When $F \leq Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

4.3.2. Tensile test with clamped ends.
Setup of the problem. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $u_2(L,t) = u_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 4.3.1 we find the equation

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2. \tag{4.8}$$

Solution in the elastic range. In the elastic range $p = 0$ everywhere in $(0, L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu} F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F.$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress is the elastic range is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}\left(1 - \frac{(d-1)\kappa-\mu}{d\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)\right)F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first.

Solution in the plastic range. From the above considerations we are let to seek for a solution where $p > 0$ in the interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \in [0, L/2]$ is a function of time.

Remark 4.4. Since the solution is symmetric with respect to $x = L/2$ from now on we focus on the $(0, L/2)$ half of the bar only.

On $(0, L/2 - \epsilon)$ the solution is given by $p = 0$ and u_2 of the form

$$u_2 = \frac{(d-1)\kappa-\mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F + C_1\sinh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}x\right) \tag{4.9}$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$
$$\frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa u_2 - \frac{d\kappa}{d-1}p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1}\frac{\mu}{\kappa+\mu}F - \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}u_2 - Y}{\frac{d^2}{(d-1)^2}\frac{\mu\kappa}{\kappa+\mu} + B}.$$

Substituting that identity into equation (4.8) we have obtain a differential equation in y of the form

$$y'' = ay + b$$

where a and b depends on time only. A solution which is symmetric with respect to $x = L/2$ is of the form

$$y = -\frac{b}{a} + C_2\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right). \tag{4.10}$$

We now look for a solution $u_2 \in C^2([0, L])$. When $\epsilon = L/2$ the solution is simply given by

$$u_2 = \frac{b}{a}\left(\frac{\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right)}{\cosh\left(\sqrt{a}\frac{L}{2}\right)} - 1\right)$$

When $\epsilon \in (0, L/2)$ we consider (4.9) on $(0, L/2-\epsilon)$ and (4.10) on $(L/2-\epsilon, L/2)$ and we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$. Solving the resultant linear system we find the constants C_1 and C_2 that we do not report.

Determination of ϵ. We are left with the determination of $\epsilon \in [0, L/2]$ as a function of time. To this aim we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the value of F for which that Mandel stress reach Y. We find

$$F_Y = \frac{d-1}{d}\frac{k+\mu}{\mu}\left(-\frac{(d-1)\kappa-\mu}{d\mu}\left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)}-1\right)+1\right)^{-1} Y.$$

For $F \leq F_Y$ we have $\epsilon = 0$. To determine ϵ when $F > F_Y$ we consider the equation $m(L/2 - \epsilon) = Y$. That equation cannot be solved analytically in ϵ. Instead we can find explicitly F as a function $f =: [0, \epsilon_0) \to \mathbb{R}$ of ϵ. Here $[0, \epsilon_0)$ is the maximal interval where f is well defined. After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load F reach the value

$$f(L/2) = \frac{d-1}{d}\frac{\kappa+\mu}{\mu}Y,$$

the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. We conclude that

$$\epsilon = \begin{cases} 0 & \text{if } F \leq F_Y \\ f^{-1}(F) & \text{if } F_Y < F \leq f(L/2) \\ L/2 & \text{if } F > f(L/2). \end{cases}$$

Absence of necking. From the analytical expression one can see that for $F > f(L/2)$ the solution u_2 depends linearly on F. In particular as $F \to \infty$ we have that $u_2 \to -\infty$ uniformly on all compact sets contained in $(0, L)$. We conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section appears to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

The limit of thin bar. To get further insight on the solution we look at the limit $l \to 0^+$.

Proposition 4.5. In the limit $l \to 0^+$ the solution u_2 that we have found converges pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 4.3.1. This convergence is also uniform on compact subsets of $(0, L) \times [0, T]$.

Proof. It is a direct computation. □

This results confirm the absence of necking in the small deformation model. Indeed it says that a thin bar deforms almost uniformly.

CHAPTER 5

Model at large deformations

As we showed in the previous chapter, the one dimensional small deformations model of a bar is not able to predict necking. This leads us to refine the one dimensional model allowing large deformations. This is the content of this chapter. The discussion is structures as in the previous chapter. In the first section we develop the model, in the second section we obtain an existence theorem of solutions to the boundary value problem produced by that model. In the third section we face the problem of studying the boundary value problem describing the tensile test: in this case we will not be able to obtain any explicit solution, so we will propose some methods to gain qualitative insight on the behavior of the solution, in particular regarding the onset of necking.

5.1. Formulation

In a similar manner to the previous chapter, before developing the strictly one-dimensional model, we examine the kinematics of a fully multidimensional bar under large strain. We assume that bending and shear are neglected; consequently, cross-sections remain plane, undergoing only axial translation and uniform transversal stretching.

5.1.1. Motivation form the multidimensional kinematics. Let $\Omega = (0, L) \times \Sigma \subset \mathbb{R}^d$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ the reference configuration of the cylindrical bar. The generic material point will be denoted as $(X, Y) \in \Omega$ with $X \in (0, L)$ and $Y \in \Sigma$. We consider deformations of the form

$$(X, Y) \mapsto \begin{pmatrix} \chi_1(X) \\ \chi_2(X)Y \end{pmatrix}$$

where $\chi_1 \colon (0, L) \to \mathbb{R}$ satisfy $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. The corresponding deformation gradient is

$$F = \begin{pmatrix} \chi_1' & 0 \\ \chi_2' Y & \chi_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$F_p = \begin{pmatrix} P & 0 \\ 0 & P^{-\frac{1}{d-1}} I \end{pmatrix}$$

where $P \colon (0, L) \to (0, \infty)$. This choice implies plastic incompressibility. The elastic distortion is then given by the Kroner–Lee multiplicative decomposition

$$F_e = F F_p^{-1} = \begin{pmatrix} \chi_1' P^{-1} & 0 \\ \chi_2' P^{-1} Y & \chi_2 P^{\frac{1}{d-1}} I \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ GY & E_2 I \end{pmatrix},$$

where we have defined $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$. To motivate the development of the one-dimensional constitutive relations, we examine the neo-Hookean elastic free-energy density under the aforementioned kinematic assumptions. This will serve as a basis for the development of the constitutive model presented in the following section. Assuming the particular kinematic assumptions above, the first principal invariant of $C_e = F_e^T F_e$ is

$$I_e = \operatorname{tr}(C_e) = E_1^2 + G^2 |Y|^2 + E_2^2$$

and the determinant of F_e is

$$J_e = \det F_e = E_1 E_2,$$

so that the neo-Hookean elastic free energy density is given by

$$\psi_e = C \left(\frac{I_e}{J_e} - 2 \right) + w(J_e) = C \left(\frac{E_1^2 + G^2 |Y|^2 + E_2^2}{E_1 E_2} - 2 \right) + w(E_1 E_2). \tag{5.1}$$

5.1.2. Kinematics. We now proceed to develop a purely one-dimensional model of the bar. This model is formulated independently of the fully multidimensional one; therefore, the discussion in the previous subsection should be viewed merely as a motivation for the specific choices made in the following derivation.

Let $L > 0$ the length of the cylindrical bar. The macroscopic configuration of the bar is described by $\chi_1 \colon (0, L) \to \mathbb{R}$ with $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. Here χ_1 describes the positions of the sections and χ_2 is the transversal deformation factor of the sections. The configuration of the microscopic structure of the bar, that is the plastic distortion, is described by $P \colon (0, L) \to (0, \infty)$. As suggested by the multidimensional kinematics, we define the elastic distortions as $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$.

The configuration space Z of the system are thus the elements of the form $z = (\chi, P) \colon (0, L) \to \mathbb{R}\times(0, \infty)\times(0, \infty)$. The virtual rates are denoted as $\tilde{w} = (\tilde{v}, \tilde{Q}) \colon (0, L) \to \mathbb{R} \times (0, \infty) \times (0, \infty)$. The right functional setting will be discussed in Section 5.2 where the existence theorem is proved.

5.1.3. Statics. The internal generalized force is chosen so that clearly appear the elastic forces and stresses, conjugate to the virtual rates of the elastic distortions, and plastic microforces and microstresses power, conjugate to the the virtual rate of the plastic distortion and its spatial derivative:

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_0^L \left(\zeta \tilde{\dot{G}} + \sigma_1 \tilde{\dot{E}}_1 + \sigma_2 \tilde{\dot{E}}_2 \right) dX + \int_0^L \left(\tau \tilde{Q} + \kappa \tilde{Q}' \right) dX.$$

Here $\zeta \colon (0, L) \to \mathbb{R}$ is the transversal elastic stress, $\sigma_1 \colon (0, L) \to \mathbb{R}$ is the longitudinal elastic stress and $\sigma_2 \colon (0, L) \to \mathbb{R}$ is the transversal elastic force, while $\tau \colon (0, L) \to \mathbb{R}$ is the microscopic force and $\kappa \colon (0, L) \to \mathbb{R}$ the microscopic stress. Moreover the virtual rates $\tilde{\dot{E}}_1$, $\tilde{\dot{E}}_2$ and $\tilde{\dot{G}}$ are defined by the relations

$$\begin{aligned} G^{-1} \tilde{\dot{G}} &= (\chi_2')^{-1} \tilde{v}_2' - P^{-1} \tilde{Q} \\ E_1^{-1} \tilde{\dot{E}}_1 &= (\chi_1')^{-1} \tilde{v}_1' - P^{-1} \tilde{Q} \\ E_2^{-1} \tilde{\dot{E}}_2 &= \chi_2^{-1} \tilde{v}_2 + (d-1)^{-1} P^{-1} \tilde{Q}, \end{aligned}$$

Taking into account these definitions the internal generalized force can be rewritten more explicitly as

$$\begin{aligned} \langle \mathcal{I}, \tilde{w} \rangle = \int_0^L G\zeta \left((\chi_2')^{-1} \tilde{v}_2' - P^{-1} \tilde{Q} \right) dX + \\ + \int_0^L \left[E_1 \sigma_1 \left((\chi_1')^{-1} \tilde{v}_1' - P^{-1} \tilde{Q} \right) + E_2 \sigma_2 \left(\chi_2^{-1} \tilde{v}_2 + (d-1)^{-1} P^{-1} \tilde{Q} \right) \right] dX + \\ + \int_0^L \left[\tau \tilde{Q} + \kappa \tilde{Q}' \right] dX. \end{aligned}$$

Remark 5.1. In the small strain model of Chapter 4 we did not included the microscopic stress κ. However, we are now including it because we expect that in the large deformation regime, it is necessary to obtain an existence result in Sobolev spaces. This is suggested by the existence theorem for three-dimensional elastoplasticity at large strains presented in [43]. It will ensure the necessary compactness needed to handle the complex nonlinear coupling implied by the present model. The insertion of the microstress κ will prevent the plastic distortion field P to exhibiting jumps, so it works as a regularization.

The external generalized force is chosen of the form

$$\langle \mathcal{X}, \tilde{w} \rangle = \int_0^L f \cdot \tilde{v} dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

As in the small strain model, $f \colon (0, L) \to \mathbb{R}^2$ is an external force distributed along the bar and $h \colon \{0, L\} \to \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

Applying the principle of virtual powers, $\langle \mathcal{I}, \tilde{w} \rangle = \langle \mathcal{X}, \tilde{w} \rangle$ for all $\tilde{w} \in Z$, we get the equilibrium equations whose strong form is given below.

PROPOSITION 5.2. The local form of the equilibrium condition is

$$\begin{gathered}(P^{-1}\sigma_1)' + f_1 = 0 \quad \text{in } (0,L)\\ h_1(X) = P(X)^{-1}\sigma_1(X) \quad \text{for } X \in \{0,L\}\\ -P^{\frac{1}{d-1}}\sigma_2 + (P^{-1}\zeta)' + f_2 = 0 \quad \text{in } (0,L)\\ h_2(X) = P(X)^{-1}\zeta(X) \quad \text{for } X \in \{0,L\}\\ \chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\zeta - \tau + \kappa' = 0 \quad \text{for } X \in \{0,L\}\\ \kappa(X) = 0 \quad \text{for } X \in \{0,L\}\end{gathered}$$

PROOF. Rearranging the terms we have

$$\langle \mathcal{J}, \tilde{w}\rangle = \int_0^L \left[P^{-1}\sigma_1\tilde{v}_1' + P^{\frac{1}{d-1}}\sigma_2\tilde{v}_2 + P^{-1}\zeta\tilde{v}_2'\right] dX + \\ + \int_0^L \left[\left(\tau - \chi_1' P^{-2}\sigma_1 + (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 - \chi_2' P^{-2}\zeta\right)\tilde{Q} + \kappa\tilde{Q}'\right] dX.$$

By integration by parts and application of the fundamental lemma of the calculus of variation we get the thesis. □

As we did also in the previous chapter, we define the Mandel stress as the elastic force felt by the microscopic configuration, that is

$$m = \chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\zeta. \tag{5.3}$$

5.1.4. Constitutive relations. To appropriately build the constitutive relations we consider a free energy $\Psi\colon Z \to \mathbb{R}$ and a dissipation potential $\Phi\colon Z \to \mathbb{R}$ of the same structure that we sow in the previous plasticity models.

$$\Psi(z) = \int_0^L (W(E_1, E_2, G) + H(P, P'))\, dX \quad \text{and} \quad \Phi(z, \dot{z}) = \int_0^L R(P, \dot{P}) dX.$$

Here $W\colon (0,\infty)\times(0,\infty)\times\mathbb{R} \to \mathbb{R}$ and $H\colon (0,\infty)\times\mathbb{R} \to \mathbb{R}$ are the smooth elastic and plastic free energy densities, and $R\colon (0,\infty)\times\mathbb{R} \to [0,\infty]$, convex in $\dot{P}$ and satisfying $R(0) = 0$, is dissipation potential per unit axial length. Consistently with that choice of Ψ and Φ we find the following constitutive relations

$$\begin{gathered}\sigma_i = \frac{\partial W}{\partial E_i} \quad \text{and} \quad \zeta = \frac{\partial W}{\partial G}\\ \tau \in \frac{\partial H}{\partial P} + P^{-1}\partial_{\dot{P}} R \quad \text{and} \quad \kappa = \frac{\partial H}{\partial P'}.\end{gathered}$$

The general discussion made in Section 1.1 of Chapter 1 implies that the dissipation inequality is satisfied by these constitutive relations.

We will see particular examples of constitutive relations in the next section, after we have clarified which assumptions are required to produce boundary value problems which are well-posed. This is indeed the content of next section.

5.2. Existence result

In this section we prove the existence of energetic solutions. A self-contained introduction to this solution concept, including an explanation of the notation employed in this section, can be found in Section 2.2 of Chapter 2. The aim of this section is to verify the hypotheses required to apply the abstract existence result provided by Theorem 2.8.

5.2.1. Formulation of the initial boundary value problem. We now describe how the model formulated in the previous section, can be treated within the framework of energetic solutions.

Dirichlet conditions. Suppose that at time $t \in [0,T]$ the following Dirichlet conditions are imposed

$$\chi_i(X) = g_i(X,t) \quad \text{for } X \in \Gamma_D^i \text{ and } i \in \{1,2\}, \tag{5.5}$$

where $\Gamma_D^i \subset \{0,L\}$ and $g_i\colon (0,L)\times[0,T] \to \mathbb{R}$ satisfy $g_1' > 0$ and $g_2 > 0$. To avoid free translations of the bar, it is required that $\Gamma_D^1 \neq \emptyset$, but Γ_D^2 can be empty instead. Conditions (5.5) can be converted into time-independent conditions by expressing χ_i as

$$\chi_1(X) = g_1(y_1(X),t) \quad \text{and} \quad \chi_2(X) = g_2(y_1(X),t)y_2(X),$$

with $y_i\colon (0,L) \to \mathbb{R}$, $y_1' > 0$ and $y_2 > 0$, and requiring that

$$y_1(X) = X \quad \text{for } X \in \Gamma_D^1 \quad \text{and} \quad y_2(X) = 1 \quad \text{for } X \in \Gamma_D^2.$$

Having made that change of variables, the unknown of the problem are y and P. We notice that the elastic distortions E_1, E_2 and G are given by

$$E_1 = g_1'(y_1,t)y_1'P^{-1}, \quad E_2 = g_2(y_1,t)y_2P \quad \text{and} \quad G = g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}.$$

Function spaces. Let $Y = W^{1,q_Y}((0,L),\mathbb{R}^2)$ with $q_Y \in (1,\infty)$ be the space of y and $Z = L^{q_H}((0,L)) \cap W^{1,r_H}((0,L))$ with $q_H \in (1,\infty)$, $r_H \in (1,q_H]$ be the space of P. Let $Q = Y \times Z$. The Dirichlet conditions are imposed requiring that y, P belong respectively to the admissible spaces

$$\mathscr{Y} = \left\{ y \in Y \,\middle|\, y_1(X) = X \text{ for } X \in \Gamma_D^1 \text{ and } y_2(X) = 1 \text{ for } X \in \Gamma_D^2 \right\} \quad \text{and} \quad \mathscr{Z} = Z.$$

Let $\mathcal{Q} = \mathscr{Y} \times \mathscr{Z}$. It is evident that $\mathcal{Q}$ is a weakly closed subset of Q.

Energy functional and dissipation distance. To complete the formulation of the initial boundary value problem, we need to specify the energy functional $\mathcal{E}\colon \mathcal{Q}\times[0,T] \to (-\infty,\infty]$ and the dissipation distance $\mathcal{D}\colon \mathscr{Z}\times\mathscr{Z} \to [0,\infty]$. The energy functional is defined by

$$\mathcal{E}(y,P,t) = \int_0^L W\left(g_1'(y_1,t)y_1'P^{-1}, g_2(y_1,t)y_2P, g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}\right) dX +$$
$$+ \int_0^L H(P,P')dX - \int_0^L \left(f_1(t)g_1(y_1,t) + f_2(t)g_2(y_1,t)y_2\right) dX.$$

The dissipation distance can be expressed in term of a dissipation distance density $D\colon \mathbb{R}^2 \to [0,\infty]$ as

$$\mathcal{D}(P_0,P_1) = \int_0^L D(P_0,P_1)dX.$$

According to Remark 2.4, D can be defined from the dissipation potential density R by

$$D(P_0,P_1) = \inf\left\{ \int_0^1 R(P(t),\dot{P}(t))dt \,\middle|\, P \in C^1([0,1]) \text{ and } P(i) = P_i \text{ for } i \in \{0,1\} \right\}. \tag{5.6}$$

5.2.2. Assumptions and preliminary results. We now discuss the assumptions under which we will prove the validity of hypotheses (E1)-(E4), (D1)-(D2) and (C1)-(C2) of Section 2.2 in Appendix 2. Subsequently, these assumptions will be implicitly supposed to be valid and will not be recalled.

Remark 5.3 (on notation). Sometimes the notation $G = E_3$ and $E = (E_1,E_2,E_3)$ will be used. Given $E, N \in \mathbb{R}^3$, then $NE \in \mathbb{R}^3$ is the component-wise multiplication between E and N, i.e., $(NE)_i = N_iE_i$ for $i \in \{1,2,3\}$.

The assumption on the Dirichlet data is

(G) $g_1 \in C^2(\mathbb{R}\times[0,T])$, $g_2 \in C^2(\mathbb{R}\times[0,T],(0,\infty))$ and there exists $0 < \gamma < \Gamma < \infty$ for $i \in \{1,2\}$ such that $\gamma < g_1' < \Gamma$, $\gamma < g_2 < \Gamma$ and $|g_2'| \le \Gamma$.

Only to prove Proposition 5.10 we will assume also the following additional requirement

($\hat{\text{G}}$) $g_2' = 0$.

We do not know whether the proof works without ($\hat{\text{G}}$). Anyway assuming ($\hat{\text{G}}$), it is not restrictive for our purposes.

The elastic free energy density is supposed to satisfy the following assumptions

(W1) there exists $\mathbb{W}\colon \mathbb{R}^4 \to (-\infty,\infty]$ continuous such that

$$W(E_1,E_2,G) = \mathbb{W}(E_1,E_2,G,E_1E_2)$$

for all $E_1,E_2,G \in \mathbb{R}$ and $\mathbb{W}$ is convex in (E_1,G,E_1E_2), (W2) there exist $C_W > 0$ and $q_W \in (1,\infty)$ such that $W(E_1,E_2,G) \geq C_W(-1+|E_1|^{q_W}+|E_2|^{q_W}+|G|^{q_W})$ for all $E_1,E_2,G \in \mathbb{R}$,

(W3) $W(E_1,E_2,G) = \infty$ if and only if $E_1 \leq 0$ or $E_2 \leq 0$,

(W4) W is differentiable in $\operatorname{dom} W$ and there exist $C_0^W \in \mathbb{R}$, $C_1^W > 0$ and a modulus of continuity ω^W such that
  (i) for all $E \in \operatorname{dom} W$ and $i \in \{1,2,3\}$ it holds $\left|E_i \frac{\partial W}{\partial E_i}(E)\right| \leq C_1^W(C_0^W + W(E))$,
  (ii) let $K(E) = E \cdot \frac{\partial W}{\partial E}(E)$, then $|K(NE) - K(E)| \leq \omega^W(|N-1|)(W(E)+C_0^W)$ for all $N \in (0,\infty)^3$.

Assumption (W1) is a sort policonvexity requirement necessary to ensure lower semicontinuity, (W2) is a growth condition, (W3) is an incompenetrability condition for the elastic distortion and (W4) is a further regularity assumption which is needed to get suitable a priori estimates. Later on we will use the following consequence of (W4).

Lemma 5.4. Fix $0 < \delta < 1$ small enough. Then there exists $C > 0$ such that

$$W(NE) + \max_{i\in\{1,2,3\}} \left|E_i \frac{\partial W}{\partial E_i}(NE)\right| \leq C(W(E) + C_0^W)$$

for all $E \in (0,\infty)\times(0,\infty)\times\mathbb{R}$ and $N \in (1-\delta,1+\delta)^3$. Here $NE \in \mathbb{R}^3$ denotes the component-wise multiplication, i.e., $(NE)_i = N_iE_i$ for all $i \in \{1,2,3\}$.

Proof. *Step 1.* For all $i \in \{1,2,3\}$, by (W4, i)

$$\left|E_i \frac{\partial W}{\partial E_i}(NE)\right| = |N_i|^{-1}\left|N_iE_i \frac{\partial W}{\partial E_i}(NE)\right| \leq |N_i|^{-1}C_1^W(C_0^W + W(NE)) \leq$$
$$\leq \frac{C_1^W}{1-\delta}(C_0^W + W(NE)).$$

Step 2. We have that

$$|W(NE) - W(E)| = \left|\int_0^1 \frac{d}{d\lambda} W(\tilde{N}(\lambda)E)d\lambda\right|,$$

where $\tilde{N}(\lambda) = (1-\lambda)+\lambda N$, i.e., $\tilde{N}_i(\lambda) = (1-\lambda)+\lambda N_i \in (1-\delta,1+\delta)$ for all $i \in \{1,2,3\}$. Then

$$|W(NE) - W(E)| = \left|\int_0^1 \frac{\partial W}{\partial E}(\tilde{N}(\lambda)E)\cdot(N-1)d\lambda\right| \leq \sum_{i=1}^3 \int_0^1 \left|\tilde{N}_i(\lambda)E_i \frac{\partial W}{\partial E_i}(\tilde{N}(\lambda)E)\right| \frac{|N_i-1|}{\tilde{N}_i(\lambda)} d\lambda \leq$$
$$\leq \frac{\delta}{1-\delta}C_1^W \int_0^1 (W(\tilde{N}(\lambda)E) + C_0^W)d\lambda.$$

Step 3. Let $\theta(E) = \sup_{N\in(1-\delta,1+\delta)^3} W(NE)$. From Step 2

$$|W(NE) - W(E)| \leq \theta(E) - W(E) \leq \frac{\delta}{1-\delta}C_1^W(\theta(E) + C_0^W).$$

This implies

$$\left(1 - \frac{\delta}{1-\delta}C_1^W\right)\theta(E) \leq W(E) + \frac{\delta}{1-\delta}C_1^WC_0^W.$$

If δ is small enough then

$$\theta(E) \leq \frac{W(E) + \frac{\delta}{1-\delta}C_1^WC_0^W}{1 - \frac{\delta}{1-\delta}C_1^W}.$$

Step 4. From Step 3

$$|W(NE) - W(E)| \leq \frac{\delta}{1-\delta}C_1^W\left(\frac{W(E) + \frac{\delta}{1-\delta}C_1^WC_0^W}{1 - \frac{\delta}{1-\delta}C_1^W} + C_0^W\right) \leq \left(\frac{1}{\frac{\delta}{1-\delta}C_1^W} - 1\right)^{-1}(W(E) + C_0^W).$$

Step 5. From Step 1

$$\max_{i\in\{1,2,3\}} \left|E_i \frac{\partial W}{\partial E_i}(NE)\right| \leq \frac{C_1^W}{1-\delta}(C_0^W + \theta(E)) \leq \frac{C_1^W}{1-\delta}\left(\frac{W(E) + \frac{\delta}{1-\delta}C_1^WC_0^W}{1 - \frac{\delta}{1-\delta}C_1^W} + C_0^W\right) =$$
$$= \frac{C_1^W}{1-\delta(1+C_1^W)}(W(E) + C_0^W).$$

Step 6. Summing together the results of Step 4 and Step 5 we get the thesis. □

The plastic free energy density satisfies

(H1) $H\colon \mathbb{R}^2 \to (-\infty, \infty]$ is continuous and $H(P, \cdot)$ is convex for all $P \in \mathbb{R}$,

(H2) there exist $C_H > 0$ and $q_H \in (1, \infty)$, $r_H \in (1, q_H]$ such that $H(P, P') \geq C_H(-1 + |P|^{q_H} + |P'|^{r_H})$ for all $P, P' \in \mathbb{R}$,

(H3) there exists $\Pi > 0$ such that $H(P, P') = \infty$ if and only if $P \leq \Pi$.

The convexity assumption (H1) is required for lower semicontinuity, (H2) is a growth condition ensuring that the hardening and the plastic gradient regularization is sufficiently strong. Assumption (H3) is a strong incompenetrability condition for the plastic distortion.

To handle correctly the nonlinear coupling between elastic and plastic distortions, we require the following compatibility condition on the Hölder exponents

(Ö) $\frac{1}{q_Y} = \frac{1}{q_W} + \frac{1}{q_H}$.

The requirements on the dissipation potential density are

(d1) $D\colon \mathbb{R}^2 \to [0, \infty]$ is lower semicontinuous,

(d2) D is a quasidistance, in the sense that $D(P_0, P_1) = 0$ if and only if $P_0 = P_1$ and $D(P_0, P_2) \leq D(P_0, P_1) + D(P_1, P_2)$ for all $P_0, P_1, P_2 \in \mathbb{R}$,

(d3) D is finite and continuous on $[\Pi, \infty) \times [\Pi, \infty)$, where Π is the constant appearing in (H3).

5.2.3. Verification the properties of the energy functional. We start with the following lemma which is necessary to handle the nonlinear coupling between elastic and plastic distortion.

Lemma 5.5. Let $a, b > 0$. Then for all $\delta > 0$ and $\rho > 1$

$$\frac{a}{b} \geq \rho\delta^{\frac{\rho-1}{\rho}} a^{\frac{1}{\rho}} - (\rho - 1)\delta b^{\frac{1}{\rho-1}}$$

Proof. By Young inequality, if $A, B > 0$ then

$$AB \leq \frac{A^\rho}{\rho} + \frac{B^{\rho^*}}{\rho^*}$$

and

$$A^{\frac{1}{\rho}} B^{\frac{1}{\rho^*}} \leq \frac{A}{\rho} + \frac{B}{\rho^*}.$$

It follows that

$$A \geq \rho A^{\frac{1}{\rho}} B^{\frac{1}{\rho^*}} - \frac{\rho}{\rho^*} B.$$

If we put $A = a/b$ and B such that

$$B^{\frac{1}{\rho^*}} = \delta^{\frac{1}{\rho^*}} b^{\frac{1}{\rho}}$$

the thesis follows. □

We now prove that (E2) is satisfied. We start with proving the boundedness of sublevels.

Proposition 5.6 (Boundedness of energy sublevels). The sublevels of $\mathcal{E}(\cdot, t)$ are bounded in Q for all $t \in [0, T]$.

Proof. From (W1) and (H1) it follows

$$\begin{aligned}\mathcal{E}(y, P, t) \geq C\Big(-1 &+ \|g_1'(y_1, t)y_1'P^{-1}\|_{q_W}^{q_W} + \|g_2(y_1, t)y_2 P\|_{q_W}^{q_W} + \\ &+ \|g_2'(y_1, t)y_1'y_2 P^{-1} + g_2(y_1, t)y_2'P^{-1}\|_{q_W}^{q_W} + \|P\|_{q_H}^{q_H} + \|P'\|_{r_H}^{r_H}\Big).\end{aligned} \tag{5.7}$$

By Hölder inequality

$$\|y_1'\|_{q_Y} = \|y_1'P^{-1}P\|_{q_Y} \leq \|y_1'P^{-1}\|_{q_W}\|P\|_{q_H}$$

so that using Lemma 5.5 with $\rho = q_W/q_Y$ one gets

$$\|y_1'P^{-1}\|_{q_W}^{q_W} \geq \frac{\|y_1'\|_q^{q_W}}{\|P\|_{q_H}^{q_W}} \geq \rho\delta^{\frac{\rho-1}{\rho}}\|y_1'\|_{q_Y}^{q_Y} - (\rho - 1)\delta\|P\|_{q_H}^{q_H}$$

for all $\delta > 0$. We have used that $q_W/(\rho - 1) = q_H$. Moreover

$$\|y_1' P^{-1}\|_{q_W} = \|g_1'(y_1,t)^{-1} g_1'(y_1,t) y_1' P^{-1}\|_{q_W} \le \gamma^{-1} \|g_1'(y_1,t) y_1' P^{-1}\|_{q_W}.$$

This implies that

$$\|g_1'(y_1,t)y_1'P^{-1}\|_{q_W}^{q_W} \ge \gamma^{q_W} \rho \delta^{\frac{\rho-1}{\rho}} \|y_1'\|_{q_Y}^{q_Y} - \gamma^{q_W}(\rho-1)\delta \|P\|_{q_H}^{q_H}. \tag{5.8}$$

Similarly by Hölder and Young inequalities, for all $\delta > 0$

$$\|y_2\|_{q_Y}^{q_Y} = \|y_2 P P^{-1}\|_{q_Y}^{q_Y} \le \|y_2 P\|_{q_W}^{q_Y} \|P^{-1}\|_{q_H}^{q_Y} \le \frac{1}{\rho\delta^\rho}\|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*}\|P^{-1}\|_{q_H}^{q_H} \le \frac{1}{\rho\delta^\rho}\|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} L \Pi^{-q_H}$$

where now $\rho = 1 + q_W/q_H$. Last inequality holds since $P > \Pi$ a.e. on the sublevels of $\mathcal{E}$. Then

$$\|g_2(y_1,t)y_2P\|_{q_W}^{q_W} \ge \gamma^{q_W}\rho\delta^\rho \|y_2\|_q^q - \gamma^{q_W}\frac{\rho}{\rho^*}\delta^{\rho+\rho^*}\Pi^{-q_H}. \tag{5.9}$$

Substituting (5.8) and (5.9) into (5.7) with δ sufficiently small, one gets that on sublevels of $\mathcal{E}$ the norms $\|y_1'\|_{q_Y}$, $\|y\|_{q_Y}$, $\|P\|_{q_H}$ and $\|P'\|_{r_H}$ are bounded. By Poincaré inequality it follows that also $\|y_1\|_{1,q_Y}$ is bounded. It remains to prove that $\|y_2'\|_{q_Y}$ is bounded. Let

$$G = g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}.$$

Then

$$y_2' = \frac{GP - g_2'(y_1,t)y_1'y_2}{g_2(y_1,t)}.$$

Il follows that

$$\begin{aligned}\|y_2'\|_{q_Y} &\le \frac{1}{\gamma}\|GP\|_{q_Y} + \frac{1}{\gamma}\|g_2'(y_1,t)y_1'y_2\|_{q_Y} \le \frac{1}{\gamma}\|G\|_{q_W}\|P\|_{q_H} + \frac{\Gamma}{\gamma}\|y_1'y_2\|_{q_Y} \le \\ &\le \frac{1}{\gamma}\|G\|_{q_W}\|P\|_{q_H} + \frac{\Gamma}{\gamma}\|y_1'\|_{q_Y}\|y_2\|_\infty.\end{aligned}$$

Since $L^\infty((0,T))$ is continuously embedded in $W^{1,q_Y}((0,L))$, it follows that for all $\epsilon > 0$ there exists $C > 0$ such that

$$\|y_2\|_\infty \le C(\|y_2\|_{q_Y} + \epsilon\|y_2'\|_{q_Y}) \quad \text{for all } y_2 \in W^{1,q_Y}((0,L)).$$

On a sublevel of $\mathcal{E}$ the quantities $\|G\|_{q_W}$, $\|P\|_{q_H}$, $\|y_1'\|_{q_Y}$ and $\|y_2\|_{q_Y}$ are bounded, so that there exists $M > 0$ such that

$$\|y_2'\|_{q_Y} \le M(1 + \epsilon\|y_2'\|_{q_Y}).$$

Choosing ϵ small enough, it follows that $\|y_2'\|_{q_Y}$ is bounded on the sublevels of $\mathcal{E}$. This concludes the proof. □

We now move to the proof of the weak lower semicontinuity of $\mathcal{E}$. We first need the following technical result.

Proposition 5.7 (convergence of elastic distortion). Let $y^k, y \in W^{1,q_Y}((0,L),\mathbb{R}\times\mathbb{R})$ and $P_k, P \in C^0([0,L])$ such that

$$\begin{aligned} y^k &\rightharpoonup y \quad \text{in } W^{1,q_Y}((0,L)) \\ P_k &\to P \quad \text{in } C^0([0,L]).\end{aligned}$$

Assume also that $P_k, P \ge \Pi > 0$ a.e. in $(0,L)$. Let also $\gamma_1^k, \gamma_1, \gamma_2^k, \gamma_2, \rho_2^k, \rho_2 \in C^0([0,L])$ such that

$$\begin{aligned} \gamma_1^k &\to \gamma_1 \quad \text{in } C^0([0,L]) \\ \gamma_2^k &\to \gamma_2 \quad \text{in } C^0([0,L]) \\ \rho_2^k &\to \rho_2 \quad \text{in } C^0([0,L])\end{aligned}$$

Then, up to extracting a subsequence,

$$\begin{aligned} \gamma_1^k (y_1^k)' P_k^{-1} &\rightharpoonup \gamma_1 y_1' P^{-1} && \text{in } L^1((0,L)) \\ \gamma_2^k y_2^k P_k &\to \gamma_2 y_2 P && \text{in } C^0([0,L]) \\ \rho_2^k (y_1^k)' y_2^k P_k^{-1} + \gamma_2^k (y_2^k)' P_k^{-1} &\rightharpoonup \rho_2 y_1' y_2 P^{-1} + \gamma_2 y_2' P^{-1} && \text{in } L^1((0,L)) \\ \gamma_1^k (y_1^k)' \gamma_2^k y_2^k &\rightharpoonup \gamma_1 y_1' \gamma_2 y_2 && \text{in } L^1((0,L))\end{aligned}$$

Proof. First notice that $P_k^{-1} \to P^{-1}$ in $L^\infty((0,L))$. Indeed
$$|P_k^{-1} - P^{-1}| = \frac{|P_k - P|}{|P_k P|} \le \frac{|P_k - P|}{\Pi^2}.$$
Notice also that, since $W^{1,q}((0,L))$ is compactly embedded in $C^0([0,L])$, up to extracting a subsequence $y_2^k \to y_2$ in $C^0([0,L])$. Now the results follows from Lemma 5.8 below. □

Lemma 5.8. Let $f_k, f \in L^q((0,L))$ with $q \in (1,\infty)$, $g_k, g \in L^\infty((0,L))$ and $h_k, h \in C^0([0,L])$ such that
$$\begin{aligned} f_k &\rightharpoonup f && \text{in } L^q((0,L)) \\ g_k &\to g && \text{in } L^\infty((0,L)) \\ h_k &\to h && \text{in } C^0([0,L]). \end{aligned}$$
Then $f_k g_k h_k \rightharpoonup fgh$ in $L^1((0,L))$.

Proof. Let $\phi \in L^\infty((0,L))$. Then
$$\left|\int_0^L (f_k g_k h_k - fgh)\phi dX\right| \le$$
$$\le \left|\int_0^L (g_k - g) f_k h_k \phi dX\right| + \left|\int_0^L g f_k (h_k - h)\phi dX\right| + \left|\int_0^L g(f_k - f) h\phi dX\right|.$$
Since f_k are bounded in $L^q((0,L))$, h_k are bounded in $C^0([0,L])$, then $f_k h_k \phi$ and $g f_k \phi$ are bounded in $L^1((0,L))$. Moreover $gh\phi \in L^{q^*}((0,L))$. Then the thesis follows. □

We are now ready to get the weakly lower semicontinuity of $\mathcal{E}$.

Proposition 5.9 (lower semicontinuity of the energy). For all $t \in [0,T]$ the map $\mathcal{E}(\cdot,t)$ is lower semicontinuous.

Proof. Let $(y^k, P_k) \rightharpoonup (y,P)$ in $Y \times Z$. We have to prove that
$$\mathcal{E}(y,P,t) \le \liminf_{k\to\infty} \mathcal{E}(y_k, P_k, t).$$
We take a (not relabeled) subsequence such that $\mathcal{E}(y_k,P_k,t)$ has a limit. Then we need to show that
$$\mathcal{E}(y,P,t) \le \lim_{k\to\infty} \mathcal{E}(y_k,P_k,t). \tag{5.10}$$
If $\mathcal{E}(y_k,P_k,t) \to \infty$ we have nothing to prove. Then we assume that instead it converges to a finite limit. In this case we can assume that $\mathcal{E}(y_k,P_k,t) < \infty$ for all k. This implies that $P_k > \Pi$ a.e. in $(0,L)$, so that also $P \ge \Pi$ a.e. in $(0,L)$.

By Rellich–Kondrachov theorem, up to extracting a subsequence, $y_i^k \to y_i$ for $i \in \{1,2\}$ and $P_k \to P$ in $C^0([0,L])$. Moreover $g_1'(y_1^k,t) \to g_1'(y_1,t)$, $g_2(y_1^k,t) \to g_2(y_1,t)$ and $g_2'(y_1^k,t) \to g_2'(y_1,t)$ in $C^0([0,L])$. Indeed if $\gamma \in C^1(\mathbb{R})$ then
$$\begin{aligned} |\gamma(y_1^k) - \gamma(y_1)| &= \left|\int_0^1 \frac{d}{d\lambda}\gamma((1-\lambda)y_1^k + \lambda y_1) d\lambda\right| = \left|\int_0^1 \gamma'((1-\lambda)y_1^k + \lambda y_1)(y_1 - y_1^k) d\lambda\right| \le \\ &\le \|y_1 - y_1^k\|_\infty \int_0^1 |\gamma'((1-\lambda)y_1^k + \lambda y_1)| d\lambda. \end{aligned} \tag{5.11}$$
Since y_1^k converges to y_1 uniformly, then $y_1^k(t)$ takes values in a bounded set as $k \in \mathbb{N}$ and $t \in [0,T]$. Then the continuity of γ' implies that the last integral in (5.11). We have proved that $\gamma(y_1^k) \to \gamma(y_1)$ in $C^0([0,T])$, and this provides the desired results.

Applying Proposition 5.7, it follows that, up to extracting a subsequence,
$$\begin{aligned} g_1'(y_1^k,t)(y_1^k)' P_k^{-1} &\rightharpoonup g_1(y_1,t) y_1' P^{-1} && \text{in } L^1((0,L)) \\ g_2(y_1^k,t) y_2^k P_k &\to g_2(y_1,t) y_2 P && \text{in } C^0([0,L]) \\ g_2'(y_1^k,t)(y_1^k)' y_2^k P_k^{-1} + g_2(y_1^k,t)(y_2^k)' P_k^{-1} &\rightharpoonup g_2'(y_1,t) y_1' y_2 P^{-1} + g_2(y_1,t) y_2' P^{-1} && \text{in } L^1((0,L)) \\ g_1'(y_1^k,t)(y_1^k)' g_2(y_1^k,t) y_2^k &\rightharpoonup g_1'(y_1,t) y_1' g_2(y_1,t) y_2 && \text{in } L^1((0,L)). \end{aligned}$$
By these convergence results, together with $P_k \to P$ in $C^0([0,L])$ and $P_k' \rightharpoonup P'$ in $L^{r_H}((0,T))$, and by the convexity assumptions (W0) and (H0), we can apply Theorem B.30 which leads to (5.10). This concludes the proof. □

Until now we have established (E1) and (E2). We now prove (E3) and (E4).

Proposition 5.10. Then there exists $\mathscr{D} \subset \mathcal{Q}$ such that $\operatorname{dom} \mathcal{E}(\cdot, t) = \mathscr{D}$ for all $t \in [0, T]$. Moreover

(i) for all $q \in \mathscr{D}$ the function $\mathcal{E}(q, \cdot)$ is $C^1([0, T])$,
(ii) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E(C_0^E + \mathcal{E}(q, t))$ for all $q \in \mathcal{Q}$ and $t \in [0, T]$,
(iii) there exists a modulus of continuity ω^E such that for all $q \in \mathscr{D}$ and $t, s \in [0, T]$,

$$|\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| \leq \omega^E(|t - s|) \min\{\mathcal{E}(q, t) + C_0^W, \mathcal{E}(q, s) + C_0^W\}.$$

Proof. Since $g_1' > 0$ and $g_2 > 0$, it is clear that

$$\mathscr{D} = \left\{(y, P) \in Y \times Z \;\middle|\; y_1', y_2, P > 0 \text{ a.e. on } (0, L)\right\}.$$

Proof of (i) Let $E_1 = g_1'(y_1, t) y_1' P^{-1}$, $E_2 = g_2(y_1, t) y_2 P$ and $G = g_2(y_1, t) y_2' P^{-1}$ as functions of $(y, y', P, t) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \times t \in [0, T]$. Let $q = (y, P) \in \mathscr{D}$. Then $(E_1, E_2, G) \in \operatorname{dom} W$ a.e. on $(0, L)$ and everywhere on $[0, T]$. Then $t \mapsto W(E_1, E_2, G)$ is differentiable in $[0, T]$ and a.e. in $(0, L)$ and

$$\frac{d}{dt} W(E_1, E_2, G) = E_1 \frac{\partial W}{\partial E_1}(E_1, E_2, G) \frac{\dot{g}_1'(y_1, t)}{g_1'(y_1, t)} + E_2 \frac{\partial W}{\partial E_2}(E_1, E_2, G) \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)} + G \frac{\partial W}{\partial G}(E_1, E_2, G) \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}$$

there. We have to prove that for all $t \in [0, T]$, there exists

$$\partial_t \mathcal{E}(q, t) = \lim_{\epsilon \to 0} \frac{\mathcal{E}(q, t + \epsilon) - \mathcal{E}(q, t)}{\epsilon} = \int_0^L \frac{d}{dt} W(E_1, E_2, G) dX. \tag{5.12}$$

By Lagrange theorem there exists $\eta \in (0, 1)$ depending on X, t and ϵ, such that

$$\frac{\mathcal{E}(q, t + \epsilon) - \mathcal{E}(q, t)}{\epsilon} = \int_0^L W(g_1'(y_1, t + \eta\epsilon) y_1' P^{-1}, g_2(y_1, t + \eta\epsilon) y_2 P, g_2(y_1, t + \eta\epsilon) y_2' P^{-1}) dX.$$

For ϵ small, $g_1'(y_1, t + \eta\epsilon)/g_1'(y_1, t)$ and $g_2(y_1, t + \eta\epsilon)/g_2(y_1, t)$ are close to 1 uniformly on $(0, L) \times [0, T]$. Thus, for ϵ small enough, we can apply Lemma 5.4 and we get

$$\begin{aligned} -C_0^W &\leq W(g_1'(y_1, t + \eta\epsilon) y_1' P^{-1}, g_2(y_1, t + \eta\epsilon) y_2 P, g_2(y_1, t + \eta\epsilon) y_2' P^{-1}) \leq \\ &\leq C(W(g_1'(y_1, t) y_1' P^{-1}, g_2(y_1, t) y_2 P, g_2(y_1, t) y_2' P^{-1}) + C_0^W) \in L^1((0, L)). \end{aligned}$$

Then, by dominated convergence theorem, we can pass to the limit $\epsilon \to 0$ and we get (5.12).

Proof of (ii). Taking into account the fact that $\frac{\dot{g}_1'}{g_1'}$ and $\frac{\dot{g}_2}{g_2}$ are bounded in $\mathbb{R} \times [0, T]$, assumption (W4, i) yields

$$|\partial_t \mathcal{E}(y, P, t)| \leq 3 C_1^W \max\left\{\left\|\frac{\dot{g}_1'}{g_1'}\right\|_\infty, \left\|\frac{\dot{g}_2}{g_2}\right\|_\infty\right\} \int_\Omega (C_0^W + W(E_1, E_2, G)) dX.$$

This implies the existence of $C_0^E \in \mathbb{R}$ and $C_1^E > 0$ such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E(C_0^E + \mathcal{E}(q, t))$ for all $(q, t) \in \mathscr{D} \times [0, T]$.

Proof of (iii). Let

$$K_i = E_i \frac{\partial W}{\partial E_i} \quad \text{for } i \in \{1, 2, 3\} \quad \text{and} \quad V = \left(\frac{\dot{g}_1'(y_1, t)}{g_1'(y_1, t)}, \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}, \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}\right)$$

as functions of $(X, t) \in (0, L) \times [0, T]$. Then

$$\frac{d}{dt} W(E) = K \cdot V.$$

Let $E \in \mathbb{R}$ and $q \in \mathscr{D}$ such that $\sup_{t \in [0, T]} \mathcal{E}(q, t)$ and let $t, s \in [0, T]$. By the regularity properties (G) of g, there exists modulus of continuity ω^V and ω^G independent of q such that

$$|V(t) - V(s)| \leq \omega^V(|t - s|) \quad \text{and} \quad \max\left\{\frac{g_1'(y_1, s)}{g_1'(y_1, t)}, \frac{g_2(y_1, s)}{g_2(y_1, t)}\right\} \leq \omega^V(|t - s|)$$

on $(0, L)$. Moreover $|K| \le 3C_1^W(W(E)+C_0^W)$ and there exists $M > 0$ independent of q, such that $\|V\|_\infty \le M$. Then by (W2, i) and (W2, ii),

$$\begin{aligned}
|\partial_t \mathcal{E}(q,t) - \partial_t \mathcal{E}(q,s)| &\le \int_0^L |K(t)\cdot V(t) - K(s)\cdot V(s)| dX \le \\
&\le \int_0^L |K(t)-K(s)||V(s)| dX + \int_0^L |K(t)||V(t)-V(s)| dX \le \\
&\le \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W \omega^V(|t-s|)\right) \int_0^L (W(E(t)) + C_0^W) dX \\
&\le \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W \omega^V(|t-s|)\right) (\mathcal{E}(q,t) + C_0^W).
\end{aligned}$$

This concludes the proof. □

5.2.4. Verification the properties of the dissipation distance. Notice that (d1) implies that $\mathcal{D}$ is well defined. Property (D1) follows immediately form (d2). We now prove (D2).

Proposition 5.11. Let $P_k \rightharpoonup P$ and $\hat{P}_k \rightharpoonup \hat{P}$ in Z. Then $\mathcal{D}(P, \hat{P}) \le \liminf_{k\to\infty} \mathcal{D}(P_k, \hat{P}_k)$.

Proof. Let $P_k, P, \hat{P}_k, \hat{P} \in Z$ such that

$$P_k \rightharpoonup P \quad \text{and} \quad \hat{P}_k \rightharpoonup \hat{P} \quad \text{in } Z.$$

By Proposition B.3 it follows that

$$P_k \to P \quad \text{and} \quad \hat{P}_k \to \hat{P} \quad \text{in } C^0([0, L]).$$

Now

$$\liminf_{k\to\infty} \mathcal{D}(P_k, \hat{P}_k) = \liminf_{k\to\infty} \int_0^L D(P_k, \hat{P}_k) dX = \lim_{k\to\infty} \inf_{j\ge k} \int_0^L D(P_j, \hat{P}_j) dX \ge \lim_{k\to\infty} \int_0^L \inf_{j\ge k} D(P_j, \hat{P}_j) dX.$$

By monotone convergence the last limit can be took inside the integral yielding

$$\liminf_{k\to\infty} \mathcal{D}(P_k, \hat{P}_k) \ge \int_0^L \liminf_{k\to\infty} D(P_k, \hat{P}_k) dX \ge \int_0^L D(P, \hat{P}) dX.$$

The last inequality holds since D is lower semicontinuous. This concludes the proof. □

5.2.5. Verification of the compatibility conditions. We now verify (C1) and (C2). In view of Proposition 2.9 we need to verify (PC1) and (PC2). Notice that (PC2) has been established in Proposition 5.10 (iii). Thus we need to verify (PC1) only. A stronger version of (PC1) is the following proposition.

Proposition 5.12. Let $(q_k, t_k), (q, t) \in Q \times [0, T]$ such that $(q_k, t_k) \rightharpoonup (q, t)$ in $Q \times [0, T]$ and $q_k \in \mathscr{D}$. Then for all $\hat{q} \in Q$,

$$\limsup_{k\to\infty} \left(\mathcal{E}(\hat{q}, t_k) + \mathcal{D}(q_k, \hat{q})\right) \le \mathcal{E}(\hat{q}, t) + \mathcal{D}(q, \hat{q}).$$

Proof. Let $(q_k, t_k) \rightharpoonup (q, t)$ in $Q \times [0, T]$. If $\hat{q} \notin \mathscr{D}$ there is nothing to prove since $\mathcal{E}(\hat{q}, t) = \infty$. Then assume that $\hat{q} \in \mathscr{D}$. By Proposition'(5.10) the map $\mathcal{E}(\hat{q}, \cdot)$ is continuous. Then $\mathcal{E}(\hat{q}, t_k) \to \mathcal{E}(\hat{q}, t)$ and it remains to prove only that

$$\limsup_{k\to\infty} \mathcal{D}(q_k, \hat{q}) \le \mathcal{D}(q, \hat{q}).$$

Recall that this is the same as

$$\limsup_{k\to\infty} \mathcal{D}(P_k, \hat{P}) \le \mathcal{D}(P, \hat{P}).$$

Since $\hat{q} \in \mathscr{D}$, then (H3) implies that $\hat{P} > \Pi$ a.e. in $(0, L)$, so that $\hat{P} \ge \Pi$ in $[0, L]$. Since also $q_k \in \mathscr{D}$ then, for the same reason, also $P_k \ge \Pi$ in $[0, L]$. Notice also that by B.3, $P_k \to P$ in $C^0([0, T])$. Then actually

$$\lim_{k\to\infty} \mathcal{D}(P_k, \hat{P}) = \lim_{k\to\infty} \int_0^L D(P_p, \hat{P}) dX = \mathcal{D}(P, \hat{P}),$$

where last inequality holds because D is finite and continuous on $[\Pi, \infty) \times [\Pi, \infty)$ by (d3). This concludes the proof. □

5.2.6. Conclusion of the existence proof. All the hypotheses of Theorem 2.8 have been established. Thus applying that theorem we get desired existence result.

Theorem 5.13. Let $q_0 = (y_0, P_0) \in \mathcal{Q}$ such that $\mathcal{E}(q_0, 0) < \infty$. Then there exists an energetic solution $q = (y, P)\colon [0, T] \to \mathcal{Q}$ such that $q(0) = q_0$.

5.2.7. Particular constitutive relations that satisfy the hypotheses of the existence theorem. We now discuss particular constitutive relations that satisfies the hypotheses required to apply Theorem 5.13. Taking inspiration from the neo-Hooken elastic free energy density (5.1), we consider

$$W(E_1, E_2, G) = \begin{cases} \mathbb{W}(E_1, E_2, G, E_1 E_2) & \text{if } E_1 > 0 \text{ and } E_2 > 0 \\ \infty & \text{otherwise,} \end{cases}$$

with

$$\mathbb{W}(E_1, E_2, G, J) = A \frac{|E_1 - 1|^p + |E_2 - 1|^p + |lG|^p}{J^\alpha} + B|J - 1|^r$$

where $A, B, l > 0$, $r > 1$, $0 < \alpha \leq p - 1$ and $p > 1 + \frac{\alpha}{r}$. The parameter $l > 0$ is connected with the diameter of the cross section Σ. We now check that this particular W satisfies (W1), (W2), (W3) and (W4) of previous section. We start with the check of the assumptions (W1) and (W2). Hypotheses (W1) is a consequence of the following lemma.

Lemma 5.14. Let $w\colon (0, \infty) \to \mathbb{R}$ strictly convex and let $g\colon (0, \infty) \times (0, \infty) \to \mathbb{R}$ given by $g(r, y) = \frac{r^p}{y^\alpha} + w(y)$ for some $p > 1$ and $0 < \alpha \leq p - 1$. Let also $n \in \mathbb{N}$ and $f\colon \mathbb{R}^n \times (0, \infty) \to \mathbb{R}$ given by $f(x, y) = g(|x|, y)$. Then f is strictly convex in $\mathbb{R}^n \times (0, \infty)$.

Proof. We first prove that g is strictly convex. The hessian matrix of g is given by

$$\nabla^2 g(r, y) = \begin{pmatrix} p(p-1)\frac{r^{p-2}}{y^\alpha} & -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} \\ -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} & \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) \end{pmatrix}.$$

Then

$$\begin{aligned} \det \nabla^2 g(r, p) &= p(p-1)\frac{r^{p-2}}{y^\alpha}\left(\alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y)\right) - p^2\alpha^2\frac{r^{2p-2}}{y^{\alpha+2}} = \\ &= p\alpha(p - \alpha - 1)\frac{r^{2p-2}}{y^{\alpha+2}} + p(p-1)\frac{r^{p-2}}{y^\alpha}w''(y) > 0. \end{aligned}$$

and

$$\operatorname{tr} \nabla^2 g(r, y) = p(p-1)\frac{r^{p-2}}{y^\alpha} + \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) > 0.$$

This proves that $\nabla^2 g$ is everywhere positive-definite. This implies that g is strictly convex. Then for any $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^n \times (0, \infty)$ and $0 < \lambda < 1$ it holds

$$\begin{aligned} f((1-\lambda)(x_0, y_0) + \lambda(x_1, y_1)) &= \frac{|(1-\lambda)x_0 + \lambda x_1|^p}{y^\alpha} + w(y) \leq \\ &\leq \frac{[(1-\lambda)|x_0| + \lambda|x_1|]^p}{\left[(1-\lambda)y_0 + \lambda y_1)\right]^\alpha} + w((1-\lambda)y_0 + \lambda y_1)) = \\ &= g((1-\lambda)(|x_0|, y_0) + \lambda(|x_1|, y_1)) < (1-\lambda)g(|x_0|, y_0) + \lambda g(|x_1|, y_1) = \\ &= (1-\lambda)f(x_0, y_0) + \lambda f(x_1, y_1). \end{aligned}$$

This is exactly the thesis. □

We now prove the growth condition (W2) for $q_W = p/\left(1 + \frac{\alpha}{r}\right) > 1$. Lemma 5.5 yields for all $\delta > 0$ and $\rho > 1$ that

$$\begin{aligned} \mathbb{W}(E_1, E_2, G, J) \geq A\rho\delta^{\frac{\rho-1}{\rho}}|E_1 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|E_2 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|lG|^{\frac{p}{\rho}} - \\ - (2A+1)(\rho-1)\delta J^{\frac{\alpha}{\rho-1}} + B|J-1|^r. \end{aligned}$$

Choosing ρ such that $\frac{\alpha}{\rho-1} = r$ we get $\frac{p}{\rho} = q_H$. Moreover choosing $\delta > 0$ sufficiency small we have

$$\inf_{J>0}\left[-(2A+1)(\rho-1)\delta J^r + B|J-1|^r\right] > -\infty.$$

This is implied immediately from the following lemma.

Lemma 5.15. Let $r \geq 1$. Then there exists $C > 0$ such that $|x - 1|^r \geq C|x|^r - 1$ for all $x \in \mathbb{R}$.

Proof. By the equivalence of the norms there exists $K > 0$ such that

$$(|x-1|^r + 1)^{1/r} \geq K(|x-1| + 1) \geq K|x|.$$

Now the thesis follows immediately. □

A possible choice of the hardening free energy density is

$$H(P,P') = \begin{cases} C(P-1)^q + \frac{D}{P-\Pi} + E|lP'|^q & \text{if } P > \Pi \\ \infty & \text{otherwise.} \end{cases}$$

where $q > 1$, $0 < \Pi < 1$ and $C, D, E > 0$. It trivially satisfies hypotheses (H1), (H2) and (H3).

Remark 5.16. In what follow we will consider only the following refined version of the hardening free energy

$$H(P,P') = \begin{cases} C\left(P^q + \frac{q}{\delta}e^{-\delta(P-1)}\right) + \frac{D}{P-\Pi} + E|lP'|^q & \text{if } P > p \\ \infty & \text{otherwise,} \end{cases} \tag{5.13}$$

where $C, D, E, \delta > 0$ and $q > 1$. This constitutive relation still satisfies the assumptions (H1), (H2) and (H3). The expression (5.13) is obtained as a modification of the Armstrong–Frederick kinematical hardening model which can reproduce better the experimentally observed relation between force and elongation. According to the Armstrong–Frederick model, the back-stress β (that in our model correspond to the conservative microforce $\beta = \frac{\partial H}{\partial P}$) is required to satisfy the condition

$$\dot\beta = \delta(C\dot P - |\dot P|\beta) \tag{5.14}$$

for some $\delta, C > 0$, see [64]. In tension $\dot P \geq 0$ and (5.14) becomes

$$\delta \dot P = \frac{\dot\beta}{C-\beta} = -\frac{d}{dt}\log(C-\beta).$$

This suggest the relation

$$\log\frac{C-\beta}{C} = -\delta(P-1) \quad \text{so that} \quad \beta = C(1-e^{-\delta(P-1)}). \tag{5.15}$$

The main feature of this relation is that β is bounded from above by C and $P \to C^-$ as $P \to \infty$. Now (5.15) corresponds to the energy density

$$H = C\left(P + \frac{1}{\delta}e^{-\delta(P-1)}\right).$$

Formula (5.13) is a modification of this as the linear growth in P has been substituted with a superlinear growth in order to satisfy (H2). The superlinear growth makes β unbounded as $\beta \to \infty$, but the main features of the original Armstrong–Frederick model are preserved.

The disipation potential density can be chosen as

$$R(P,\dot P) = \begin{cases} Y\frac{|\dot P|}{P} & \text{if } P \geq \Pi \\ \infty & \text{otherwise,} \end{cases}$$

where $Y > 0$ is the yield strength. The corresponding dissipation distance density $D\colon \mathbb{R} \to [0,\infty]$ can be calculated by mean of formula (5.6). We find that

$$D(P_0,P_1) = \begin{cases} Y|\log P_1 - \log P_0| & \text{if } P_0, P_1 \geq p \\ \infty & \text{otherwise.} \end{cases} \tag{5.16}$$

Indeed notice that for each $P \in C^1([0,1],(0,\infty))$ connecting P_0 with P_0, it holds

$$\int_0^1 R(P(t),\dot P(t))dt \geq Y\int_0^1 \left|\frac{\dot P(t)}{P(t)}\right| dt \geq Y\left|\int_0^1 \frac{\dot P(t)}{P(t)}dt\right| = Y|\log P_1 - \log P_0|.$$

This implies $D(P_0,P_1) \geq Y|\log P_1 - \log P_0|$. If $P_i < \Pi$ for some $i \in \{0,1\}$, then every C^1 curve P connecting P_0 with P_1, has $\int_0^1 R(P(t),\dot P(t))dt = \infty$. If instead $P_0, P_1 \geq \Pi$ then we can take $P(t) = P_0^t P_1^{1-t} \geq \Pi$ and in this case

$$\int_0^1 R(P(t),\dot P(t))dt = Y\int_0^1 \left|\frac{\dot P(t)}{P(t)}\right| dt = Y|\log P_1 - \log P_1|.$$

This implies $D(P_0,P_1) = Y|\log P_1 - \log P_0|$. Now it is easy to see that D in (5.16) satisfies the assumptions (d1), (d2) and (d3).

5.3. Qualitative analysis of the tensile test problem

We consider the following boundary value problem describing the tensile test: the bar in not subject to any distributed load, that is $f = 0$. Moreover the end sections are left free to slide in the transversal direction, this means that $h_2(X,t) = 0$ for $X \in \{0, L\}$ and $t \in [0,T]$ and $\Gamma_D^2 = \emptyset$. The longitudinal position of the end sections is assigned, this means that $\Gamma_D^1 = \{0, L\}$. In particular we consider the boundary conditions $\chi_1(X,t) = \eta(t)X$ for $X \in \{0, L\}$ and $t \in [0,T]$, where $\eta \in C^\infty([0,T],(0,\infty))$ is the imposed elongation. We suppose that $\eta(0) = 1$, $\eta \geq 1$ and η is nondecreasing. To handle the boundary conditions we define $g\colon \mathbb{R}^2 \times [0,T] \to \mathbb{R}^2$ by $g(y,t) = (\eta(t)y_1, y_2)$ so that $\chi(X,t) = \eta(t)y_1(X,t)$ and $\chi_2(X,t) = y_2(X,t)$. Then the unknowns of the problem are $y\colon (0,L)\times[0,T] \to \mathbb{R}^2$ and $P\colon (0,L)\times[0,T] \to \mathbb{R}$ with the constrain $y_1(X,t) = X$ for $X \in \{0, L\}$.

We will see that a trivial strong solution to this problem is the spatially homogeneous one that is $\bar{y}_1(X,t) = X$ for all $X \in (0,L)$ and $t \in [0,T]$, $\bar{y}_2(\cdot,t)$ and $\bar{P}(\cdot,t)$ constant for all $t \in [0,T]$. However we expect that this trivial solution is unstable when η is sufficiently large, in the sense that it does not satisfy the global stability condition

$$\mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}) \quad \text{for all } (\hat{y}, \hat{P}) \in \mathcal{Q}.$$

The instability of the trivial solution would correspond with the onset of necking, when a non uniform localized solution becomes energetically favorable.

We will discuss first the stability of the trivial solution when the elastic response only is considered. In this case we will be able to obtain some explicit results. Then we will show our efforts to extend the analysis also in the elastoplastic case. In this second case we cannot obtain explicit results, however we can gain some insight about the phenomenology of necking.

Remark 5.17. Most of the stability analysis below is only formal. Some additional effort should be made in order to get fully rigorous results. The mathematical theory that should be employed to reach a full justification of the procedure below is that of bifurcation theory in Banach spaces, probably with some nontrivial adjustments needed to treat the nonsmooth character of plasticity theory. Some results regarding this theory in the smooth context and applications to instability phenomena in nonlinear elasticity can be found in our review [4].

5.3.1. Purely elastic model. In the purely elastic model, equations (5.2) reduces to

$$\begin{aligned} \sigma_1' &= 0 \quad \text{in } (0,L) \\ -\sigma_2 + \zeta' &= 0 \quad \text{in } (0,L) \\ \zeta(X) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

From the first equation it follows that $\sigma_1 = F$, where F is a constant, representing the axial reaction force needed to impose the boundary conditions on χ_1. The constitutive relations yields $\sigma_1 = W_1(E_1, E_2, G)$, $\sigma_2 = W_2(E_1, E_2, G)$ and $\zeta = W_3(E_1, E_2, G)$, where $E_1 = \chi_1'$, $E_2 = \chi_2$ and $G = \chi_2'$. We have denoted the partial derivatives of W with respect to its arguments E_1, E_2 and G with the subscripts $1, 2$ and 3 respectively. Inserting these constitutive relations into (5.17), we get

$$\begin{aligned} \frac{d}{dX} W_1(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0,L) \\ -W_2(\chi_1', \chi_2, \chi_2') + \frac{d}{dX} W_3(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0,L) \\ W_3(\chi_1'(X), \chi_2(X), \chi_2'(X)) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

We first look for homogeneous solutions of the form $\bar{\chi}_1(X,t) = \eta(t)X$, $\bar{\chi}_2(X,t) = \bar{E}_2(t)$ where $\bar{E}_2\colon (0,T) \to (0,\infty)$ is the only unknown. Notice that $\bar{E}_1(t) = \bar{\chi}_1(X,t) = \eta(t)$ and $\bar{G}(t) = \bar{\chi}_2'(X,t) = 0$ for all $X \in (0,L)$ and $t \in [0,T]$. To find $\bar{E}_2$ we need only to solve the nonlinear equation

$$W_2(\eta(t), \bar{E}_2(t), 0) = 0.$$

This yields $\bar{E}_2$ as a function of η.

We now turn to the study of the stability of the homogeneous solution $(\bar{\chi}, \bar{P})$. To this aim we should check whether the second variation of the energy becomes indefinite for η large enough. This can be done as follows: we insert a perturbation of the homogeneous solution $\chi_1 = \bar{\chi}_1 + v_1$ and $\chi_2 = \bar{\chi}_2 + v_2$ into problem (5.18), then we linearize the resulting problem around the homogeneous solution and we look

(A) Reference configuration

(B) Bifurcation mode

FIGURE 5.1. Bifurcation mode of the elastic bar under traction.

for nontrivial solutions. The linearized problem reads as follows

$$\bar{W}_{11}v_1'' + \bar{W}_{12}v_2' + \bar{W}_{13}v_2'' = 0 \quad \text{in } (0, L)$$
$$\begin{aligned} &- \bar{W}_{12}v_1' - \bar{W}_{22}v_2 - \bar{W}_{23}v_2' + \\ &+ \bar{W}_{13}v_1'' + \bar{W}_{23}v_2' + \bar{W}_{33}v_2'' = 0 \end{aligned} \quad \text{in } (0, L)$$

together with the boundary conditions

$$v_1(X) = 0 \quad \text{for } X \in \{0, L\}$$
$$v_2'(X) = 0 \quad \text{for } X \in \{0, L\}.$$

We have denoted with a superimposed bar, the evaluation at $(\eta, \bar{E}_2, 0)$. From mechanical ground, it is reasonable to assume that $\sigma_i = W_i$ for $i \in \{1, 2\}$ is even in G and $\zeta = W_3$ is odd in G. Then $\bar{W}_{i3} = 0$ for $i \in \{1, 2\}$. Moreover we assume that $\bar{W}_{33} > 0$, $\bar{W}_{11} > 0$, $\bar{W}_{22} > 0$. Then (5.19) simplifies to

$$\bar{W}_{11}v_1'' + \bar{W}_{12}v_2' = 0 \quad \text{in } (0, L)$$
$$\bar{W}_{33}v_2'' - \bar{W}_{12}v_1' - \bar{W}_{22}v_2 = 0 \quad \text{in } (0, L).$$

The first equation yields

$$v_1' = -\frac{\bar{W}_{12}}{\bar{W}_{11}}v_2 + K_1 \tag{5.21}$$

for some constant $K_1 \in \mathbb{R}$. Substituting into the second equation we obtain

$$\bar{W}_{33}v_2'' + \left(\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}\right)v_2 = \bar{W}_{12}K_1 \quad \text{in } (0, L).$$

Let

$$\lambda = \frac{\frac{(\bar{W}_{12})^2}{\bar{W}_{11}} - \bar{W}_{22}}{\bar{W}_{33}}.$$

Taking into account the boundary condition $v_2'(X) = 0$ for $X \in \{0, L\}$, we find that a nontrivial solution exists if and only if $\lambda = \frac{k^2\pi^2}{L^2}$ for $k \in \{1, 2, \dots\}$ and is of the form

$$v_2(X) = \frac{\bar{W}_{12}K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + K_2 \cos\left(\sqrt{\lambda}X\right).$$

Here $K_2 \in \mathbb{R}$ is a constant. Inserting in (5.21) we get that

$$v_1(X) = \left(-\frac{\bar{W}_{12}}{\bar{W}_{11}}\frac{\bar{W}_{12}K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + 1\right)K_1X - \frac{\bar{W}_{12}}{\bar{W}_{11}}\frac{K_2}{\sqrt{\lambda}}\sin\left(\sqrt{\lambda}X\right) + K_3$$

with $K_3 \in \mathbb{R}$. Boundary condition imposes that $K_1 = 0$ and $K_3 = 0$. From these considerations we conclude that the homogeneous solution becomes unstable when $\lambda = \pi^2/L^2$ for the first time. The shape of the instability is showed schematically in Figure 5.1

Now let $\bar{F}(t) = W_1(\eta(t), \bar{E}_2(t), 0)$ be the axial force along the homogeneous solution. We now show that the instability can occur only after that F reaches its maximum. Indeed let us view $\bar{F}$ as a function of η, so that

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} + \bar{W}_{12}\frac{d\bar{E}_2}{d\eta}.$$

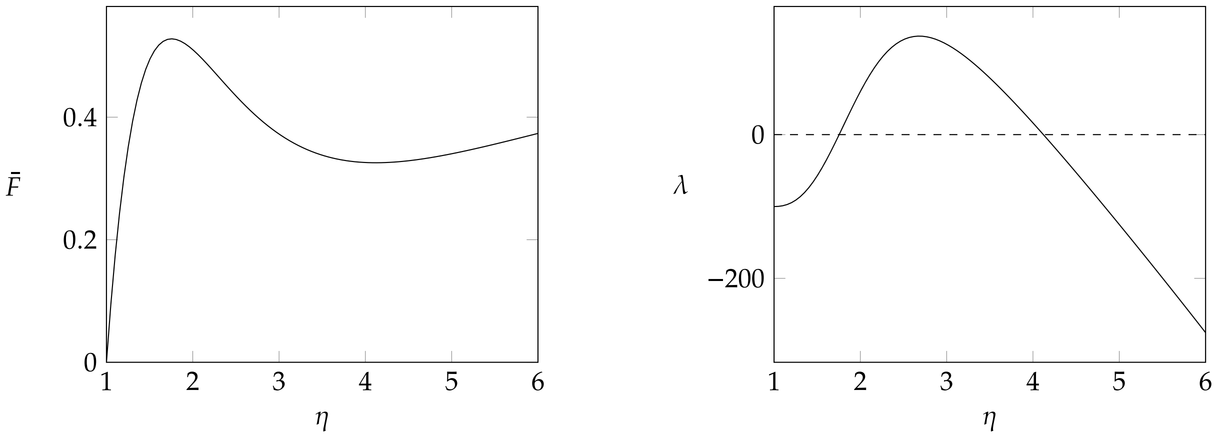


FIGURE 5.2. Graphs of $\bar{F}$ and λ as functions of the imposed elongation η for the homogeneous deformation in the purely elastic model with the choice of the value of the parameters in 5.22.

On the other end we know that $\bar{W}_2 = W_2(\eta, \bar{E}_2, 0) = 0$ so that, by differentiating with respect to η,

$$\bar{W}_{12} + \bar{W}_{22}\frac{d\bar{E}_2}{d\eta} = 0 \quad \text{that implies} \quad \frac{d\bar{E}_2}{d\eta} = -\frac{\bar{W}_{12}}{\bar{W}_{22}}.$$

Then

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} - \frac{\bar{W}_{12}^2}{\bar{W}_{22}} = -\bar{W}_{33}\frac{\bar{W}_{11}}{\bar{W}_{22}}\lambda.$$

Now, at the first stages of deformation the force increases with elongation so $d\bar{F}/d\eta > 0$ that implies $\lambda < 0$ and no instability can occur. It $\bar{F}$ has a (local) maximum then λ become positive and instability occur if it reach the value π^2/L^2. This clearly happen if the bar is sufficiently long.

Consider now the particular constitutive model described in Section 5.2.7. If we choose

$$A = 1, B = 10^{-4}, p = 2, \alpha = 1 \text{ and } r = 2, \tag{5.22}$$

we get the graphs of $\bar{F}$ and λ as functions of η reported in Figure 5.2. Notice that F has a maximum and λ became positive so that instability occur if L is sufficiently large.

REMARK 5.18. We notice that the particular choice of the parameters in (5.22) is not mechanically reasonable for metals, and was motivated only by the need to show that the mechanical model that we have developed in this chapter can exhibit instability phenomena. The inaccuracy of (5.22) stems to the fact that, since B is small compared to A, the nearly-compressibility behavior of metals is not correctly reproduced. Moreover (5.22) implies that cross-sections growth in traction, this is clearly wrong for metals.

5.3.2. Elastoplastic model. We start with the computation of the trivial solution of the form $\bar{y}_1(X,t) = \eta(t)X$, $\bar{y}_2(X,t) = \bar{y}_2(t)$ and $\bar{P}(X,t) = \bar{P}(t)$ for all $X \in (0,L)$ and $t \in [0,T]$. In the elastic range, $\bar{y}_2(t)$ can be found as we showed previously while $\bar{P}(t) = 1$. When the Madel stress (5.3) overcomes the yield strength Y, then $\dot{\bar{P}} > 0$ and the plastic force is everywhere given by $\partial_{\dot{P}}R(\bar{P}, \dot{\bar{P}}) = Y/\bar{P}$. Then the balance of microforces coupled with the balance in the transversal direction gives a nonlinear algebraic equation that allows to compute $\bar{y}_2$ and $\bar{P}$.

To study the stability of the trivial solution we have to check whether the global stability condition

$$\mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}) \quad \text{for all } (\hat{y}, \hat{P}) \in \mathcal{Q}. \tag{5.23}$$

is satisfied. Since $\mathcal{D}$ is a quasidistance, and in particular $\mathcal{D}(P_0, P_1) \geq 0$ and equality holds if and only if $P_0 = P_1$, the global stability condition is equivalent to say that $(\bar{y}(t), \bar{P}(t))$ is a global minimizer of $\mathcal{J}_t \colon \mathcal{Q} \to \mathbb{R} \cup \{\infty\}$ defined by

$$\mathcal{J}_t(\hat{y}, \hat{Q}) = \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}).$$

The aim of what follows would be to identify conditions that ensure that for sufficiently large elongations (5.23) is violated, and to understand qualitatively the feature of the stable localized solution. We were actually not able to get conclusive results in this sense, so we now propose some heuristic considerations that suggest that instability in the form of necking is likely to occur if suitable constitutive relation for the hardening free energy density are chosen.

Remark 5.19. Our study of stability will be local, meaning that we will study the behavior of $\mathcal{J}_t$ near $(\bar{y}(t), \bar{P}(t))$ only, taking into account first and second order variations. In particular our analysis will not be able to identify the cases in which $(\bar{y}(t), \bar{P}(t))$ is a local but not a global minimum of $\mathcal{J}_t$: if this happen, according to the definition of energetic solution, the configuration must jump from the trivial to the energetically favorable one. On the other hand the experimental observations and the theoretical and numerical studies mentioned in the introduction of Chapter 3 suggest that necking happen without jumps. Thus we expect that the estimates we will find of the threshold of instability detected by our local analysis corresponds with the true condition for the beginning of necking. See also the discussion in Chapter 2 regarding the potential limitations of energetic solutions in identifying the exact onset of instability.

We start our analysis assuming that the stability condition is not violated in the elastic range. This is certainly true in the case of metals since yielding starts when the geometric nonlinearities have still a negligible effect. So fix $t \in [0, T]$ and suppose that the yield point has been already reached at that time. Given the difficulties in fully characterizing the exact stability condition (5.23), we define instead two simplified stability conditions that can however give us some information about the occurrence of necking. We call them *stability under force control* and *regularized stability*.

Stability under force control. To introduce this first alternative stability condition, let us define

$$\bar{F}(t) = \bar{P}(t)^{-1} W_1(\eta(t)\bar{P}(t)^{-1}, \bar{y}_2(t)\bar{P}(t), \bar{y}_2'(t)\bar{P}(t)^{-1}),$$

the reaction tensile force acting on the ends of the bar along the trivial solution. Consider the function $J_t\colon (0,\infty)^3 \to \mathbb{R} \cup \{\infty\}$ given by

$$J_t(\rho, y_2, P) = W(\rho P^{-1}, y_2 P, 0) + H(P, 0) + D(\bar{P}(t), P) - \bar{F}(t)\rho.$$

The quantity $J_t(\rho, y_2, P) - J_t(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is the energy per unit length that must be supplied to the system to bring it from the configuration $\bar{y}_1(t)$, $\bar{y}_2(t)$ and $\bar{P}(t)$ to the configuration $y_1(X) = \rho X$, $y_2(X) = y_2$ and $P(X) = P$ if the ends are now left free to move and the tensile force is kept equal to $\bar{F}(t)$. It includes both the energy stored by the system in the form of free energy and the energy dissipated as the plastic distortion is changed. Now our first simplified stability condition is the following.

Definition 5.20 (stability under force control). We say that the trivial uniform solution at time $t \in (0, T]$, is stable under force control if $(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is a local minimizer of J_t.

The mechanical meaning of this definition is the following: the trivial uniform solution $(\eta, \bar{y}_2(t), \bar{P}(t))$ is stable under force control when it is the most energetically favored configuration among all other uniform configurations and under the condition of fixed tensile force is applied. We now study whether the stability of the trivial uniform solution under force control is lost for sufficiently large elongations when the particular constitutive relations proposed in Section 5.2.7 are considered. In this particular case J_t is not differentiable at $P = \bar{P}(t)$ but $J_t \in C^1(H_t^+)$ and $J_t \in C^1(H_t^-)$, where

$$H_t^+ = \{(\rho, y_2, P) \in (0,\infty)^3 \mid P \geq \bar{P}(t)\} \quad \text{and} \quad H_t^- = \{(\rho, y_2, P) \in (0,\infty)^3 \mid \Pi < P \leq \bar{P}(t)\}.$$

We easily find that

$$\nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^-) = \left(0, 0, -\frac{2Y}{\bar{P}(t)}\right) \quad \text{and} \quad \nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^+) = (0,0,0).$$

Then, wether $(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is a local minimum of J_t depends on the eigenvalues of the hessian matrix

$$h(t) = \nabla^2 J_t(\eta(t), \bar{y}_2(t), \bar{P}(t)^+).$$

It is clear that at $t = 0$ the trivial solution is stable and that $h(0)$ is positive definite. Then instability occur when $h(t)$ become singular. We now show that this happen exactly when $\bar{F}$ reaches its maximum, in agreement with Considère criterion. Indeed the following lemma holds.

Lemma 5.21. Let

$$a(t) = h_{11}(t), \quad b(t) = \begin{pmatrix} h_{22}(t) & h_{23}(t) \\ h_{32}(t) & h_{33}(t) \end{pmatrix} \quad \text{and} \quad c(t) = \begin{pmatrix} h_{21}(t) \\ h_{31}(t) \end{pmatrix}$$

so that

$$h(t) = \begin{pmatrix} a(t) & c(t)^T \\ c(t) & b(t) \end{pmatrix}.$$

Then it holds that

$$\frac{d\bar{F}}{d\eta}(t) := \frac{d\bar{F}}{dt}(t) \left(\frac{d\eta}{dt}(t)\right)^{-1} = \frac{\det h(t)}{\det b(t)}.$$

Proof. Notice that

$$\bar{F}(s) = \frac{\partial J_t}{\partial \rho}(\eta(s), \bar{y}_2(s), \bar{P}(s)) + \bar{F}(t).$$

Then, differentiating with respect to s, computing for $s = t$ and dividing by $\frac{d\eta}{dt}(t)$, we get

$$\frac{d\bar{F}}{d\eta} = \frac{\partial J_t}{\partial\rho\partial\rho}(\eta, \bar{y}_2, \bar{P}) + \frac{\partial J_t}{\partial\rho\partial y_2}(\eta, \bar{y}_2, \bar{P})\frac{d\bar{y}_2}{d\eta} + \frac{\partial J_t}{\partial\rho\partial P}(\eta, \bar{y}_2, \bar{P})\frac{d\bar{P}}{d\eta}. \tag{5.24}$$

On the other hand, the balance in the transversal direction and the balance of microforces can be written respectively as

$$\frac{\partial J_t}{\partial y_2}(\eta, \bar{y}_2, \bar{P}) = 0 \quad \text{and} \quad \frac{\partial J_t}{\partial P}(\eta, \bar{y}_2, \bar{P}) = 0.$$

Differentiating with respect to η we get

$$\begin{cases} \dfrac{\partial J_t}{\partial\rho\partial y_2}(\eta, \bar{y}_2, \bar{P}) + \dfrac{\partial J_t}{\partial y_2\partial y_2}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{y}_2}{d\eta} + \dfrac{\partial J_t}{\partial y_2\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{P}}{d\eta} = 0 \\ \dfrac{\partial J_t}{\partial\rho\partial P}(\eta, \bar{y}_2, \bar{P}) + \dfrac{\partial J_t}{\partial y_2\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{y}_2}{d\eta} + \dfrac{\partial J_t}{\partial P\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{P}}{d\eta} = 0. \end{cases}$$

Then

$$\begin{pmatrix} \dfrac{d\bar{P}}{d\eta} \\ \dfrac{d\bar{P}}{d\eta} \end{pmatrix} = -b^{-1}c.$$

Substituting into (5.24) we get

$$\frac{d\bar{F}}{d\eta} = a - c \cdot b^{-1}c.$$

But since

$$\det h = \det b \left(a - c \cdot b^{-1}c\right)$$

the thesis follows. □

It is easy to see that the constitutive model described in Section 5.2.7 implies that $\det b(t) > 0$. Then previous lemma shows indeed that instability occur when $\bar{F}$ reaches its maximum. In Figure 5.3 the result is demonstrated numerically: we adopted the Armstrong–Frederick hardening function of Remark 5.16 and we tuned the amount of hardening (in particular the parameters C and δ) so that $\bar{F}$ exhibited a maximum. By a numerical computation one can find also that at the instant $t_0 \in (0, T]$ when $\det h(t_0) = 0$, the kernel of $h(t_0)$ is the span of a vector of the form (θ, v_2, Q) with $\theta > 0$, $v_2 < 0$ and $Q > 0$. This means that, if at time t_0 the bar is constrained to remains uniform, allowed to change its length and with the tensile $\bar{F}(t_0)$ kept fixed, it tends to shrink and increase its plastic distortion. This behavior constitutes the expected mechanism for the initiation of necking. Returning to the actual boundary value problem where the elongation, rather than the tensile force, is controlled, our findings suggest that when the total elongation is sufficiently large, the bar tends to shrink and elongate more than what is predicted by the trivial uniform solution. However, since the global boundary conditions prevent this additional deformation from occurring everywhere, the bar can only satisfy this tendency at a specific point along its length, thus initiating the necking process.

Regularized stability. The feature that makes inequality (5.23), or equivalently the minimality of $\mathcal{J}_t$ at $(\bar{y}(t), \bar{P}(t))$ difficult to analyze, is the lack of smoothness of $\mathcal{D}$. This indeed prevents us from considering the first and second variation of $\mathcal{J}_t$ to check whether $(\bar{y}(t), \bar{P}(t))$ is a (local) minima. The idea to circumvent these difficulties would be to consider only variations in the directions where the one-sided directional derivatives exists. Specifically we restrict to the particular constitutive relation for the dissipation potential discussed in Section 5.2.7, we define the two smooth functionals $\mathcal{J}_t^\varsigma \colon \mathcal{Q} \to \mathbb{R} \cup \{\infty\}$ for $\varsigma \in \{\pm 1\}$ defined by

$$\mathcal{J}_t^\varsigma(\hat{y}, \hat{P}, t) = \int_0^L W(\eta(t)\hat{y}_1'\hat{P}^{-1}, \hat{y}_2\hat{P}, \hat{y}_2'\hat{P}^{-1})dX + \int_0^L H(\hat{P}, \hat{P}')dX + \int_0^L D^\varsigma(\bar{P}(t), \hat{P})dt,$$

with $D^\varsigma(P_0, P_1) = \varsigma Y \log(P_1/P_0)$. This definitions allows to write $\mathcal{J}_t(\hat{q}) = \mathcal{J}_t^\pm(\hat{q})$ if $\hat{q} \in \mathcal{Q}_t^\varsigma$, where

$$\mathcal{Q}_t^\varsigma = \{\hat{q} = (\hat{y}, \hat{P}) \in \mathcal{Q} \mid \varsigma(\hat{P} - \bar{P}(t)) \geq 0 \text{ in } (0, L)\}.$$

Then the following lemma express a relation between the minimality of $\mathcal{J}_t$ and similar conditions for $\mathcal{J}_t^\varsigma$ that can be useful.

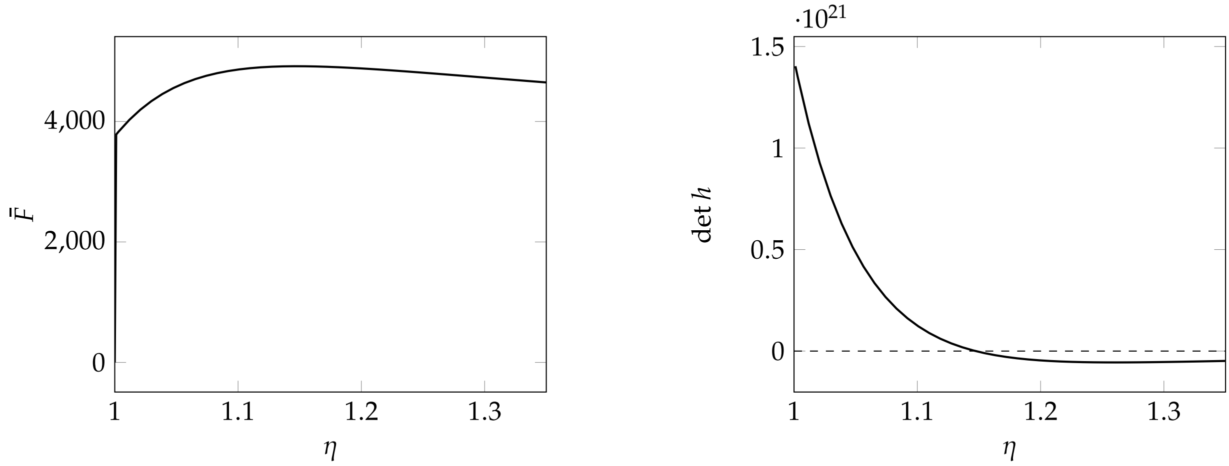


FIGURE 5.3. Graphs of $\bar{F}$ and $\det h$ as functions of the elongation η along the trivial solution in the plastic model. Notice that the maximum of $\bar{F}$ is reached when h vanishes at $\eta_{\mathrm{m}} = 1.1474$.

LEMMA 5.22. The global stability condition

$$\mathcal{J}_t(\bar{q}(t)) \le \mathcal{J}_t(\hat{q}) \quad \text{for all } \hat{q} \in \mathcal{Q}$$

implies the following couple of one-sided global stability conditions

$$\mathcal{J}_t^{\varsigma}(\bar{q}(t)) \le \mathcal{J}_t^{\varsigma}(\hat{q}) \quad \text{for all } \hat{q} \in \mathcal{Q}_t^{\varsigma} \text{ and } \varsigma \in \{\pm 1\}. \tag{5.25}$$

PROOF. Since $\mathcal{J}_t(\hat{q}) = \mathcal{J}_t^{\pm}(\hat{q})$ if $\hat{q} \in \mathcal{Q}_t^{\varsigma}$, the results follows immediately. □

Thus a sufficient condition for instability is that at least one among (5.25) is violated, so we now try to study whether this happen. Considering the way in which the trivial solution is computed (see the beginning of Section‘5.3.2) it is immediate to check that $\partial\mathcal{J}_t^{+}(\bar{q}(t)) = 0$ for all $t \in [0, T]$. This implies that for all $w = (v, Q) \in Q$ with $w_1(0) = w_1(L) = 0$ and $Q \le 0$ in $(0, L)$, it holds that

$$\langle \partial\mathcal{J}_t^{-}(\bar{q}(t)), w\rangle = \langle \partial\mathcal{J}_t^{+}(\bar{q}(t)), w\rangle - 2Y\int_0^L \log\frac{Q}{\bar{P}(t)}dX = -2Y\int_0^L \frac{Q}{\bar{P}(t)}dX \ge 0.$$

This means that (5.25) with $\varsigma = -1$ is satisfied, at least locally for $\hat{q}$ near $\bar{q}(t)$, for all $t \in [0, T]$. This means that instability modes in which the plastic distortion P diminish everywhere with respect to $\overline{P}(t)$ are excluded. This result is expected: as the bar remains under tension, there is no driving force that could justify reversal of plastic flow. So we turn to study whether (5.25), for $\varsigma = 1$, is violated. To this aim we could study the second variation of $\mathcal{J}_t^{+}$ at $(\bar{y}(t), \bar{P}(t))$: if we are able to find $w = (v, Q) \in Q$ with $w_1(0) = w_1(L) = 0$ and $Q \ge 0$ in $(0, L)$ such that $\langle \partial^2\mathcal{J}_t^{+}(\bar{q}(t))w, w\rangle < 0$, we have done. Clearly the critical condition would be the existence of a nontrivial $w = (y, Q)$ with $Q \ge 0$ such that $\partial^2\mathcal{J}_t^{+}(\bar{q}(t))w = 0$. This last equation is the linerization of the perturbed equation $\partial\mathcal{J}_t^{+}(\bar{q}(t) + w) = 0$ for w small. Since our efforts to find a nontrivial solution w satisfying the constraint $Q \ge 0$ were unsuccessful (see Remark **??** below), in what follows we presents a simplified analysis in which that constraint is disregarded. This leads us to the following second simplified notion of stability.

DEFINITION 5.23 (regularized stability). We say that the trivial uniform solution satisfies the regularized stability condition at $t \in (0, T]$ if $\partial^2\mathcal{J}_t^{+}(\bar{y}(t), \bar{P}(t))$ is positive definite.

REMARK 5.24. We have already established that instability modes where P decreases everywhere in $(0, L)$ are excluded. Furthermore, if the trivial uniform solution satisfies the regularized stability condition, instability modes where P increases everywhere in $(0, L)$ are likewise excluded. Consequently, if we assume a priori that instability modes cannot exhibit a plastic increment that is negative in some regions and positive in others (as is expected for a bar under tension, where P should not decrease at any point) then the regularized stability condition would imply true local stability.However, if the trivial uniform solution becomes unstable in the regularized sense, true local stability is not necessarily compromised. Indeed, even if the second variation $\partial^2\mathcal{J}_t^{+}(\overline{y}(t), \overline{P}(t))$ is not positive definite, no actual instability occurs as long as there is no direction $w = (v, Q) \in Q$ with $Q \ge 0$ such that $\langle \partial^2\mathcal{J}_t^{+}(\overline{y}(t), \overline{P}(t))w, w\rangle < 0$. These considerations imply that the instant at which regularized stability is lost represents a lower bound for the onset of necking.

We now seek the instant at which the regularized stability condition is lost. This can be achieved by determining whether there exists a non-trivial direction $w \in \mathcal{Q}$ such that $\partial^2\mathcal{J}_t^+(\bar{q}(t))w = 0$. It is easy to prove that

$$\langle \partial^2\mathcal{J}_t^+(\bar{q}(t))w, \tilde{w}\rangle = \int_0^L \left(M(t)w' \cdot \tilde{w}' + N(t)w \cdot \tilde{w}' + N^T w' \cdot \tilde{w} + K(t)w \cdot \tilde{w}\right) dX$$

where

$$M = \begin{pmatrix} \bar{W}_{y_1'y_1'} & \bar{W}_{y_1'y_2'} & 0 \\ \bar{W}_{y_1'y_3} & \bar{W}_{y_2'y_2'} & 0 \\ 0 & 0 & \bar{H}_{P'P'} \end{pmatrix}, \quad N = \begin{pmatrix} 0 & \bar{W}_{y_1'y_2} & \bar{W}_{y_1'P} \\ 0 & \bar{W}_{y_2'y_2} & \bar{W}_{y_2'P} \\ 0 & 0 & \bar{H}_{PP'} \end{pmatrix} \quad \text{and} \quad K = \begin{pmatrix} \bar{W}_{y_1y_1} & \bar{W}_{y_1y_2} & \bar{W}_{y_1y_3} \\ \bar{W}_{y_2y_1} & \bar{W}_{y_2y_2} & \bar{W}_{y_2y_3} \\ \bar{W}_{y_3y_1} & \bar{W}_{y_3y_2} & \bar{W}_{y_3y_3} + \bar{H}_{PP} + \bar{D}_{22} \end{pmatrix}.$$

Here the bar denotes quantities computed along the trivial solution. It is understood that W is computed in $E_1 = \eta y_1' P^{-1}$, $E_2 = y_2 P$ and $G = y_2' P^{-1}$ so it can be considered as a function of $\eta, y_1', y_2, y_2', y_3$. Moreover $\overline{D}_{22}$ denotes the partial derivatives of D with respect to its second argument. Now, the strong form of $\partial^2\mathcal{J}_t^+(\bar{q}(t))w = 0$ is

$$Aw'' + Bw' + Cw = 0 \quad \text{in } (0, L), \tag{5.26}$$

together with the boundary conditions $w_1(X) = 0$, $w_2'(X) = 0$ and $w_3'(X) = 0$ for $X \in \{0, L\}$. We have used the notation $v = (w_1, w_2)$ and $Q = w_3$ and we have defined $A = M$, $B = (N - N^T)/2$ and $C = -K$. After some tedious computations one finds

$$A = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} = \begin{pmatrix} \bar{W}_{y_1'y_1'} & \bar{W}_{y_1'y_2'} & 0 \\ \bar{W}_{y_1'y_2'} & \bar{W}_{y_2'y_2'} & 0 \\ 0 & 0 & \bar{H}_{22} \end{pmatrix} = \begin{pmatrix} \eta^2 \frac{\bar{W}_{11}}{\bar{P}^2} & 0 & 0 \\ 0 & \frac{\bar{W}_{33}}{\bar{P}^2} & 0 \\ 0 & 0 & \bar{H}_{P'P'} \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & b_1 & b_2 \\ -b_1 & 0 & 0 \\ -b_2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \bar{W}_{y_1'y_2} & \bar{W}_{y_1'P} \\ -\bar{W}_{y_1'y_2} & 0 & \bar{W}_{y_2'P} \\ -\bar{W}_{Py_1'} & -\bar{W}_{Py_2'} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \eta\bar{W}_{12} & \eta\frac{\bar{y}_2}{\bar{P}^2}\bar{W}_{12} - \eta^2\frac{\bar{W}_{11}}{\bar{P}^3} - \eta\frac{\bar{W}_1}{\bar{P}^2} \\ -\eta\bar{W}_{12} & 0 & 0 \\ -\left(\eta\frac{\bar{y}_2}{\bar{P}^2}\bar{W}_{12} - \eta^2\frac{\bar{W}_{11}}{\bar{P}^3} - \eta\frac{\bar{W}_1}{\bar{P}^2}\right) & 0 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & c_1 & c_2 \\ 0 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\bar{W}_{y_2y_2} & -\bar{W}_{y_2\bar{P}} \\ 0 & -\bar{W}_{Py_2} & -\bar{W}_{PP} - \bar{H}_{PP} - \bar{D}_{22} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\bar{P}^2\bar{W}_{22} & -\bar{W}_2 + \frac{\eta}{\bar{P}}\bar{W}_{12} - \bar{y}_2\bar{P}\bar{W}_{22} \\ 0 & -\bar{W}_2 + \frac{\eta}{\bar{P}}\bar{W}_{12} - \bar{y}_2\bar{P}\bar{W}_{22} & -\eta^2\frac{\bar{W}_{11}}{\bar{P}^4} - \bar{y}_2^2\bar{W}_{22} - 2\eta\frac{\bar{W}_1}{\bar{P}^3} + 2\eta\frac{\bar{y}_2}{\bar{P}^2}\bar{W}_{12} - \bar{H}_{PP} - \bar{D}_{22} \end{pmatrix}$$

where the bar denotes quantities computed along the trivial solution. The system of second order partial differential equations (5.26) can be written more explicitly as

$$\begin{aligned} a_1 w_1'' + b_1 w_2' + b_2 w_3' &= 0, \\ a_2 w_2'' - b_1 w_1' + c_1 w_2 + c_2 w_3 &= 0, \\ a_3 w_3'' - b_2 w_1' + c_2 w_2 + c_3 w_3 &= 0 \end{aligned}$$

in $(0, L)$. Taking into account the boundary conditions we write w_1, w_2 and w_3 as the following formal series

$$w_1 = \sum_{k=1}^{\infty} w_1^k \sin(\gamma_k x), \quad w_2 = w_2^0 + \sum_{k=1}^{\infty} w_2^k \cos(\gamma_k x) \quad \text{and} \quad w_3 = w_3^0 + \sum_{k=1}^{\infty} w_3^k \cos(\gamma_k x),$$

where $\gamma_k = \frac{\pi k}{L}$. Substituting into (5.28) we get

$$\sum_{k=1}^{\infty}\left(-a_1\gamma_k^2 w_1^k - b_1\gamma_k w_2^k - b_2\gamma_k w_3^k\right)\sin(\gamma_k x) = 0,$$

$$c_1 w_2^0 + c_2 w_3^0 + \sum_{k=1}^{\infty}\left((c_1 - a_2\gamma_k^2)w_2^k - b_1\gamma_k w_1^k + c_2 w_3^k\right)\cos(\gamma_k x) = 0,$$

$$c_2 w_2^0 + c_3 w_3^0 + \sum_{k=1}^{\infty}\left(c_2 w_2^k - b_2\gamma_k w_1^k + (c_3 - a_3\gamma_k^2)w_3^k\right)\cos(\gamma_k x) = 0$$

in $(0, L)$, where. Due to the orthogonality relations

$$\int_0^L \sin(\gamma_k x)\sin(\gamma_j x)dx = \begin{cases} \frac{L}{2} & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \quad \text{and} \quad \int_0^L \cos(\gamma_k x)\cos(\gamma_j x)dx = \begin{cases} L & \text{if } k = j = 0 \\ \frac{L}{2} & \text{if } k = j \neq 0 \\ 0 & \text{if } k \neq j, \end{cases}$$

we find that (5.29) is equivalent to $M_k w^k = 0$ for all $k \in \mathbb{N}_0$ where $w^0 = (w_2^0, w_3^0) \in \mathbb{R}^2$, $w^k = (w_1^k, w_2^k, w_3^k) \in \mathbb{R}^3$ for $k \in \mathbb{N}$ and

$$M_0 = \begin{pmatrix} c_1 & c_2 \\ c_2 & c_3 \end{pmatrix} \quad \text{and} \quad M_k = \begin{pmatrix} -a_1\gamma_k^2 & -b_1 & -b_2 \\ -b_1 & c_1 - a_2\gamma_k^2 & c_2 \\ -b_2 & c_2 & c_3 - a_3\gamma_k^2 \end{pmatrix} \quad \text{for } k \in \mathbb{N}.$$

Thus the existence of a nontrivial solution is equivalent to the existence of some $k \in \mathbb{N}_0$ such that $\det M_k = 0$. In Figure 5.4, on the left, the numerically computed values of $\det M_k$ as functions of η are shown. We notice that $\det M_0$ never vanish and the first determinant to vanish is $\det M_1$ slightly after the tensile force reaches its maximum. In the same figure, on the right, is shown the value of the elongation

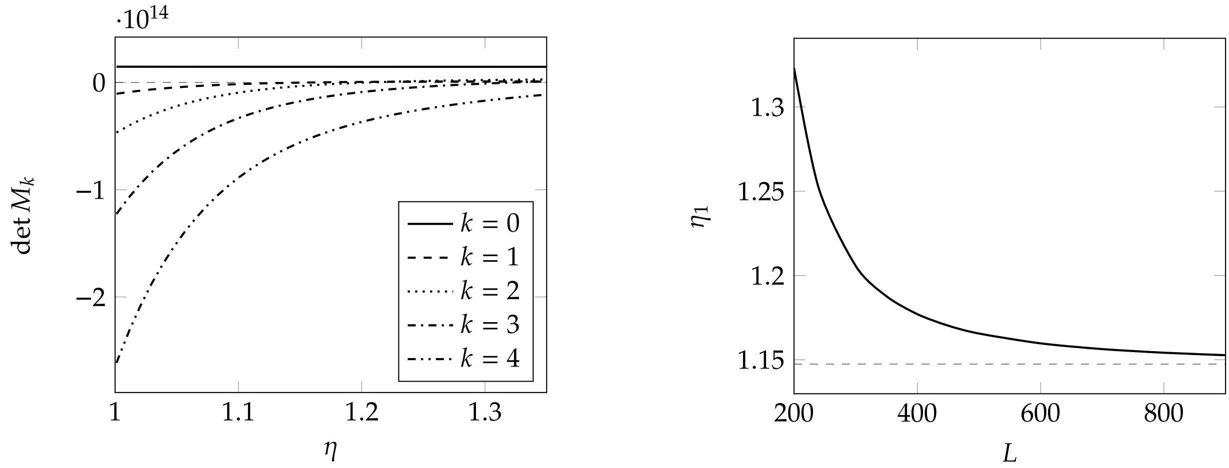


FIGURE 5.4. On the left, value of $\det M_k$ for $k \in \{0, 1, 2, 3, 4\}$. Notice the $\det M_0$ never vanishes while $\det M_1$ is the first determinant to vanish at elongation $\eta_1 = 1.1597$, slightly greater than the elongation $\eta_{\mathrm{m}} = 1.1474$ corresponding to the maximum of $\bar{F}$. On the right is η_1 as L changes. Notice that η_1 tends to η_{m} as $L \to \infty$.

η_1 when $\det M_1$ vanish as a function of the length of L of the bar. In the limit $L \to \infty$ it results that η_1 converges to η_{m}, the elongation corresponding to the maximum tensile force. This is indeed justified by the following lemma.

LEMMA 5.25. Assume that as $L \to \infty$ the value η_k corresponding to vanishing $\det M_k$ has a finite limit. Then that limit coincides to η_{m} for all $k \in \mathbb{N}$.

PROOF. Notice that $\gamma_k \to 0^+$ when $L \to \infty$. After some computations it results that for each η

$$\det M_k = \gamma_k^2 (a_1(c_1c_3 - c_2^2) + b_1^2 c_3 + b_2^2 c_1 - 2b_1b_2c_2) + o(\gamma_k^2)$$

where $o(\gamma_k^2)$ is uniform for η constrained in a compact set. Now let $\tilde{\eta}_k \colon (0,\infty) \to [1,\infty)$ be the curve that assigns to each L the corresponding value of η_k at which $\det M_k$ vanish. Computing $\det M_k$ for $\eta = \tilde{\eta}_k(L)$, dividing it for γ_k^2, and letting $L \to \infty$, we get that

$$a_1(c_1c_3 - c_2^2) + b_1^2 c_3 + b_2^2 c_1 - 2b_1b_2c_2 = 0 \quad \text{for } \eta = \lim_{L\to\infty} \hat{\eta}_k(L).$$

This last condition is equivalent to $\det M = 0$, where

$$M = \begin{pmatrix} a_1 & -b_1 & -b_2 \\ -b_1 & c_1 & c_2 \\ -b_2 & c_2 & c_3 \end{pmatrix}.$$

On the other and the equilibrium of the trivial solution and the definition of $\bar{F}$ give

$$\begin{aligned} \bar{F} &= \bar{W}_{y_1'} \\ -\bar{W}_{y_2} + \frac{d}{dx}\bar{W}_{y_2'} &= 0 \\ -\bar{W}_{y_3} + \frac{d}{dx}\bar{W}_{y_3'} &= 0. \end{aligned}$$

Differentiating these equations, solving for $\left(\frac{d\bar{y}_2}{d\eta}; \frac{d\bar{P}}{d\eta}\right)$ from the second and the third and substituting in the first, we get

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{y_1'y_2}\frac{d\bar{y}_2}{d\eta} + \bar{W}_{y_1'\bar{P}}\frac{d\bar{P}}{d\eta} + \bar{W}_{y_1'\eta} = \begin{pmatrix}\bar{W}_{y_1'y_2}\\ \bar{W}_{y_1'\bar{P}}\end{pmatrix}\cdot\begin{pmatrix}\frac{d\bar{y}_2}{d\eta}\\ \frac{d\bar{P}}{d\eta}\end{pmatrix} + \bar{W}_{y_1'\eta} = $$
$$= -\begin{pmatrix}\bar{W}_{y_1'y_2}\\ \bar{W}_{y_1'\bar{P}}\end{pmatrix}\cdot\begin{pmatrix}\bar{W}_{y_2y_2} & \bar{W}_{y_2\bar{P}}\\ \bar{W}_{\bar{P}y_2} & \bar{W}_{\bar{P}\bar{P}} + \bar{H}_{11} + \bar{D}_{22}\end{pmatrix}^{-1}\begin{pmatrix}\bar{W}_{y_2\eta}\\ \bar{W}_{\bar{P}\eta}\end{pmatrix} + \bar{W}_{y_1'\eta}.$$

Now reasoning as in the proof of Lemma 5.21 and using (5.27), it follows that $\frac{d\bar{F}}{d\eta}$ vanish if and only if $\det M$ vanish. This concludes the proof. □

We have thus proved that when $\eta \geq \eta_1$ the trivial solution does not satisfy the regularized stability condition. When $\det M_1$ vanishes, the kernel of $\partial^2\mathfrak{J}_t^+(\overline{q}(t))$ becomes nontrivial and is spanned by w

$$w_1(x) = w_1^1\sin(\gamma_1 x),\quad w_2(x) = w_2^1\cos(\gamma_1 x)\quad\text{and}\quad w_3(x) = w_3^1\cos(\gamma_1 x)$$

with $w^1 = (w_1^1, w_2^1, w_3^1)$ satisfying $M_1 w^1 = 0$. If we compute numerically w^1 we find that $w^1 > 0$, $w^2 < 0$ and $w^3 > 0$, so interpreted as a variation of the configuration of the bar it appears qualitatively as the instability mode we found in the purely elastic case and reported in Figure 5.1. As previously observed in Remark 5.24, although w belongs to the kernel of $\partial^2\mathfrak{J}_t^+(\bar{q}(t))$, it does not represent a true instability mode since it fails to satisfy the condition $w_3 \geq 0$. Consequently, we have not proven that necking necessarily occurs at this stage; rather, we have established that if necking were to occur, it must do so for $\eta > \eta_1$.

Remark 5.26. We have proved that the kernel of $\partial\mathfrak{J}_t^+(\bar{q}(t))$ never contains $w \in Q$ with $Q \geq 0$. However to prove that $\bar{q}(t)$ is not a minimizer of $\mathfrak{J}_t^+(\overline{q}(t))$ for some $\eta > \eta_1$ it would be sufficient to show that such w exists with the property that $\langle\mathfrak{J}_t^+(\bar{q}(t))w, w\rangle < 0$. Here we describe our attempts in that direction. Consider a finite dimensional subspace of Q spanned by $\{w^1, \dots, w^m\}$ that we collect in the matrix field $O\colon (0, L) \to \mathbb{R}^{3\times m}$ given by $O = (w_1, \dots, w_m)$. Then generic element of that subspace is $w = Oc$ where $c \in \mathbb{R}^m$. Thus

$$\langle\mathfrak{J}_t^+(\overline{q}(t))w, w\rangle = c^T G(t)c,$$

where the symmetric matrix $G \in \mathbb{R}^{d\times d}$ is given by

$$G = \int_0^L \left((O')^T M O' + 2(O')^T N O + O^T K O\right) dX.$$

Thus out aim is to find O and $\eta > \eta_1$ such that there exists $c \in \mathbb{R}^m$ with $c^T G c < 0$ and satisfying the constrain $(Oc)_3 \geq 0$ in $(0, L)$. We first consider the case

$$O(x) = \begin{pmatrix}\sin(\gamma_1 x) & 0 & 0\\ 0 & 1 & 0\\ 0 & \cos(\gamma_1 x) & 0\\ 0 & 0 & 1\\ 0 & 0 & \cos(\gamma_1 x)\end{pmatrix}.$$

The condition $(Oc)_3 \geq 0$ in this case reads as $|c_5| < c_4$. We find that

Part 2

A model describing pressure sensitive plastic materials

CHAPTER 6

Introduction

Until the 1950s, the application of plasticity theory remained confined to metals. In 1952, Drucker and Prager [21] used the same framework of metal plasticity to describe the mechanical behavior of soils, interpreting the classic Mohr-Coulomb [17, 50] failure criterion as a characterization of the yield limit. They also proposed a new yield criterion, now known as the Drucker-Prager criterion.

The characteristic of the Mohr-Coulomb and the Drucker-Prager yield criteria, absent in the classical plasticity of metals, is that the yield strength increases with increasing pressure. This behavior is appropriate to describe the plasticity of materials such as soils due to their granular nature. In fact in these cases the plastic behavior is due to the sliding of the grains of which the material is composed. The sliding is opposed by the internal friction between the grains and increases as the grains are pressed against each other. As we saw in Section 1.2 of Chapter 1, in the classical formulation of plasticity theory, once the admissibility set for the stress is fixed, the flow rule of the plastic strain is automatically determined by the normality condition of the plastic flow law (see Remark 1.14 and the discussion reported below). In this case the flow rule is said to be *associated*. In the case in which the admissibility criterion also takes pressure into account, the normality condition dictates that the plastic stain cannot remain deviatoric, so the hypothesis of plastic incompressibility must be abandoned. This actually corresponds to a real mechanical phenomenon called *dilatancy* and has a precise micromechanical interpretation: when the plastic flow occurs the grains slide over each other and in doing so they pass from a more optimal packing configuration to a less optimal one, causing an increase in volume. However, the associated flow rule has the drawback of overestimating the dilatancy. To overcome this, non-associated flow rules have been developed, i.e., the plastic flow law is chosen independently of the set of admissible stresses.

The use of the non-associated flow law involves additional mathematical complications because it breaks the standard variational structure of classical plasticity theory. However recently [26, 65] it has been demonstrated that non-associated plasticity actually admits a variational formulation and existence results have been obtained.

This second part of the thesis is dedicated to the development of an alternative approach to describe plastic materials exhibiting pressure sensitivity and dilatancy. The aim is to build a mechanical theory, coherent with the general view of a mechanical systems described in Section 1.1 of Chapter 1, where the idea the internal plastic friction depends on pressure is made clearer than it appears in the original formulation of Drucker–Prager and its non-associated modification. In fact, we believe that the procedure proposed by Drucker and Prager of using the same conceptual apparatus of the plasticity of metals, modifying only the yield criterion to include the influence of pressure and dilatancy, is a mostly formal approach and not directly corresponding to the mechanical intuition that the internal friction of the material depends on pressure. Moreover the fact of having to artificially alter the plastic flow rule making it non-associated leads us to find a more elegant and unitary description. In addition to the modeling benefits just mentioned, the formulation we propose also brings benefits from the point of view of the development of the existence theory. In fact, being by construction based on the balance of generalized forces (the macroscopic ones together with the plastic microforces), it admits a clearer variational formulation.

The discussion is organized as follows. This introductory chapter reviews the standard extension of the plasticity theory presented in Section 1.2 to account for plastic compressibility and non-associated flow rules. Furthermore, we describe the Drucker–Prager yield criterion and provide an overview of the known existence results. For the sake of simplicity, this review is confined to the case of small deformations. The following chapters are dedicated to the formulation of our alternative approach.

6.1. Extension of plasticity theory incorporating plastic compressibility and non-associated flow rules

We now describe how the theory presented in Section 1.2 is extended to include plastic compressibility and non-associated flow rules. The formulation presented here is the standard way used to perform that extension. In the following chapters we will propose an alternative approach.

Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz, be the reference configuration of a body. Its configurations are assumed to be the couples $z = (u, E_p) \in Z$ where $u \colon \Omega \to \mathbb{R}^d$ is the displacement and $E_p \colon \Omega \to \mathrm{Sym}$ is the plastic strain. Notice that plastic incompressibility is not assume since E_p is not required to be deviatoric. The elastic strain is $E_e = Eu - E_p$. Analogously of what we did in Section 1.2 of Chapter 1, we require that the internal and external virtual power functionals are of the form

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_\Omega \left(T_e : \tilde{V}_e dx + T_p : \tilde{V}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v}$$

for all $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$. For simplicity we have assumed that $K_p = 0$, i.e., we are not considering plastic strain gradient effects since they are not relevant in the present discussion. Then the strong form of the equilibrium condition $\mathcal{I} = \mathcal{X}$ is

$$\begin{aligned} \operatorname{div} T_e + f &= 0 \quad \text{in } \Omega \\ T_e n &= h \quad \text{in } \partial\Omega \\ T_e - T_p &= 0 \quad \text{in } \Omega \end{aligned}$$

where the first two equations describe the balance of macroforces while the last equation represents the balance of plastic microforces. The main difference with respect to equations (1.8) is that in the microforce balance the whole elastic stress tensor T_e, and not only is deviator, appears.

We now discuss the choice of constitutive relations. We assume that the free energy $\Psi \colon Z \to \mathbb{R}$ is of the form

$$\Psi(z) = \int_\Omega \left(W_e(E_e) + W_p(E_p) \right) dx,$$

where W_e and W_p form Sym to $\mathbb{R}$ are the elastic and plastic free energy which are supposed to be differentiable. This gives the constitutive relation for the elastic stress $T_e = W_e'(E_e)$. If we split the plastic force T_p as the sum of a conservative back-stress B_p and a dissipative plastic force A_p, then $B_p = W_p'(E_p)$. Notice that, having written $T_p = A_p + B_p$, the equilibrium of microforces can be written as

$$A_p = T_e - B_p = -\frac{\partial W}{\partial E_p}(Eu, E_p) \quad \text{in } \Omega, \tag{6.1}$$

where $W(Eu, E_p) = W_e(Eu - E_p) + W_p(E_p)$. To characterize the plastic response

(i) we require the admissibility condition $A_p \in C$ where $C = \{f \leq 0\} \subset \mathrm{Sym}$, with $f \colon \mathrm{Sym} \to \mathbb{R}$ a convex *yield function* satisfying $f(0) < 0$, and

(ii) we impose the flow rule $\dot{E}_p = 0$ if $A_p \in \operatorname{int} C$ and $\dot{E}_p$ belongs to the normal outer cone to $L(A_p) = \{A \in \mathrm{Sym} \mid g(A) \leq g(A_p)\}$ at A_p if $A_p \in \partial C$, where $g \colon \mathrm{Sym} \to \mathbb{R}$ is a convex *plastic potential*.

When the yield function f coincides with the plastic potential g, then the flow rule reduces to $\dot{E}_p \in N_C(A_p)$ which is the normality rule. When $f = g$ we say that the flow rule is associated, while when $f \neq g$ we say that the flow rule is non-associated. This is the way in which the plastic response is characterized in the classical presentations of non-associated plasticity models. However this formulation does not fit directly into the framework of Section 1.1 of Chapter 1 and is not suitable for proving existence theorems. A better reformulation is described in [40, 26, 65] and is reproduced in the following lemma.

Lemma 6.1. The admissibility condition and the flow rule are equivalent to

$$A_p \in \hat{C}(A_p) \quad \text{and} \quad \dot{E}_p \in N_{\hat{C}(A_p)}(A_p), \tag{6.2}$$

where

$$\hat{C}(A_p) = \left\{ A \in \mathrm{Sym} \mid g(A) \leq g(A_p) - f(A_p) \right\}.$$

In turn, the inclusions in (6.2) are also equivalent to

$$A_p \in \partial \hat{R}(A_p, \dot{E}_p), \tag{6.3}$$

where $\hat{R} \colon \mathrm{Sym} \times \mathrm{Sym} \to [0, \infty]$ is defined as

$$\hat{R}(A_p, \dot{E}_p) = \sup_{A \in \mathrm{Sym}} A : \dot{E}_p = \sigma_{\hat{C}(A_p)}(\dot{E}_p). \tag{6.4}$$

Proof. It is trivial to see that $A_p \in \hat{C}(A_p)$ if and only if $f(A_p) \leq 0$, or equivalently $A_p \in C$. Moreover $A_p \in \operatorname{int} \hat{C}(A_p)$ if and only if $A_p \in \operatorname{int} C$ and $A_p \in \partial \hat{C}(A_p)$ if and only if $A_p \in \partial C$. This implies also that if $A_p \in \operatorname{int} C$, then $N_{\hat{C}(A_p)}(A_p) = \{0\}$. Instead, if $A_p \in \partial \hat{C}(A_p)$ then $f(A_p) = 0$ so that $\hat{C}(A_p) = L(A_p)$ and $N_{\hat{C}(A_p)}(A_p) = N_{L(A_p)}$. This proves the first equivalence.

The second equivalence follows from a well known results of complex analysis [23, Corollary 5.2]. □

Now in order to write the inclusion (6.3) as a true constitutive relation, i.e., expressing the dependence of A_p on kinematical quantities alone, one can use (6.1). Thus we define $R\colon \mathrm{Sym}\times\mathrm{Sym}\times\mathrm{Sym} \to [0,\infty]$ by

$$R(Eu, E_p, \dot{E}_p) = \hat{R}\left(-\frac{\partial W}{\partial E_p}(Eu, E_p), \dot{E}_p\right) \tag{6.5}$$

and we require that $A_p \in \partial R(Eu, E_p, \dot{E}_p)$. This correspond to the choice of the dissipation potential $\Psi\colon Z\times Z \to [0,\infty]$ defined as

$$\Psi(z,\dot{z}) = \int_\Omega R(Eu, E_p, \dot{E}_p)dx.$$

Remark 6.2. If the dependence on plastic strain gradients is included, the balance of microforces (6.1) becomes

$$A_p = T_e - B_p + \operatorname{div} K_p.$$

The quantity $\operatorname{div} K_p$ is not defined as a function in general. This prevents the possibility of defining R from $\hat{R}$ by a simple substitution. In this case, see [65], the definition of R kept as in (6.5).

6.2. The Mohr–Coulomb and Drucker–Prager yield criterion

In this section we show how the Mohr–Coulomb and Drucker–Prager yield criterion can be formulated within the notation of previous section.

6.2.1. The Mohr–Coulomb criterion. The Mohr–Coulomb criterion was the first criterion failure of cohesive materials such as rocks and soil. The original formulation states that failure happen at a material point where the Cauchy stress tensor $T \in \mathrm{Sym}$ there exists a direction $n \in \partial B(0,1)$ such that

$$\tau = -\sigma\tan\phi + c, \tag{6.6}$$

where $\sigma = Tn\cdot n$ and $\tau = |Tn - \sigma n|$ are the normal and shear stress respectively acting on the plane orthogonal to n. Here $\phi \in [0,\pi/2)$ is the angle of internal friction and $c \geq 0$ is the cohesion. The idea behind (6.6) is that the resistance of the material to shear stresses increases with compression. One can show (see [42]) that failure occur on the two hyperplanes whose normal form an angle of $\pi/2 + \phi/2$ with respect to the eigenspace corresponding to the greatest eigenvalue; this is the reason why ϕ is called the angle of internal friction. After some computations one finds that the failure criterion can be written directly in term of T as

$$f(T) = F(\operatorname{dev} T) + \frac{2}{d}(\operatorname{tr} T)\sin\phi - 2c\cos\phi = 0. \tag{6.7}$$

with $f\colon \mathrm{Sym} \to \mathbb{R}$ and $F\colon \mathrm{DevSym} \to \mathbb{R}$ given by

$$F(S) = S_{\max} - S_{\min} + (S_{\max} + S_{\min})\sin\phi, \tag{6.8}$$

where $S_{\min}$ and $S_{\max}$ are the maximum and minimum eigenvalues of S.

Now the Mohr–Coulomb criterion is included in the theory described in Section 6.1 by using f in (6.7) as the yield function.

6.2.2. The Drucker–Prager yield criterion. When $d = 2$ it is easy to check that F in (6.8) can be also written as $F(S) = \sqrt{2}|S|$. In this case the yield function becomes

$$f(T) = \sqrt{2}|\operatorname{dev} T| + (\operatorname{tr} T)\sin\phi - 2c\cos\phi.$$

In the general case, this formula suggest to consider the yield function $f\colon \mathrm{Sym} \to \mathbb{R}$ of the form

$$f(A) = |\operatorname{dev} A| - \left(Y_0 - \alpha\frac{\operatorname{tr} A}{d}\right)$$

where $Y_0 > 0$ and $\alpha > 0$. The corresponding criterion is known as Drucker–Prager yield criterion and can be seen as an approximation of the Mohr–Coulomb criterion. Since the hydrostatic pressure is $p = -\operatorname{tr} T/d$, the Drucker–Prager criterion can be formulated as $|\operatorname{dev} T| \leq Y(p) = Y_0 + \alpha p$. Thus it is a generalization of the Von Mises criterion where the yield strength is made dependent on the hydrostratic pressure.

6.3. An existence result for non-associative plasticity

An existence theorem for non-associative compressible plasticity at small deformations has been derived in [26] and is reported below. It is obtained assuming no hardening and adding a viscous plastic microforce.

Theorem 6.3. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz and $\Gamma_D \partial\Omega$ non-empty and relatively open. Let $\mathfrak{C} \in \mathscr{L}(\mathrm{Sym}, \mathrm{Sym})$ the elasticity tensor and suppose that there exists $c > 0$ such that $\mathfrak{C}E : E \geq c|E|^2$ for all $E \in \mathrm{Sym}$. Let $\eta > 0$ a plastic viscosity. Let $K \subset \mathrm{DevSym}$ be bounded, closed and convex and such that $0 \in \operatorname{int} K$, and define the admissibility set

$$C = \left\{ A \in \mathrm{Sym} \mid A \in \alpha(\Pi - \operatorname{tr} A)K \text{ and } \operatorname{tr} A \geq -M \right\}$$

for some $\alpha > 0$, $\Pi > 0$ and $M > 0$. Let also

$$G = \left\{ A \in \mathrm{Sym} \mid A \in \beta(\Theta - \operatorname{tr} A)K \text{ and } \operatorname{tr} A \geq -M \right\},$$

where $\beta > 0$ and $\Theta > 0$. Define f and g as the signed distance from C and G respectively, and let $\hat{R}$ defined as in (6.4). Let $\bar{u} \in H^1((0,T), H^1(\Omega, \mathbb{R}^d))$ and $(u_0, E_p^0) \in H^1(\Omega, \mathbb{R}^d) \times L^2(\Omega, \mathrm{Sym})$ such that $u_0 = \bar{u}(0)$ in Γ_D in the sense of the trace. Then there exists a unique couple (u, E_p) with

$$u \in H^1((0,T), H^1(\Omega, \mathbb{R}^d)) \quad \text{and} \quad E_p \in W^{1,\infty}((0,T), L^2(\Omega, \mathrm{Sym}))$$

such that, setting $T = \mathfrak{C}(Eu - E_p)$, the following conditions hold:

(i) the initial condition $u(0) = u_0$,

(ii) the macroscopic equilibrium

$$\operatorname{div} T(t) = 0 \quad \text{in the sense of distributions for all } t \in [0,T],$$

(iii) the flow rule

$$\eta \dot{E}_p(t) - T(t) + \partial_{\dot{E}_p} \hat{R}(T_p(t), \dot{E}_p(t)) \ni 0 \quad \text{for all } t \in [0,T].$$

Remark 6.4. Using a time rescaling approach it is possible to prove also the existence of a limit solution (u, E_p) in the limit $\eta \to 0^+$ of vanishing viscosity, with $u \in C^0([0,T], \mathrm{BD}(\Omega))$ and $E_p \in W^{1,\infty}((0,T), \mathscr{M}(\bar{\Omega}, \mathrm{Sym}))$, where $\mathrm{BD}(\Omega)$ is the space of functions of bounded variations and $\mathscr{M}(\bar{\Omega}, \mathrm{Sym})$ the space of bounded Radon measures.

Notice that the original Drucker–Prager model is not covered by the hypotheses of Theorem 6.3 since it does not involve the condition $\operatorname{tr} A \geq -M$ in the definition of the sets C and G. However the addition of this constrain is common in the modeling of soil and is known as the cap model. It is added to better the describe the volumetric behavior of the material under compressive loads.

CHAPTER 7

Small deformation model

7.1. Formulation

7.1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u\colon \Omega \to \mathbb{R}^d$. We set $E = \operatorname{sym}\operatorname{grad} u$ and $M = \operatorname{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $u = \bar{u}$. The configuration of the microscopic structure is described by the plastic distortion $H_p\colon \Omega \to \mathrm{Lin}$. The skew symmetric part $W_p = \operatorname{skew} H_p$ describes the rotation of the microscopic structure and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \operatorname{sym} H_p\colon \Omega \to \mathrm{Sym}$ which describes the deformation of microscopic structure. We can regard E_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$. The volumetric deformation of the microscopic structure is described by $\lambda_p = \operatorname{tr} E_p$ while the deviatoric deformation by $U_p = \operatorname{dev} E_p$. We call $M_p = \operatorname{grad} U_p$ and we set $\mathbb{U}_p = (U_p, l\, M_p) \in \mathrm{DevPla}$ where $l > 0$ is an internal length scale. Here $\mathrm{DevPla} = \mathrm{Sym} \otimes (\mathrm{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic strains and their gradients. In what follows we will denote with $\circ$ the natural scalar product on DevPla. We assume that the material is not dilatant (this is a simplification and should be generalized to describe the dilatant behavior observed in most materials exhibiting a pressure dependent yield strength) so we add the constrain $\lambda_p = 0$. To impose this constrain we add the Lagrange multiplier $p\colon \Omega \to \mathbb{R}$ with the meaning of a plastic pressure. The constrain can be written in weak form as

$$\int_\Omega \lambda_p \tilde{q}\, dx = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

7.1.2. Statics. We denote with $\tilde{v}$, $\tilde{V}_p$ and $\tilde{\mu}_p$ the virtual rates of u, U_p and λ respectively. The virtual rate of E_e is $\tilde{L}_e = \operatorname{grad}\tilde{v} - \tilde{\rho} - \frac{\tilde{\mu}}{d} I$. Due to the presence of the Dirichlet condition we require that $\tilde{v}$ vanish on Γ_D. Instead we do not require that $\tilde{\mu}_p = 0$ because the constrain $\lambda_p = 0$ is imposed by mean of the Lagrange multiplier p. To describe the coupling between the macroscopic deformation and the microscopic configuration we choose an internal generalized force of the form

$$\tilde{\mathfrak{I}} = \int_\Omega \left(T : \tilde{L} + K \,\vdots\, \operatorname{grad}^2 \tilde{v} \right) dx + \int_\Omega T_e : \tilde{L}_e dx + \int_\Omega \left(T_p : \tilde{\rho} + K_p \,\vdots\, \operatorname{grad}\tilde{\rho} \right) dx.$$

The first integral is related to the macroscopic response. Here $T \in \mathrm{Sym}$ is the macrostress and $K \in \mathrm{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order of $\tilde{v}$ is due to the need of a sufficient regularization in view of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e \in \mathrm{Sym}$ is the elastic stress. The third integral is associated with the microscopic response. Here $T_p \in \mathrm{DevSym}$ is the microforce and $K_p \in \mathrm{DevSym} \otimes \mathbb{R}$ is the microstress. In Figure 7.1 we report a picture that represents schematically the meaning of the internal generalized force that we have chosen. We define $\tilde{\mathbb{V}}_p = (\tilde{V}_p, l \operatorname{grad} \tilde{V}_p)$ and $\mathbb{T}_p = (T_p, l^{-1} K_p)$ so that

$$T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad}\tilde{V}_p = \mathbb{T}_p \circ \tilde{\mathbb{V}}_p.$$

The action of the plastic pressure is obtained by considering the reaction virtual power

$$\tilde{\mathfrak{R}} = -\int_\Omega p \tilde{\mu}_p dx.$$

We notice that this power vanish on virtual rates that satisfy the constrain that is $\tilde{\mu}_p = 0$. The external actions of the volume force f on Ω and the traction h on Γ_N are included choosing the external generalized forces

$$\tilde{\mathfrak{X}} = \int_\Omega f \cdot \tilde{v} dx + \int_{\Gamma_N} h \cdot \tilde{v} dx.$$

For simplicity we assume that no external torque is applied to the body and that there is no external action on the microscopic configuration. These external actions could be added to describe more accurately the boundary conditions imposed on the body.

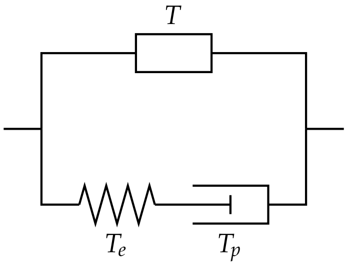


FIGURE 7.1. Rheological model describing the choice of the internal virtual power $\tilde{\mathcal{W}}$. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the microscopic response. We notice that the microscopic response besides the usual dissipative plastic microforce force can include conservative microforces to model hardening. A more detailed discussion on constitutive relations is in Section 7.1.3.

LEMMA 7.1. The local form of the equilibrium condition is

$$\begin{aligned}
&\operatorname{div}(T + T_e - \operatorname{div} K) + f = 0 \quad \text{in } \Omega \\
&(T + T_e - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) = h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega \\
&\{\!\{(Kn)\tau\}\!\} = 0 \quad \text{in } \partial^{d-2}\Omega \\
&-\operatorname{dev} T_e + T_p - \operatorname{div} K_p = 0 \quad \text{in } \Omega \\
&p = -\frac{1}{d} \operatorname{tr} T_e \quad \text{in } \Omega \\
&K_p n = 0 \quad \text{in } \partial^{d-1}\Omega.
\end{aligned}$$

PROOF. By integration by parts we can write

$$\int_\Omega \left(T : \operatorname{grad} \tilde{v} + K \vdots \operatorname{grad}^2 \tilde{v} \right) dx = \\
= -\int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} \left[(T - \operatorname{div} K)n \cdot \tilde{v} + Kn : \operatorname{grad} \tilde{v} \right] dx.$$

Now denoting with $P = I - n \otimes n$ the projector on $\partial^{d-1}\Omega$ we have

$$Kn : \operatorname{grad} \tilde{v} = Kn : \left(\frac{\partial \tilde{v}}{\partial n} \otimes n + \operatorname{grad}_P \tilde{v} \right) = (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} + Kn : \operatorname{grad}_P \tilde{v}$$

and

$$Kn : \operatorname{grad}_P \tilde{v} = (Kn)P : \operatorname{grad}_P \tilde{v} = \operatorname{div}_P(P(Kn)^T \tilde{v}) - \operatorname{div}_P((Kn)P) \cdot \tilde{v}.$$

But by integration by parts

$$\int_{\partial^{d-1}\Omega} \operatorname{div}_P(P(Kn)^T \tilde{v}) dx = \int_{\partial^{d-2}\Omega} \{\!\{P(Kn)^T \tilde{v} \cdot \tau\}\!\} dx = \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.$$

Here $\{\!\{\cdot\}\!\}$ denotes sum of its argument evaluated on both sides of $\partial^{d-2}\Omega$ and τ denotes the unit normal vector to $\partial^{d-2}\Omega$ and outer tangent to $\partial^{d-1}\Omega$. We have obtained

$$\int_\Omega \left(T : \operatorname{grad} \tilde{v} + K \vdots \operatorname{grad}^2 \tilde{v} \right) dx = \\
= -\int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n - \operatorname{div}_P((Kn)P)] \cdot \tilde{v} dx + \\
+ \int_{\partial^{d-1}\Omega} (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} dx + \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.$$

Moreover the remaining terms of the virtual power can be written as

$$-\int_\Omega \operatorname{div} T_e : \tilde{v} dx + \int_{\partial^{d-1}\Omega} T_e n \cdot \tilde{v} dx + \int_\Omega (T_p - \operatorname{dev} T_e - \operatorname{div} K_p) : \tilde{V}_p dx +$$
$$+ \int_{\partial^{d-1}\Omega} K_p n : \tilde{V}_p dx - \int_\Omega \frac{1}{d} \operatorname{tr} T_e \tilde{\mu} dx.$$

Now the results follows from the fundamental lemma of the calculus of variations. □

7.1.3. Constitutive relations. We assume a constitutive relation for the free energy density of the form

$$\psi = \frac{1}{2}\mathfrak{C}E_e : E_e + \frac{1}{2}\mathfrak{A}M \vdots M + \frac{1}{2}\mathfrak{B}\mathbb{U}_p \circ \mathbb{U}_p.$$

with $\mathfrak{C}$, $\mathfrak{A}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym, Lin⊗$\mathbb{R}$ and DevPla respectively. Moreover we choose a dissipation potential density of the form

$$\phi = \frac{1}{2}\mathfrak{D}\dot{E} : \dot{E} + \frac{1}{2}\mathfrak{E}\dot{M} \vdots \dot{M} + \frac{1}{2}\mathfrak{F}\dot{\mathbb{U}}_p \circ \dot{\mathbb{U}}_p + Y(p)G(\dot{U}_p).$$

Here $\mathfrak{D}$, $\mathfrak{E}$, $\mathfrak{F}$ are symmetric and positive definite linear operators on Sym, Lin⊗$\mathbb{R}$ and DevPla respectively. Moreover G: DevSym $\to [0,\infty]$ is the convex and subdifferentiable plastic dissipation potential satisfying $G(0) = 0$ and $Y: \mathbb{R} \to [0,\infty)$ is the pressure dependent yield strength. Consistently with these choices of the free energy density and of the dissipation potential density we impose the following constitutive relations. The macroscopic response is described by

$$T = \mathfrak{D}\dot{E} \quad \text{and} \quad K = \mathfrak{A}M + \mathfrak{E}\dot{M}.$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite linear operators on Sym and Lin ⊗ $\mathbb{R}$ respectively. The elastic response is given by

$$T_e = \mathfrak{C}E_e$$

and the viscoplastic response by

$$\mathbb{T}_p \in \mathfrak{B}\mathbb{U}_p + \mathfrak{F}\dot{\mathbb{U}}_p + Y(p)\partial_{\dot{U}_p} G(\dot{U}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla, G: DevSym $\to \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p \geq 0$ and $Y: \mathbb{R} \to [0,\infty)$ is the pressure dependent yield strength.

Remark 7.2. In the above expression we are viewing $Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) \in$ DevSym as an element of DevPla by considering DevSym as a subspace of DevPla.

7.1.4. Refinements of the model. To include the dilatancy in the model we could do the following. We put the unilateral constrain $\lambda_p \geq F(U_p)$, meaning that when the microscopic structure suffer shear distortions it must suffer also an expansion. This constrain is still imposed through the Lagrange multiplier p representing the microscopic pressure. To make the expansion irreversible (once the microscopic structure expanded it can not be compacted again) we add a microscopic force ξ conjugate to $\tilde{\mu}_p$ in the internal virtual power and in the dissipation potential density we add $\iota_{[0,\infty)}(\dot{\xi})$. In this way ξ opposes to the compaction of the microscopic structure.

7.2. Existence result

In what follows we write simply U and λ in place of U_p and λ_p respectively. Moreover for simplicity we assume $\Gamma_D = \partial\Omega$ and $\bar{u} = 0$ since the extension to the general case in trivial.

7.2.1. Hypotheses. To get the global existence result we assume the following hypotheses.

- (S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2,3\}$ is open, bounded and Lipschitz
  - (b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$
- (Y) (a) $Y \in C^{0,1}(\mathbb{R}, [0,\infty))$
- (G) (a) $G \in C^{1,1}(\text{DevSym}, \mathbb{R})$ is convex
  - (b) $\partial G \in L^\infty(\text{DevSym}, \text{DevPla})$
  - (c) $\partial G(\dot{U}) : \dot{U} \geq 0$ for all $\dot{U} \in$ DevSym
- (L) (a) $f \in C^1([0,T], L^2(\Omega)^d)$
  - (b) $h \in C^1([0,T], L^2(\Gamma_N)^d)$
- (E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in Sym⊗Sym, $\mathfrak{B}$ and $\mathfrak{F}$ are in DevPla⊗DevPla, $\mathfrak{A}$ and $\mathfrak{E}$ are in (Lin⊗$\mathbb{R}$)⊗(Lin⊗$\mathbb{R}$)

(b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geq \gamma|E|^2$ and $\mathfrak{D}E : E \geq \delta|E|^2$ for all $E \in \mathrm{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}\mathbb{U} \circ \mathbb{U} \geq \beta|\mathbb{U}|^2$ and $\mathfrak{F}\mathbb{U} \circ \mathbb{U} \geq \phi|\mathbb{U}|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M \,\vdots\, M \geq \alpha|\operatorname{sym} M|^2$ and $\mathfrak{E}M \,\vdots\, M \geq \epsilon|\operatorname{sym} M|^2$ for all $M \in \mathrm{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

7.2.2. Function spaces. We now introduce the function spaces that we use to formulate the problem. Let

$$W = \left\{ z = (u, U, \lambda) \middle| \begin{smallmatrix} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \mathrm{DevSym}) \\ \lambda \in L^2(\Omega) \end{smallmatrix} \right\} \quad \text{and} \quad \mathcal{Q} = \left\{ p \in L^2(\Omega) \right\}.$$

The rates of $z = (u, U, \lambda) \in W$ will be denoted by $\dot{z} = (\dot{u}, \dot{U}, \dot{\lambda}) \in W$. The variations of $\dot{z} = (\dot{u}, \dot{\xi}, \dot{\lambda}) \in W$ and $p \in Q$ will be denoted as $w = (v, V, \mu) \in W$ and $q \in Q$. The test functions will be denoted as $\tilde{w} = (\tilde{v}, \tilde{V}, \tilde{\mu}) \in W$.

7.2.3. Formulation of the problem. We define $\mathcal{F}\colon W \times Q \times Z \times [0, T] \to Z' \times Q'$ by

$$\begin{aligned} \langle \mathcal{F}(p, y, \dot{y}, t), (\tilde{z}, \tilde{q}) \rangle = & \int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ & + \int_\Omega (\mathfrak{A} \operatorname{grad}^2 u + \mathfrak{E} \operatorname{grad}^2 \dot{u}) \,\vdots\, \operatorname{grad}^2 \tilde{v} dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v} dx + \\ & + \int_\Omega \left[(\mathfrak{B}\mathbb{U} + \mathfrak{F}\dot{\mathbb{U}}) \circ \tilde{\mathbb{V}} + Y(p)\partial G(\dot{U}) : \tilde{V} \right] dx - \\ & - \int_\Omega p\tilde{\mu} dx - \int_\Omega \lambda \tilde{q} dx - \\ & - \int_\Omega f(t) \cdot \tilde{v} dx - \int_{\Gamma_N} h(t) \cdot \tilde{v} dx. \end{aligned}$$

Lemma 7.3. The map $\mathcal{F}$ is well defined.

Proof. The linear terms are trivially well defined so we need to check only

$$\int_\Omega Y(p)\partial G(\dot{U}) : \tilde{V} dx.$$

But

$$\int_\Omega |Y(p)||\partial G(\dot{\mathbb{U}})||\tilde{\mathbb{V}}| dx \leq \|\partial G\|_\infty \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\tilde{\mathbb{V}}| dx \lesssim (1 + \|p\|_2)\|\tilde{V}\|_{1,2}.$$

This concludes the proof. □

Now we can formulate the evolution problem as follows.

Problem 7.4. Find $z \in H^1((0, T), W)$ and $p \in L^2([0, T], Q)$ such that $z(0) = 0$ and

$$\mathcal{F}(p(t), y(t), \dot{y}(t), t) = 0$$

for a.e. $t \in (0, T)$.

7.2.4. Global existence. This section is devoted to prove the following existence of a global in time solution to Problem 7.4.

Theorem 7.5. Assume the hypotheses of Section 7.2.1. Then there exists $z \in H^1((0, T), W)$ and $p \in C^0([0, T], Q)$ solution to Problem 7.4.

The proof is based on the Rothe method so we first prove the existence to a discrete in time version of the problem and then we pass to the vanishing time step limit.

Existence of the discrete in time solution. Let $N \in \mathbb{N}$. We define $\tau_N = N^{-1}T$ and $t_n^N = n\tau_N$ for $n \in \{0, \dots, N\}$. We consider the discrete in time version of Problem 7.4.

Problem 7.6 (Discrete problem). Let $y_0^N = 0 \in \mathscr{Z}$. Find $y_n^N \in \mathscr{Z}$ and $p_n^N \in \mathcal{Q}$ for $n \in \{1, \dots, N\}$ such that

$$\mathcal{F}\left(\frac{y_n^N - y_{n-1}^N}{\tau_N}, p_n^N, y_n^N, t_n^N \right) = 0 \tag{7.1}$$

for all $n \in \{1, \dots, N\}$.

We first prove the existence of a solution to Problem 7.6. Reasoning by induction we reduce to find $y_n^N \in \mathscr{Z}$ and $p_n^N \in \mathscr{Q}$ that satisfy (7.1) given $y_{n-1}^N \in \mathscr{Z}$. We immediately observe that in view of the constrain $\lambda_n^N = 0$ for all n. To simplify the notation we call $y = y_{n-1}^N$, $\dot{y} = \tau^{-1}(y_n^N - y_{n-1}^N)$ and $t = t_n^N$. Then $y_n^N = y + \tau\dot{y}$. It is enough to prove that there exists $\dot{y} \in \mathscr{Z}$ and $p \in Q$ such that

$$\mathcal{F}(\dot{y}, p, y + h\dot{y}, t) = 0. \tag{7.2}$$

We define the operator $\mathcal{A}\colon \mathscr{Q} \times \mathscr{Z} \to \mathscr{Z}'$ by

$$\langle \mathcal{A}(p,\dot{y}), \tilde{z}\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx +$$
$$+ \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})\operatorname{grad}^2 \dot{u} \mathbin{\vdots} \operatorname{grad}^2 \tilde{v}dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}} \circ \tilde{\mathbb{V}} + Y(p)\partial G(\dot{U}) : \tilde{V}\right]dx$$

the linear operator $\mathcal{B}\colon \mathscr{Q} \to \mathscr{Z}'$ by

$$\langle \mathcal{B}p, \tilde{z}\rangle = -\int_\Omega p\tilde{\mu}dx$$

and the operator $\mathcal{C}\colon \mathscr{Z} \to \mathscr{Q}'$ by

$$\langle \mathcal{C}(\dot{y}), \tilde{q}\rangle = \int_\Omega (\lambda + \tau\dot{\lambda})\tilde{q}dx = 0.$$

Moreover we introduce the linear form $l \in \mathscr{Z}'$

$$\langle l, \tilde{z}\rangle = \int_\Omega f(t)\cdot\tilde{v}dx + \int_{\Gamma_N} h(t)\cdot\tilde{v}dx -$$
$$- \int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx -$$
$$- \int_\Omega \mathfrak{A}\operatorname{grad}^2 u \mathbin{\vdots} \operatorname{grad}^2 \tilde{v}dx - \int_\Omega \mathfrak{B}\mathbb{U} \circ \tilde{\mathbb{V}}dx.$$

Now equation (7.2) can be written as

$$\begin{cases} \mathscr{A}(p,\dot{y}) + Bp = l(t) \\ \mathscr{C}(\dot{y}) = 0. \end{cases}$$

We have already noticed that $\lambda = 0$ and $\dot{\lambda} = 0$. We call

$$\mathscr{N} = \left\{ \dot{y} = (\dot{u}, \dot{U}, 0) \;\middle|\; \begin{smallmatrix} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \mathrm{DevSym}) \end{smallmatrix} \right\}.$$

Now we study the first equation in (7.2). We first study the solvability of that equation in $\dot{y} \in \mathscr{N}$ given $p \in \mathscr{Q}$. If we test the first equation with $\tilde{z} \in \mathscr{N}$ we get

$$\langle \mathcal{A}(p,\dot{y}), \tilde{z}\rangle = \langle l, \tilde{z}\rangle \quad \text{for all } \tilde{z} \in \mathscr{N}.$$

Now we define the space

$$\mathcal{X} = \left\{ \xi = (\dot{u}, \dot{U}) \;\middle|\; \begin{smallmatrix} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \mathrm{DevSym}) \end{smallmatrix} \right\}$$

and we can consider the natural bijection $\mathcal{J}$ from $\mathcal{X}$ to $\mathscr{N}$. In particular if $\xi = (\dot{u}, \dot{U})$ we define

$$\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{u}, \dot{U})$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V})$ in $\mathcal{X}$ we set

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle \mathscr{A}(p, \mathscr{J}(\xi)), \mathscr{J}(\tilde{\zeta})\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right) : \left(E\tilde{v} - \tilde{V}\right)dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx +$$
$$+ \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})\operatorname{grad}^2 \dot{u} \mathbin{\vdots} \operatorname{grad}^2 \tilde{v}dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}} \circ \tilde{\mathbb{V}} + Y(p)\partial G(\dot{U}) : \tilde{V}\right]dx.$$

Now for each $p \in \mathscr{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle l, \mathcal{J}(\tilde{\zeta})\rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

We want to apply the Browder–Minty theorem (Theorem B.21). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

Lemma 7.7. The operator $\mathcal{T}_p\colon \mathcal{X} \to \mathcal{X}'$ is bounded and continuous.

Proof. We have

$$|\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle| \lesssim \left(\|E\dot{u}\|_2 + \|\dot{U}\|_2\right)\left(\|E\tilde{v}\|_2 + \|\tilde{V}\|_2\right) + \\ + \|\operatorname{grad}^2 \dot{u}\|_2 \|\operatorname{grad}^2 \tilde{v}\|_2 + \|\dot{\mathbb{U}}\|_2 \|\tilde{\mathbb{V}}\|_2 + (1 + \|p\|_2)\|\tilde{\mathbb{V}}\|_2.$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \to \xi$ in $\mathcal{X}$. Then $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2. We have

$$\left|\int_\Omega Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}})) \circ \tilde{\mathbb{V}} dx\right| \leq \|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2 \|\tilde{\mathbb{V}}\|_2$$

and

$$\|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2^2 = \int_\Omega Y(p)^2 (\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))^2 dx \to 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2 and we can apply the dominated convergence theorem to pass to the limit. □

Lemma 7.8. The operator $\mathcal{T}_p \colon \mathcal{X} \to \mathcal{X}'$ is coercive uniformly in p.

Proof. For any $\xi \in \mathcal{X}$ we have

$$\langle \mathcal{T}_p(\xi), \xi\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right) : \left(E\dot{u} - \dot{U}\right) dx + \int_\Omega \mathfrak{D}E\dot{u} : E\dot{u}dx + \\ + \int_\Omega (\mathfrak{E} + \tau\mathfrak{A}) \operatorname{grad}^2 \dot{u} \;\vdots\; \operatorname{grad}^2 \dot{u} dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}} \circ \dot{\mathbb{U}} + Y(p)\partial G(\dot{U}) \circ \dot{U}\right] dx.$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{U} : I = 0$ since $\dot{U} \in \mathrm{Dev}$ we have

$$\langle \mathcal{T}_p(\xi), \xi\rangle \geq \tau\gamma \left\|E\dot{u} - \dot{U}\right\|_2^2 + \tau\alpha \|\operatorname{grad} E\dot{u}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2.$$

By the Young inequality

$$\left\|E\dot{u} - \dot{U}\right\|_2^2 \geq (1-\theta)\|E\dot{u}\|_2^2 + \left(1 - \frac{1}{\theta}\right)\|\dot{U}\|_2^2$$

for any $\theta > 0$. Then

$$\langle \mathcal{T}_p(\xi), \xi\rangle \geq (1-\theta)\tau\gamma\|E\dot{u}\|_2^2 + \tau\alpha\|\operatorname{grad} E\dot{u}\|_2^2 + \\ + \left(1 - \frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2.$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1 - \frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2 \geq C_1\|\dot{\mathbb{U}}\|_2^2 \geq C_2\|\dot{U}\|_{1,2}^2.$$

The thesis follows by application of Korn inequality. □

Lemma 7.9. The operator $\mathcal{T}_p \colon \mathcal{X} \to \mathcal{X}'$ is strongly monotone uniformly in p, i.e., there exists $M > 0$ such that

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle \geq M\|\xi_2 - \xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$.

Proof. For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle = \\ = \int_\Omega \tau\mathfrak{C}\left(E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\right) : (E(\dot{u}_2 - \dot{u}_1) - (U_2 - U_1))\, dx + \\ + \int_\Omega \mathfrak{D}(E(\dot{u}_2 - \dot{u}_1)) : (E(\dot{u}_2 - \dot{u}_1))dx + \\ + \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})(\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1) \;\vdots\; (\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1)dx + \\ + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})(\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1) \circ (\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1) + Y(p)(\partial G(\dot{U}_2) - \partial G(\dot{U}_1)) : (\dot{U}_2 - \dot{U}_1)\right] dx \geq \\ \geq (\delta + \tau\gamma)\|E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1\|_2^2.$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 7.8 we have

$$\begin{aligned}\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle \geq (1-\theta)(\delta + \tau\gamma)\|E(\dot{u}_2 - \dot{u}_1)\|_2^2 + (\alpha + \tau\epsilon)\|\operatorname{grad} E(\dot{u}_2 - \dot{u}_1)\|_2^2 + \\ + \left(1 - \frac{1}{\theta}\right)(\delta + \tau\gamma)\|\dot{U}_2 - \dot{U}_1\|_2^2 + \\ +(\phi + \tau\beta)\|\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1\|_2^2.\end{aligned}$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using second order Korn inequality we conclude. □

Now applying Theorem B.21 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle l, \mathcal{J}(\tilde{\zeta})\rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique. Indeed by strong monotonicity if ξ_1 and ξ_2 are two solutions we have

$$M\|\xi_2 - \xi_1\|_{\mathcal{X}}^2 \leq \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle = 0.$$

We call $\Xi\colon \mathcal{Q} \to \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ. Now to conclude the existence of a solution to Problem 7.6 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0,0,\tilde{\mu})\rangle + \langle Bp, (0,0,\tilde{\mu})\rangle = \langle l, (0,0,\tilde{\mu})\rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_\Omega \operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right]\frac{\tilde{\mu}}{d}dx - \int_\Omega p\tilde{\mu}dx = \int_\Omega \operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right]\frac{\tilde{\mu}}{d}dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$p = \frac{1}{d}\operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right] - \frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] = \mathcal{S}(p) \tag{7.3}$$

where we have introduced the operator $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem B.16). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ. We first need the following lemma.

Lemma 7.10. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

Proof. By (Y, a) and (G, c) we have

$$\begin{aligned}|\langle \mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y}), \tilde{z}\rangle| &\leq \int_\Omega |Y(p) - Y(q)||\partial G(\dot{U})||\tilde{V}|dx \leq \\ &\leq \operatorname{Lip}(Y)\|\partial G\|_\infty \int_\Omega |p - q||\tilde{V}|dx \leq L\|p - q\|_2\|\tilde{V}\|_2\end{aligned}$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y})\|_{\mathcal{X}'} \leq L\|p - q\|_{\mathcal{Q}}$$

This concludes the proof. □

Now we can prove the continuity of Ξ.

Lemma 7.11. The operator Ξ is continuous and has bounded range.

Proof. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geq n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \leq \|\langle l, \mathcal{J}(\cdot)\rangle\|_{\mathcal{X}'}.$$

But by Lemma 7.8

$$\frac{\langle \mathcal{T}_p(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \to \infty$$

and this is a contradiction. Now we prove that Ξ is Lipschitz continuous. Indeed let $p_i \in \mathcal{Q}$ and $\xi_i = \Xi(p_i)$ with i equals to 1 or 2. Since $\mathcal{T}_{p_1}(\xi_1) = \mathcal{T}_{p_2}(\xi_2)$, $\mathcal{T}_p$ is strictly monotone uniformly in p and $\mathcal{A}$ is uniformly Lipschitz in p we have

$$\begin{aligned}M\|\xi_2 - \xi_1\|^2 &\leq \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_1}(\xi_1), \xi_2 - \xi_1\rangle = \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_2}(\xi_2), \xi_2 - \xi_1\rangle \leq \\ &\leq L\|p_2 - p_1\|_{\mathcal{Q}}\|\xi_2 - \xi_1\|_{\mathcal{X}}.\end{aligned}$$

This concludes the proof. □

The following lemma establish the regularity of $\mathcal{J}$.

LEMMA 7.12. The map $\mathscr{J}\colon \mathscr{X} \to \mathscr{Z}$ is continuous and bounded.

PROOF. It is trivial since $\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0)$. □

Now we are ready to prove the desired properties of $\mathcal{S}$.

LEMMA 7.13. The map $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$ is continuous and has a bounded and compact range.

PROOF. We notice that since Ξ is continuous and has compact range then also $\mathcal{S}$ is continuous and has compact range. We notice that by Rellich–Kondarchov theorem then $\dot{u} \mapsto E\dot{u}$ from H^2 to L^2 and $\dot{U} \mapsto \dot{U}$ from H^1 to L^2 are compact. Then $\mathcal{S}$ is compact. Since $\mathcal{S}$ has bounded range and is compact then it has compact range. □

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

PROPOSITION 7.14. Problem 7.6 admits at least a solution.

Now we redefine the operator $\mathcal{A}\colon \mathscr{Z} \to \mathscr{Z}'$ by

$$\langle \mathcal{A}y, \tilde{z}\rangle = \int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right) dx + \int_\Omega \mathfrak{A}\operatorname{grad} u \,\vdots\, \operatorname{grad}\tilde{v}\,dx + \int_\Omega \mathfrak{B}\mathbb{U}\circ\tilde{\mathbb{V}}dx$$

and we define operator $\mathcal{E}\colon \mathscr{Z} \to \mathscr{Z}'$ by

$$\langle \mathcal{E}\dot{y}, \tilde{z}\rangle = \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx + \int_\Omega \mathfrak{E}\operatorname{grad}\dot{u} \,\vdots\, \operatorname{grad}\tilde{v}dx + \int_\Omega \mathfrak{F}\dot{\mathbb{U}}\circ\tilde{\mathbb{V}}dx.$$

We define also $\mathcal{D}\colon \mathcal{Q}\times\mathscr{Z} \to \tilde{\mathcal{Z}}'$ by

$$\langle \mathcal{D}(p,\dot{y}), \tilde{z}\rangle = \int_\Omega Y(p)\partial G(\dot{U}) : \tilde{V}dx.$$

The operator $\mathcal{B}$ is defined exactly as before and and $\mathcal{C}\colon \mathscr{Z} \to \mathcal{Q}'$ is redefined by

$$\langle \mathcal{C}y, \tilde{q}\rangle = \int_\Omega \lambda\tilde{q}dx = 0.$$

Let also $l\colon [0,T] \to \mathscr{Z}'$ redefined by

$$\langle l(t), \tilde{z}\rangle = \int_\Omega f(t)\cdot\tilde{v}dx + \int_{\Gamma_N} h(t)\cdot\tilde{v}dx.$$

Proposition 7.14 ensures that that for each N there exists $\left\{(\dot{y}_n^N, p_n^N, y_n^N)\right\}_{n=0}^N$ such that $(\dot{y}_0^N, p_0^N, y_0^N) = 0$, $y_n^N = y_{n-1}^N + \tau\dot{y}_n^N$ and

$$\begin{cases} \mathcal{A}y_n^N + \mathcal{E}\dot{y}_n^N + \mathcal{D}(p_n^N, \dot{y}_n^N) + \mathcal{B}p_n^N = l_n^N \\ \mathcal{C}y_n^N = 0 \end{cases} \tag{7.4}$$

for all $n \in \{1,\dots,N\}$. Here $l_n^N = l(t_n^N)$.

Construction of the continuous in time solution. Now we want to construct the continuous in time solution passing to the limit $N \to \infty$. We define

$$\begin{aligned}
l^N &\in L^2((0,T),\mathscr{Z}') && \text{piecewise constant interpolation of } \{l_n^N\}_{n=1}^N \\
\bar{y}^N &\in L^2((0,T),\mathscr{Z}) && \text{piecewise constant interpolation of } \{y_n^N\}_{n=1}^N \\
y^N &\in H^1((0,T),\mathscr{Z}) && \text{piecewise linear interpolation of } \{y_n^N\}_{n=1}^N \\
\dot{y}^N &\in L^2((0,T),\mathscr{Z}) && \text{piecewise constant interpolation of } \{\dot{y}_n^N\}_{n=1}^N \\
p^N &\in L^2((0,T),\mathcal{Q}) && \text{piecewise constant interpolation of } \{p_n^N\}_{n=1}^N.
\end{aligned}$$

We will need the following lemma.

LEMMA 7.15. We have that $l^N \to l$ in $L^2((0,T),\mathscr{Z}')$ and in particular

$$\|l^N\|^2_{L^2((0,T),\mathscr{Z}')} = \sum_{n=1}^N \tau_N\|l_n^N\|^2_{\mathscr{Z}'}$$

is bounded.

Proof. For all $t \in [t_{n-1}^N, t_n^N]$ we have

$$l^N(t) = l_n^N = l(t_n^N) = l(t) + \int_t^{t_n^N} \dot{l}(s)ds.$$

By Hölder inequality

$$\|l^N(t) - l(t)\|_{\mathcal{X}'} \leq \int_t^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}ds \leq \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}ds \leq \sqrt{\tau_N}\left[\int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}^2 ds\right]^{1/2}.$$

Taking the square and integrating for $t \in [t_{n-1}^N, t_n^N]$ we have

$$\int_{t_{n-1}^N}^{t_n^N} \|l^N(t) - l(t)\|_{\mathcal{X}'}^2 dt \leq \tau_N^2 \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}^2 ds.$$

Adding this inequalities for $n \in \{1, \ldots, N\}$ we have

$$\|l^N - l\|_{L^2((0,T),\mathcal{X}')}^2 \leq \tau_N^2 \|\dot{l}\|_{L^2((0,T),\mathcal{X}')}^2 \to 0.$$

This concludes the proof. □

We now obtain a priori estimates on $\bar{y}^N$, y^N and $\dot{y}^N$.

Lemma 7.16. The functions $\bar{y}^N$ and y^N are bounded in $L^\infty((0,T),\mathcal{X})$ and the function $\dot{y}^N$ is bounded in $L^2((0,T),\mathcal{X})$.

Proof. We first observe that y_n^N and $\dot{y}_n^N$ belongs to $\mathcal{N}$ and that $\mathcal{A}$ and $\mathcal{E}$ are coercive there. Next we test the first equation in (7.4) with $\tilde{z} = \dot{y}_n^N$. We notice that $\langle \mathcal{D}(p_n^N, \dot{y}_n^N), \dot{y}_n^N\rangle \geq 0$ and $\langle \mathcal{B}p_n^N, \dot{y}_n^N\rangle = 0$. Then

$$\frac{1}{\tau_N}\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle + \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \leq \langle l_n^N, \dot{y}_n^N\rangle.$$

Multiplying by $2\tau_N$ we have

$$2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle + 2\tau_N\langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \leq 2\tau_N\langle l_n^N, \dot{y}_n^N\rangle.$$

Now

$$\begin{aligned} 2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle &= \langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle + \langle \mathcal{A}(y_n^N - y_{n-1}^N), (y_n^N - y_{n-1}^N)\rangle \geq \\ &\geq \langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle. \end{aligned}$$

Let $E > 0$ be its coercivity constant of $\mathcal{E}$ in $\mathcal{N}$. Then

$$2\tau_N\langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \geq 2\tau_N E\|\dot{y}_n^N\|_{\mathcal{X}}^2$$

and using Young inequality

$$2\tau_N\langle l_n^N, \dot{y}_n^N\rangle \leq \frac{\tau_N}{E}\|l_n^N\|_{\mathcal{X}'}^2 + \tau_N E\|\dot{y}_n^N\|_{\mathcal{X}}^2.$$

Then

$$\langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle + \tau_N E\|\dot{y}_n^N\|_{\mathcal{X}}^2 \leq \frac{\tau_N}{E}\|l_n^N\|_{\mathcal{X}'}^2. \tag{7.5}$$

Let $A > 0$ the coercivity constant of $\mathcal{A}$ on $\mathcal{N}$. Then adding the inequalities (7.5) we have

$$A\|y_n^N\|_{\mathcal{X}}^2 + E\sum_{k=1}^n \tau_N\|\dot{y}_k^N\|_{\mathcal{X}}^2 \leq \langle \mathcal{A}y_n^N, y_n^N\rangle + E\sum_{k=1}^n \tau_N\|\dot{y}_k^N\|_{\mathcal{X}}^2 \leq \frac{1}{E}\sum_{k=1}^n \tau_N\|l_k^N\|_{\mathcal{X}'}^2.$$

Using Lemma 7.15 we have that

$$\|y_n^N\|_{\mathcal{X}} \quad \text{and} \quad \sum_{n=1}^N \tau_N\|\dot{y}_n^N\|_{\mathcal{X}}^2$$

are bounded uniformly in $N \in \mathbb{N}$ and $n \in \{0, \ldots, N\}$. This implies immediately the thesis. □

Using that lemma, up to extracting a subsequence we have that

$$\begin{aligned} \bar{y}^N &\overset{*}{\rightharpoonup} \bar{y} \quad \text{in } L^\infty((0,T),\mathcal{X}) \\ y^N &\overset{*}{\rightharpoonup} y \quad \text{in } L^\infty((0,T),\mathcal{X}) \\ \dot{y}^N &\rightharpoonup z \quad \text{in } L^2((0,T),\mathcal{X}). \end{aligned}$$

We notice that from $\lambda_n^N = 0$ it follows immediately that $\lambda = 0$.

LEMMA 7.17. We have that $\bar{y} = y \in H^1((0,T),\mathcal{X})$ and $\dot{y} = z$. Moreover $y(0) = 0$.

PROOF. We prove that $\bar{y}^N - y^N \to 0$ in $L^\infty((0,T),\mathcal{X})$. Indeed let $t \in [t_{n-1}^N, t_n^N]$. Then

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{X}} = \|y_n^N - y_n^N - (t_n^N - t)\dot{y}_n^N\|_{\mathcal{X}} \le \tau_N\|\dot{y}_n^N\|_{\mathcal{X}}.$$

It follows that

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{X}}^2 \le \tau_N^2\|\dot{y}_n^N\|_{\mathcal{X}}^2 \le \tau_N\|\dot{y}^N\|_{L^2((0,T),\mathcal{X})}^2 \to 0.$$

This establish the equality $\bar{y} = y$. Now let $\phi \in C_c^\infty([0,T))$ and $\zeta \in \mathcal{X}'$ arbitrary. Then

$$\int_0^T \phi\langle\zeta, \dot{y}^N\rangle dt = -\phi(0)\langle\zeta, y^N(0)\rangle - \int_0^T \dot{\phi}\langle\zeta, y^N\rangle dt.$$

Choosing first $\phi \in C_c^\infty((0,T))$ and exploiting (7.6) we have

$$\int_0^T \phi\langle\zeta, \dot{y}\rangle dt = -\int_0^T \dot{\phi}\langle\zeta, y\rangle dt.$$

This means exactly that $y \in H^1((0,T),\mathcal{X})$ and $\dot{y} = z$. In particular for $\phi \in C_c^\infty([0,T))$ generic we have

$$\int_0^T \phi\langle\zeta, \dot{y}\rangle dt = -\phi(0)\langle\zeta, y(0)\rangle - \int_0^T \dot{\phi}\langle\zeta, y\rangle dt.$$

Then we get also

$$\langle\zeta, y^N(0)\rangle \to \langle\zeta, y(0)\rangle.$$

Since $y^N(0) = 0$ also $y(0) = 0$. □

We now discuss the convergence of p^N.

LEMMA 7.18. Up to extracting a further subsequence $u^N \to u$ in $C([0,T], H^1(\Omega)^d)$ and $U^N \to U$ in $C([0,T], L^2(\Omega,\mathrm{DevSym}))$. Moreover $p^N \to p$ in $C([0,T],\mathcal{Q})$ for some p.

PROOF. Since $y^N \rightharpoonup y$ in $L^\infty((0,T),\mathcal{X})$ and $\dot{y}^N \rightharpoonup \dot{y}$ in $L^2((0,T),\mathcal{X})$ then also

$$u^N \rightharpoonup u \text{ in } L^\infty(\Omega, H^2) \quad\text{and}\quad \dot{u}^N \rightharpoonup \dot{u} \text{ in } L^2(\Omega, H^2),$$
$$U^N \rightharpoonup U \text{ in } L^\infty(\Omega, H^1) \quad\text{and}\quad \dot{U}^N \rightharpoonup \dot{U} \text{ in } L^2(\Omega, H^1).$$

By Aubin–Lions–Simon lemma (Theorem B.6)

$$u^N \to u \text{ in } C([0,T], H^1) \quad\text{and}\quad U^N \to U \text{ in } C([0,T], L^2).$$

We recall that by (7.3) we have

$$p_n^N = \frac{1}{d}\,\mathrm{tr}\left[\tau_N\mathfrak{C}\left(E\dot{u}_n^N - \dot{U}_n^N\right)\right] - \frac{1}{d}\,\mathrm{tr}\left[\mathfrak{C}\left(Eu_n^N - U_n^N\right)\right].$$

Since $\tau_N \to 0$ and $\dot{y}_n^N$ is bounded in $L^2((0,T),\mathcal{X})$ we have

$$p^N \to p = -\frac{1}{d}\,\mathrm{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] \quad \text{in } L^2((0,T),\mathcal{X}).$$

This concludes the proof. □

REMARK 7.19. The fact that $p \in C^0([0,T],\mathcal{Q})$ suggest that there are not bifurcation modes of pressure.

Now we notice that (7.4) implies that for all $\tilde{z} \in L^2((0,T),\mathcal{X})$ we have

$$\int_0^T \langle\mathcal{A}\bar{y}^N, \tilde{z}\rangle dt + \int_0^T \langle\mathcal{E}\dot{y}^N, \tilde{z}\rangle dt + \int_0^T \langle\mathcal{D}(P^N, \dot{y}^N), \tilde{z}\rangle dt + \int_0^T \langle\mathcal{B}p^N, \tilde{z}\rangle dt = \int_0^T \langle l^N, \tilde{z}\rangle dt.$$

By Lemma 7.15 and (7.6) we have

$$\int_0^T \langle l^N, \tilde{z}\rangle dt \to \int_0^T \langle l, \tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{A}\bar{y}^N, \tilde{z}\rangle dt \to \int_0^T \langle\mathcal{A}y, \tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{E}\dot{y}^N, \tilde{z}\rangle dt \to \int_0^T \langle\mathcal{E}\dot{y}, \tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{B}p^N, \tilde{z}\rangle dt \to \int_0^T \langle\mathcal{B}p, \tilde{z}\rangle dt.$$

Moreover we have

$$\int_0^T \|\mathcal{D}(p^N,\dot y^N)\|^2_{\mathcal{Z}'}dt \le \int_0^T\int_\Omega \Upsilon(p^N)^2|\partial G(\dot U^N)|^2 dxdt \le$$
$$\le \int_0^T\int_\Omega (\Upsilon(0)+\mathrm{Lip}(\Upsilon)|p^N|)^2\|\partial G\|^2_\infty dxdt \lesssim$$
$$\lesssim (1+\|p^N\|^2_{L^2((0,T),\mathcal{Q})}).$$

But since p^N converges in $C([0,T],\mathcal{Q})$ then $\mathcal{D}(p^N,\dot y^N)$ is bounded in $L^2((0,T),\mathcal{Z}')$. Then up to extracting a further subsequence

$$\mathcal{D}(p^N,\dot y^N) \rightharpoonup \delta \text{ in } L^2((0,T),\mathcal{Z}')$$

for some $\delta \in L^2((0,T),\mathcal{Z}')$. Then we have

$$\int_0^T \langle \mathcal{A}y,\tilde z\rangle dt + \int_0^T \langle \mathcal{E}\dot y,\tilde z\rangle dt + \int_0^T \langle \delta,\tilde z\rangle dt + \int_0^T \langle \mathcal{B}p,\tilde z\rangle dt = \int_0^T \langle l,\tilde z\rangle dt \tag{7.7}$$

for all $\tilde z \in L^2((0,T),\mathcal{Z})$. This implies also

$$\langle \mathcal{A}y,\tilde z\rangle + \langle \mathcal{E}\dot y,\tilde z\rangle + \langle \delta,\tilde z\rangle + \langle \mathcal{B}p,\tilde z\rangle = \langle l,\tilde z\rangle$$

for almost every $t \in (0,T)$. If we prove that $\delta = \mathcal{D}(p,\dot y)$ we have that (y,p) is a global in time solution to Problem 7.4 and we have concluded. To this aim let us define $\mathcal{Z}_T = L^2((0,T),\mathcal{Z})$ and $\mathcal{Q}_T = L^2((0,T),\mathcal{Q})$. Let $\mathcal{J}\colon \mathcal{Q}\times\mathcal{Z}\to\mathbb{R}$ be defined by

$$\mathcal{J}(p,\dot y) = \int_\Omega \Upsilon(p)G(\dot U)dx.$$

Let also $\mathcal{J}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T\to\mathbb{R}$ and $\mathcal{D}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T\to\mathcal{Z}_T'$ defined by

$$\mathcal{J}_T(p,\dot y) = \int_0^T \mathcal{J}(p,\dot y)dt \quad\text{and}\quad \langle \mathcal{D}_T(p,\dot y),\tilde z\rangle = \int_0^T \langle \mathcal{D}(p,\dot y),\tilde z\rangle dt$$

for all $\tilde z \in L^2((0,T),\mathcal{Z})$. We notice that for all $y \in \mathcal{Z}_T$ and $p \in \mathcal{Q}_T$ we can identify $\mathcal{D}_T(p,\dot y)\in\mathcal{Z}_T'$ with $\mathcal{D}(p,\dot y)\in L^2((0,T),\mathcal{Z}')$.

Lemma 7.20. Let $\dot y \in L^2((0,T),\mathcal{Z})$, $p\in L^2((0,T),\mathcal{Q})$ and $\delta\in L^2((0,T),\mathcal{Z}')$. Then $\delta = \mathcal{D}(p,\dot y)$ if and only if

$$\mathcal{J}_T(p,\tilde z) \ge \mathcal{J}_T(p,\dot y)dt + \int_0^T \langle \delta,\tilde z-\dot y\rangle dt \tag{7.8}$$

for all $\tilde z\in L^2((0,T),\mathcal{Z})$.

Proof. It is easy to see that $\mathcal{J}$ is convex in $\dot y$ and is Gateaux differentiable in $\dot y$ with Gateaux differential

$$\langle D_{\dot y}\mathcal{J}_T(p,\dot y),\tilde z\rangle = \int_0^T \langle \mathcal{D}(p,\dot y),\tilde z\rangle dt = \langle \mathcal{D}_T(p,\dot y),\tilde z\rangle$$

for all $\tilde z \in L^2((0,T),\mathcal{Z})$. By [23, Proposition 5.3] then $\mathcal{J}_T$ is sub differentiable and $D^{\mathrm{sub}}_{\dot y}\mathcal{J}_T(p,\dot y) = \{D_{\dot y}\mathcal{J}_T(p,\dot y)\}$. Then $\delta = \mathcal{D}(p,\dot y)$ if and only if $\delta \in D^{\mathrm{sub}}_{\dot y}\mathcal{J}_T(p,\dot y)$ but this last inclusion is equivalent to (7.8). ◻

By that lemma we have

$$\mathcal{J}_T(p^N,\tilde z) \ge \mathcal{J}_T(p^N,\dot y) + \langle \mathcal{D}_T(p^N,\dot y^N),\tilde z-\dot y^N\rangle. \tag{7.9}$$

Now we want to pass to pass to the limit in this inequality. We will need the following lemma.

Lemma 7.21. For each $\dot y\in\mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot,\dot y)$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot y\|_{\mathcal{Z}_T}$. Moreover for each $p\in\mathcal{Q}_T$ we have that $\mathcal{J}_T(p,\cdot)$ is convex and continuous. In particular $\mathcal{J}_T(p,\cdot)$ is weakly lower semicontinuous.

Proof. We have that

$$|\mathcal{J}_T(p_2,\dot y)-\mathcal{J}_T(p_1,\dot y)| \le \int_0^T\int_\Omega |\Upsilon(p_2)-\Upsilon(p_1)||G(\dot U)|dx \le$$
$$\le \|\partial G\|_\infty \mathrm{Lip}(\Upsilon)\int_0^T\int_\Omega |p_2-p_1||\dot U|dtdx \lesssim$$
$$\lesssim \|p_2-p_1\|_{\mathcal{Q}_T}\|\dot y\|_{\mathcal{Z}_T}.$$

This proves that for each $\dot{y} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot, \dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{Z}_T}$. The convexity of $\mathcal{J}_T(p, \cdot)$ follows immediately from the convexity of G. Moreover

$$\begin{aligned}|\mathcal{J}_T(p, \dot{y}_2) - \mathcal{J}_T(p, \dot{y}_1)| &\leq \int_0^T \int_\Omega |Y(p)||G(\dot{U}_2) - G(\dot{U}_1)| dx \leq \\ &\leq \|\partial G\|_\infty \int_0^T \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\dot{U}_2 - \dot{U}_1| dt dx \lesssim \\ &\lesssim (1 + \|p\|_{\mathcal{Q}_T}) \|\dot{y}_2 - \dot{y}_1\|_{\mathcal{Z}_T}.\end{aligned}$$

This proves the continuity of $\mathcal{J}_T(p, \cdot)$ for each $p \in \mathcal{Q}_T$ fixed. □

Now we have

$$|\mathcal{J}_T(p^N, \tilde{z}) - \mathcal{J}_T(p, \tilde{z})| \leq C\|p^N - p\|_{\mathcal{Q}_T} \to 0$$

for some constant $C > 0$. Moreover

$$\mathcal{J}_T(p^N, \dot{y}^N) = [\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)] + \mathcal{J}_T(p, \dot{y}^N).$$

But since $p^N \top$ in $\mathcal{Q}_T$ and y^N are bounded in $\mathcal{Z}_T$

$$|\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)| \leq C\|y^N\|_{\mathcal{Z}_T} \|p^N - p\|_{\mathcal{Q}_T} \to 0$$

and

$$\liminf_{N\to\infty} \mathcal{J}_T(p, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Then

$$\liminf_{N\to\infty} \mathcal{J}_T(p^N, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Now by construction

$$\begin{aligned}\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{E}\dot{y}^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{D}(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N\rangle dt + \\ + \int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt = \int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt.\end{aligned}$$

But by Lemma 7.15, Lemma 7.18 and (7.6) we have

$$\begin{aligned}\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y}\rangle dt \\ \int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y}\rangle dt \\ \int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle l, \tilde{z} - \dot{y}\rangle dt\end{aligned}$$

since the convergence of y^N, p^N and l^N is strong. Moreover since

$$\dot{y} \to \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt$$

is continuous and convex in $L^2((0,T), \mathcal{Z})$ it is weakly lower semicontinuous and

$$\liminf_{N\to\infty} \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt \geq \int_0^T \langle \mathcal{E}\dot{y}, \dot{y} - \tilde{z}\rangle dt.$$

This implies that

$$\begin{aligned}&\liminf_{N\to\infty} \langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N\rangle \geq \\ &\geq \int_0^T \langle l, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y}\rangle dt = \\ &= \int_0^T \langle \delta, \tilde{z} - \dot{y}\rangle dt\end{aligned}$$

where the last equality holds for (7.7). Collecting these results when we take the inferior limit for $N \to \infty$ of (7.9) and we get

$$\mathfrak{J}_T(p,\tilde{z}) \geq \mathfrak{J}_T(p,\dot{y}) + \int_0^T \langle \delta, \tilde{z}-\dot{y}\rangle dt.$$

Since now δ satisfy (7.8) we have by Lemma 7.8 that $\delta = \mathfrak{D}(p,\dot{y})$ as desired.

7.2.5. Example of constitutive relations. In this section we give examples of constitutive relations satisfying the hypotheses of Section 7.2.1. Let $g\colon [0,\infty) \to \mathbb{R}$ be defined by

$$g(r) = \begin{cases} \frac{r^2}{4c} & \text{if } r \leq c \\ r - \frac{3}{4}c - \frac{c}{2}\log\frac{r}{c} & \text{if } r > c \end{cases}$$

where $c > 0$ so that

$$g'(r) = \begin{cases} \frac{r}{2c} & \text{if } r \leq c \\ 1 - \frac{c}{2r} & \text{if } r > c. \end{cases}$$

We set $G(\dot{U}) = g(|\dot{U}|)$. We let the yield strength be given by

$$Y(p) = \begin{cases} Y_0 & \text{if } p < -p_0 \\ Y_0 + \alpha(p+p_0) & \text{if } p \geq -p_0 \end{cases}$$

where $Y_0 \geq 0$, $p_0 \geq 0$ and $\alpha > 0$. We set $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{D}$, $\mathfrak{E}$ and $\mathfrak{F}$ positive constants and $\mathfrak{C}$ and $\mathfrak{D}$ positive definite isotropic fourth order tensors.

7.2.6. Further regularity hypotheses. The local existence and uniqueness theorem is obtained under the following additional hypotheses.

(Y′) $Y \in C^{1,1}(\mathbb{R})$
(G′) $G \in C^{2,1}(\mathrm{DevSym})$

7.2.7. Local existence and uniqueness and global uniqueness.

Proposition 7.22. The map $\mathcal{F}$ is of class C^1 and the Fréchet derivative is given by

$$\begin{aligned}
\langle D_{\dot{y}}\mathcal{F}(\dot{y},p,y,t)z,(\tilde{z},\tilde{q})\rangle &= \int_\Omega \mathfrak{D}Ev : E\tilde{v}dx + \int_\Omega \mathfrak{E}\,\mathrm{grad}^2 v : \mathrm{grad}^2\tilde{v}dx + \\
&\quad + \int_\Omega \mathfrak{F}\mathbb{V}\circ\tilde{\mathbb{V}}dx + \int_\Omega Y(p)\partial^2 G(\dot{U})V : \tilde{V}dx \\
\langle D_p\mathcal{F}(\dot{y},p,y,q)q,(\tilde{z},\tilde{q})\rangle &= \int_\Omega Y'(p)q\partial G(\dot{U}) : \tilde{V}dx - \int_\Omega q\tilde{\mu}dx \\
\langle D_y\mathcal{F}(\dot{y},p,y,t)z,(\tilde{z},\tilde{q})\rangle &= \int_\Omega \mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \\
&\quad + \int_\Omega \mathfrak{A}\,\mathrm{grad}^2 v \,\vdots\, \mathrm{grad}^2\tilde{v}dx + \int_\Omega \mathfrak{B}\mathbb{V}\circ\tilde{\mathbb{V}}dx \\
\langle D_t\mathcal{F}(\dot{y},p,y,t),(\tilde{z},\tilde{q})\rangle &= -\int_\Omega \dot{f}\cdot\tilde{v}dx - \int_\Omega \dot{h}\cdot\tilde{v}dx.
\end{aligned}$$

Proof. I still need to report in LaTeXthe boring check but it should be true. First we prove the Gateaux differentiability everywhere that ensure also the continuity. Then we prove that the Gateaux differential is continuous. To this aim we use Lemma B.13, the following lemma and the embedding of H^1 in L^6 for $d = 2$ or $d = 3$.

Lemma 7.23. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi\colon \mathbb{R}^m \to \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \to f$ in $L^p(\Omega,\mathbb{R}^m)$ for some $p \in [1,\infty)$. Then $\phi(f_n) \to \phi(f)$ in $L^q(\Omega,\mathbb{R}^l)$ for all $q \in [p,\infty)$.

Proof. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \to 0$. Now let $q \in [p,\infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m}|\phi|$ we have

$$\begin{aligned}
\|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega\setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leq \\
&\leq \int_{A_n} L|f_n - f|^q dx + 2C|\Omega \setminus A_n|.
\end{aligned}$$

But on A_n we have $|f_n - f|^q \le |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \le \int_{A_n} L|f_n - f|^p q\,dx + 2C|\Omega \setminus A_n| \le L\|f_n - f\|_p^p + 2C|\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \to \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. □

□

We notice that $\mathcal{F}$ does not depend on λ. We call $\xi = (\dot{u}, \dot{U}, \lambda, p)$, $\zeta = (v, V, \mu, q) = (z, q)$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V}, \tilde{\mu}, \tilde{q}) = (\tilde{z}, \tilde{q})$ all belonging to $\mathcal{Z} \times \mathcal{Q}$.

Proposition 7.24. For any $\dot{y} \in \mathscr{Z}$, $p \in \mathscr{Q}$, $y \in \mathscr{Z}$ and $t \in [0, T]$ we have $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \operatorname{Iso}(\mathscr{Z} \times \mathscr{Q}, \mathscr{Z}' \times \mathscr{Q}')$.

Proof. We define the bilinear form $a\colon \mathscr{Z} \times \mathscr{Z} \to \mathbb{R}$ by

$$a(z, \tilde{z}) = \int_\Omega \mathfrak{D} E v : E\tilde{v}\,dx - \int_\Omega \mathfrak{C}\frac{\mu}{d} I : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I\right) dx + \int_\Omega \mathfrak{E} \operatorname{grad}^2 v \,\vdots\, \operatorname{grad}^2 \tilde{v}\,dx +$$
$$+ \int_\Omega \mathfrak{F} \mathbb{V} \circ \tilde{\mathbb{V}}\,dx + \int_\Omega Y(p) \partial^2 G(\dot{U}) V : \tilde{V}\,dx.$$

We define also the bilinear forms $b\colon \mathscr{Q} \times \mathscr{Z} \to \mathbb{R}$ and $c\colon \mathscr{Q} \times \mathscr{Z} \to \mathbb{R}$ by

$$b(p, \tilde{z}) = \int_\Omega (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q\,dx$$

and

$$c(\tilde{q}, z) = -\int_\Omega \mu \tilde{q}\,dx = 0.$$

To prove that $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \operatorname{Iso}(\mathscr{Z} \times \mathscr{Q}, \mathscr{Z}' \times \mathscr{Q}')$ we have to prove that the mixed elliptic problem

$$a(z, \tilde{z}) + b(p, \tilde{z}) = \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \mathscr{Z}$$
$$c(\tilde{q}, z) = \langle m, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \mathscr{Q}$$

is well posed for all $l \in \mathscr{Z}'$ and $m \in \mathscr{Q}'$. To this aim it is sufficient to check the hypotheses of Theorem B.11. We first check the hypotheses on b. We have

$$\tilde{\mathcal{N}} = \left\{ \tilde{z} \in \mathcal{Z} \;\middle|\; b(p, \tilde{z}) = 0 \text{ for all } p \in \mathscr{Q} \right\} = \left\{ \tilde{z} \in \mathcal{Z} \;\middle|\; \tilde{\mu} = Y'(p) \partial G(\dot{U}) : \tilde{V} \right\}.$$

Notice that $|Y'(p) \partial G(\dot{U}) : \tilde{V}| \le \operatorname{Lip}(Y) \|\partial G\|_\infty |\tilde{V}| \in L^2$. Now

$$\sup_{q \in \mathscr{Q} \setminus \{0\}} \frac{\int_\Omega (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q\,dx}{\|q\|_Q} = \|Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}\|_2.$$

On the other hand since $\tilde{z}_0 = (\tilde{v}_0, \tilde{V}_0, \tilde{\mu}_0) \in \tilde{\mathcal{N}}$ if and only if $\tilde{\mu}_0 = Y'(p) \partial G(\dot{U}) : \tilde{V}_0$ we have

$$\begin{aligned} \|\tilde{z}\|_{\mathscr{Z}/\tilde{\mathcal{N}}} &= \inf_{\tilde{z}_0 \in \tilde{\mathcal{N}}} \left[\|\tilde{v} + \tilde{v}_0\|_{1,2} + \|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2 \right] = \\ &= \inf_{\tilde{V}_0 \in H^1} \left[\|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2 \right] \le \\ &\le \|\tilde{\mu} - Y'(p) \partial G(\dot{U}) : \tilde{V}\|_2 \end{aligned}$$

where the last inequality holds since we can choose $\tilde{V}_0 = -\tilde{V}$. Then we have

$$\sup_{q \in \mathscr{Q} \setminus \{0\}} \frac{\int_\Omega (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q\,dx}{\|q\|_Q} \ge \|\tilde{z}\|_{\mathscr{Z}/\tilde{\mathcal{N}}}.$$

Moreover if $q \in \mathscr{Q}$ is such that

$$b(p, \tilde{z}) = \int_\Omega (Y'(p) \partial G(\dot{U}) \circ \tilde{V} - \tilde{\mu}) q\,dx = 0 \quad \text{for all } \tilde{z} \in \mathscr{Z}$$

then choosing $\tilde{V} = 0$ and $\tilde{\mu} \in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on b. The check of the hypotheses on c is similar. In particular we have

$$\mathcal{N} = \left\{ z \in \mathcal{Z} \;\middle|\; c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \mathscr{Q} \right\} = \left\{ z \in \mathcal{Z} \;\middle|\; \mu = 0 \right\}.$$

Now we check the hypotheses on a. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We notice that with that choice of z and $\tilde{z}$ there exist $K > 0$ and $C > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leq K\|z\|_{\mathcal{Z}}$ and

$$a(z,\tilde{z}) \geq \delta\|Ev\|_2^2 + \epsilon\|\mathrm{grad}^2 v\|_2^2 + \phi\|\mathbb{V}\|_2^2 \geq C\|z\|_{\mathcal{Z}}^2 \geq \frac{C}{K^2}\|\tilde{z}\|_{\mathcal{Z}}^2.$$

This immediately implies that for all $z \in \mathcal{N}$ we have

$$\sup_{\tilde{z}\in\mathcal{Z}} \frac{a(z,\tilde{z})}{\|\tilde{z}\|_{\mathcal{Z}}} \geq \frac{C}{K}\|z\|_{\mathcal{Z}}$$

and if $\tilde{z} \in \tilde{\mathcal{N}}$ satisfy $a(\cdot, \tilde{z}) = 0$ then $\tilde{z} = 0$. Then also the hypotheses on a are verified. This completes the proof. □

By Proposition 7.24 in a neighborhood of every $(\dot{y}_0, p_0, y_0, t_0)$ with $\lambda_0 = 0$ we can explicit $\xi = (\dot{u}, \dot{U}, \lambda, p)$ from the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0.$$

We get

$$(\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \quad \text{and} \quad (\lambda, p) = \mathcal{H}(u, U, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1. Then we can apply Theorem B.17 to get locally in time a unique solution (u, U) of class C^2 to the Cauchy problem

$$\begin{cases} (\dot{u}, U) = \mathscr{G}(u, U, t) \\ u(0) = u_0 \\ U(0) = U_0. \end{cases}$$

Then we set $(\lambda, p) = \mathscr{H}(u, U, t)$ which is of class C^1. It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

Theorem 7.25. Let $(\dot{y}_0, p_0, y_0, t_0) \in \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T]$ with $\lambda_0 = 0$. Then there exists a unique C^1 solution (y, p) to Problem 7.4 defined locally around t_0.

Considering also Theorem 7.5 we obtain global uniqueness.

Corollary 7.26. Assume the hypotheses of Section 7.2.1 and of Section 7.2.6. Then there exists a unique $y \in C^1([0, T], \mathcal{Z})$ and $p \in C^1([0, T], \mathcal{Q})$ solution to Problem 7.4.

7.2.8. Bifurcation analysis.

Remark 7.27. This section is not definitive but contains only the ideas that we have collected in the hope of being able to reach a more complete result regarding the phenomenon of localization of deformation.

Some considerations. Since we have proved in Corollary 7.26 that the solution is unique no bifurcation can occur with nonzero viscosity. As the viscosity approaches to zero we expect that the evolution of the system reveals it extreme sensibility on the problem data when strain localization occur. To explain better the ideas we present the following example that describe what is the behavior that we expect. Consider a perfectly symmetrical specimen under perfectly symmetrical compressive loads. By uniqueness and symmetry in presence of viscosity the specimen will experience a perfectly symmetrical deformation pattern during the entire evolution (a distributed deformation or at most a symmetrical pattern of localized shear bands). However if the loads, the geometry or the material properties of the specimen are not perfectly symmetric then the system experience an unsymmetrical and completely different deformation pattern (for example an unsymmetrical pattern of shear bands). When viscosity decreases, the threshold beyond which perturbations from the condition of perfect symmetry trigger a localization with a significantly altered pattern is reduced. In some sense in the case of null viscosity the system is sensible also to vanishing perturbations. This suggest a lost of uniqueness at a stage of the loading program also in case of perfect symmetry.

We also expect that the bifurcation is actually associated with a jump in the time evolution. The reasons are the following.

(i) When passing to the limit $N \to \infty$ in the Rothe method in Section 7.2.4 we used the viscosity to get the a priori estimates on $\dot{y}^N$. We where not able to get them without viscosity, using only the $H^1((0, T))$ regularity of the load, as can be done for Von Mises plasticity [35]. This suggest that without viscosity the solution of our model cannot not lies in $H^1((0, T))$ but must lie in $BV((0, T))$.

(ii) Similar models of materials with pressure dependent yield strength exhibit jumps. It is the case of Cam–Clay model studied in [19] or the model studied analytically in [18] coupling the associative Drucker–Prager model with damage.
(iii) We expect that when the viscosity is small the evolution became very fast where the deformation occur. In the limit of null viscosity this became a jump.

Remark 7.28. My desire is to understand better the phenomenon of localization when the viscosity if small but nonzero trying to understand how the sensitivity of the system to imperfection increases with decreasing viscosity. Anyway, if it is useful in some way, I will try to study also the case of zero viscosity. But I would like to avoid the use of the notion of energetic solution. Indeed the choice made by the system at the jump point in the energetic solution approach is determined by a global minimizing approach that seems nonphysical to us. This view is confirmed in [46]. The approach that I could accept is that of balanced viscosity solutions where is the viscosity that drive the evolution at the jumps.

Study on the invertibility of the stiffness operator. In what follows are reported our efforts to understand better the phenomenon of localization. In particular we have confined ourself to the time discrete problem (based on the implicit Euler scheme). We computed the "stiffness operator" (the one that would be assembled in the numerical approximation of the model with Newton method). It seems to became singular during plastic flow in absence of viscosity suggesting that the system reaches indeed a bifurcation point (where the jump occur). This analysis should be intended only as a first trail and not as a final result. We hope that the analysis can be extended also to the the time continuous case.

We consider $\mathfrak{D}_\eta$, $\mathfrak{E}_\eta$ and $\mathfrak{F}_\eta$ as dependent continuously a viscosity parameter $\eta > 0$ and vanishing for $\eta = 0$. We call $\mathcal{F}_\eta$ the corresponding functional. We consider the discrete in time problem (Problem 7.6). We focus on a single time step and to simplify the notation we set $y = y^N_{n-1}$, $t = t^N_{n-1}$, $p = p^N_n$ and $\dot{y} = \tau_N^{-1}(y^N_n - y^N_{n-1})$ and $\tau = \tau_N$. Then $y^N_n = y + \tau\dot{y}$. The incremental problem reduces to find $\dot{y} \in \mathscr{Z}$ and $p \in \mathscr{Q}$ such that

$$\mathcal{F}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p) = \mathcal{F}_\eta(\dot{y}, p, y + \tau\dot{y}, t + \tau) = 0.$$

We compute the Fréchet differential $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ of $\mathcal{F}^{(y,t)}_{(\tau,\eta)}$ with respect to $(\dot{y}, p)$. It can be written as

$$\langle \mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)(z, q), (\tilde{z}, \tilde{q})\rangle = a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z, \tilde{z}) + b^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q, \tilde{z}) + c^{(\dot{y},p,y,t)}_{(\tau,\eta)}(\tilde{q}, z).$$

We have introduced the bilinear form

$$\begin{aligned}
a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z, \tilde{z}) = &\int_\Omega \mathfrak{D}_\eta Ev : E\tilde{v}dx + \int_\Omega \tau\mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right) dx + \\
&+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2\, v : \mathrm{grad}^2\, \tilde{v}dx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V} \circ \tilde{\mathbb{V}}dx + \\
&+ \int_\Omega Y(p)\partial^2 G(\dot{U})V : \tilde{V}dx \\
b^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q, \tilde{z}) = &\int_\Omega q(Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu})dx \\
c^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q, \tilde{z}) = &-\int_\Omega \tau\mu\tilde{q}dx.
\end{aligned}$$

We want to study if $\mathscr{K}^{(y,t)}_{(\tau,\eta)}$ is in $\mathrm{Iso}(\mathscr{Z} \times \mathscr{Q}, \mathscr{Z}' \times \mathscr{Q}')$. To this aim we can apply Theorem B.11 to the bilinear forms $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$, $b^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot{y},p,y,t)}_{(\tau,\eta)}$. Reasoning as in the proof of Proposition 7.24 we have that $b^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ satisfies the hypotheses. Moreover the sets $\mathscr{N}$ and $\tilde{\mathscr{N}}$ are defined exactly in the same way. We need to check the hypotheses on $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$. We reason again as in the proof of Proposition 7.24. For any $z = (v, V, 0) \in \mathscr{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathscr{N}}$ and conversely. We have that with that choice of z and $\tilde{z}$ there exist $K > 0$ such that $\|\tilde{z}\|_{\mathscr{X}} \leq K\|z\|_{\mathscr{X}}$. Then

$$\begin{aligned} a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) = &\int_\Omega \mathfrak{D}_\eta Ev : Ev dx + \int_\Omega \tau\mathfrak{C}(Ev - V) : \left(Ev - V - \frac{Y'(p)\partial G(\dot{U}) : V}{d} I\right) dx + \\ &+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2 v : \mathrm{grad}^2 v dx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V} \circ \mathbb{V} dx + \\ &+ \int_\Omega Y(p)\partial^2 G(\dot{U})V : V dx \\ \leq\ & \delta_\eta\|Ev\|_2^2 + \tau\gamma\|Ev - V\|_2^2 - 2\tau\kappa\|Ev - V\|_2\|V\|_2 + \\ &+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot{U})\|V\|_2^2. \end{aligned}$$

Here

$$\kappa = \frac{|\mathfrak{C}|\mathrm{Lip}(Y)\|\partial G\|_\infty}{2\sqrt{d}} \quad \text{and} \quad \sigma(p,\dot{U}) = \mathrm{essinf}_\Omega\left[Y(p)\ \mathrm{se}(\partial^2 G(\dot{U}))\right] \geq 0$$

where se denotes the smallest eigenvalue. Then by Young inequality for all $\theta_1 \in (0, \kappa^{-1}\gamma)$ and $\theta_2 \in (0,\infty)$ we have

$$\begin{aligned} a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) \leq\ & \delta_\eta\|Ev\|_2^2 + \tau(\gamma - \kappa\theta_1)\|Ev - V\|_2^2 - \tau\frac{\kappa}{\theta_1}\|V\|_2^2 + \\ &+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot{U})\|V\|_2^2. \\ \leq\ & \left[\delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2)\right]\|Ev\|_2^2 + (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + \\ &+ \left\{\phi_\eta + \sigma(p,\dot{U}) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right]\right\}\|V\|_2^2 + \\ &+ (\phi_\eta + \tau\beta)l^2\|\mathrm{grad}\, V\|_2^2. \end{aligned}$$

The hypotheses on $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ are satisfied if we can find $\theta = (\theta_1,\theta_2) \in (0,\kappa^{-1}\gamma) \times (0,\infty)$ such that

$$\begin{cases} \delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2) > 0 \\ \phi_\eta + \sigma(p,\dot{U}) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right] > 0. \end{cases}$$

It $\eta > 0$ then for each θ fixed there exists $\tau > 0$ sufficiently small such that these conditions are satisfied. This connected to the fact that the solution is unique as already established in Section 7.2.7. When $\eta = 0$ then $\delta_\eta = 0$ and $\phi_\eta = 0$. If $\sigma(p,\dot{U}) > 0$ then we can choose $\theta_2 < 1$ so that the first inequality is satisfied and for $\tau > 0$ sufficiently small also the second inequality is satisfied. If instead $\sigma(p,\dot{U}) = 0$ then the inequalities does not depend on τ and can be solved only if $\beta\gamma > \kappa^2$. Indeed from the first we have $\theta_2 < 1$ and from the second we get

$$\frac{1}{\theta_2} < 1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1}.$$

Then we can find $\theta_2 \in (0,1)$ if and only if

$$1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1} > 1 \quad \text{or equivalently} \quad \beta - \frac{\kappa}{\theta_1} > 0.$$

Then we can find $\theta_1 \in (0,1)$ if and only if $\beta\gamma > \kappa^2$. This means that hardening must be strong enough compared to the sensitivity of the yield strength on pressure (notice that κ is proportional to $\mathrm{Lip}(Y)$).

Remark 7.29. In the case of Von Mises plasticity the stiffness operator is an isomorphism for any positive hardening. This due to the fact that the additional term that comes from the dependence of the yield strength on the plastic pressure is absent.

Intuition about the localization process. We now assume that $\eta = 0$ and that G is a smooth and convex dissipation potential that satisfy $G(\dot{U}) = |\dot{U}| + c$ for all $\dot{U} \in \mathrm{DevSym}$ with $|\dot{U}| > R$ where $c \in \mathbb{R}$ and $R > 0$ are two constants. This gives a plastic force that is independent on the rate and equal to the yield strength when the rate is sufficiently hight. So it is a regularization of the rate independent dissipation potential that preserve the existence of a yield limit. We notice that

$$\partial^2 G(\dot{U}) = \frac{|\dot{U}|^2 I - \dot{U} \otimes \dot{U}}{|\dot{U}|^2} \quad \text{for } |\dot{U}| > R$$

so that $\mathrm{se}(\partial^2 G(\dot{U})) = 0$ there. On the contrary $\mathrm{se}(\partial^2 G(\dot{U}))$ must be very hight for $\dot{U}$ near the origin if G is a very close approximation of the rate independent dissipation potential $|\cdot|$. We now consider a loading process. When the load is still small the plastic force is under the yield limit and $\sigma(p, \dot{U})$ is hight. By the analysis we have just performed this ensure that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible and this suggest that the incremental problem is uniquely solved. Moreover we expect that the deformation rate it is not localized in some region but distributed all over the body. As the load increases $\mathrm{se}(\partial^2 G(\dot{U}))$ decreases (uniformly in all the body). Then $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible only for small time steps τ. Assuming that the hardening is sufficiently small, we can conclude that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ became singular (or is nonsingular only for very small time increments so that the evolution stagnates). This is the signal that we have reached the bifurcation point. Now the system to restore the invertibility of $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ and overcome the singularity point makes $\dot{U}$ small everywhere except that in a small shear band. This cause an elastic unloading elsewhere. Then out of the shear band $\mathrm{se}(\partial^2 G(\dot{U}))$ becomes big again. This makes $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ coercive because out of the shear band the plastic strain is held back by the coercivity of $\partial^2 G(\dot{U})$. In the shear band the plastic strain is held back by the surrounding regions through the plastic strain gradient.

7.3. Numerical approximation

7.3.1. Formulation.

7.3.2. Validation. To validate the numerical scheme we solve the following boundary value problem that admits an analytical solution. Consider the domain $\Omega = (0, L) \times (-H, H)$ and for all $t \in [0, T]$ impose the boundary conditions $u_2(x, t) = \epsilon(t) \operatorname{sign} x_2$ for $x_2 \in \{-H, H\}$ and $u_1(x, t) = 0$ for $x_1 = 0$. Here $\epsilon(t) = \frac{t}{T}\epsilon_0$ where $\epsilon_0 > 0$. Assume that no external volume forces or external tractions are imposed. We look for uniform solutions of the form

$$u(x, t) = \begin{pmatrix} \zeta(t)x_1 \\ \epsilon(t)x_2 \end{pmatrix}, \quad U_p(x, t) = \begin{pmatrix} -\gamma(t) & 0 \\ 0 & \gamma(t) \end{pmatrix} \quad \text{and} \quad p(x, t) = p(t).$$

The unknowns are ζ, γ, p from $[0, T]$ to $\mathbb{R}$. We easily compute

$$Eu = \begin{pmatrix} \zeta & 0 \\ 0 & \epsilon \end{pmatrix}, \quad \text{and} \quad E_e = Eu - U_p = \begin{pmatrix} \zeta + \gamma & 0 \\ 0 & \epsilon - \gamma \end{pmatrix}.$$

Assuming an isotropic elastic response, then

$$\begin{aligned} T_e = 2\mu E_e + \lambda \operatorname{tr} E_e I &= \begin{pmatrix} 2\mu(\zeta + \gamma) + \lambda(\zeta + \epsilon) & 0 \\ 0 & 2\mu(\epsilon - \gamma) + \lambda(\zeta + \epsilon) \end{pmatrix} = \\ &= \begin{pmatrix} (2\mu + \lambda)\zeta + \lambda\epsilon + 2\mu\gamma & 0 \\ 0 & \lambda\zeta + (2\mu + \lambda)\epsilon - 2\mu\gamma \end{pmatrix}. \end{aligned}$$

The condition that the lateral traction vanish, impose that $(2\mu + \lambda)\zeta + \lambda\epsilon + 2\mu\gamma = 0$ so that

$$\zeta = -\frac{\lambda\epsilon + 2\mu\gamma}{2\mu + \lambda}.$$

The micropressure can be immediately computed as

$$p = -\frac{1}{2} \operatorname{tr} T_e = -(\lambda + \mu)(\zeta + \epsilon) = -E(\epsilon - \gamma),$$

where $E = \frac{2\mu(\lambda+\mu)}{2\mu+\lambda}$. Now we consider the microforce balance. Since the solution is uniform the microstress vanish and we get $-\operatorname{dev} T_e + T_p = 0$. We assume that

$$Y(p) = \begin{cases} 0 & \text{if } p \le -\frac{Y_0}{\alpha} \\ Y_0 + \alpha p & \text{if } p > -\frac{Y_0}{\alpha} \end{cases}$$

Now

$$\operatorname{dev} T_e = 2\mu \operatorname{dev} E_e = \begin{pmatrix} \mu(\zeta - \epsilon + 2\gamma) & 0 \\ 0 & \mu(\epsilon - \zeta - 2\gamma) \end{pmatrix} = E(\epsilon - \gamma) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Before the yield point is reached, $\gamma = 0$ and the yield point is reached when $|\operatorname{dev} T_e| = Y(p)$. This can be written explicitly as $\sqrt{2}E\epsilon = Y_0 - \alpha E\epsilon$, so that at yield point

$$\epsilon = \frac{Y_0}{(\sqrt{2} + \alpha)E}.$$

After that the yield point is reached $\dot{\gamma} > 0$ and the microforce balance reads

$$-\operatorname{dev} T_e + \mathfrak{B} U_p + \frac{1}{\sqrt{2}} Y(p) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = 0,$$

or equivalently

$$-\sqrt{2}E\epsilon + \sqrt{2}E\gamma + \sqrt{2}\mathfrak{B}\gamma + Y_0 - \alpha E\epsilon + \alpha E\gamma = 0.$$

This allows to compute γ as

$$\gamma = \frac{(\sqrt{2} + \alpha)E\epsilon - Y_0}{(\sqrt{2} + \alpha)E + \sqrt{2}\mathfrak{B}}.$$

This completes the computation of the solution.

CHAPTER 8

Large deformation model

This chapter is not updated and incomplete. If I have time I will try to get an existence theorem also for the large strain model but I give the priority to understand better the localization of deformation in the small strain model.

8.1. Formulation

8.1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic deformation is a map $\chi \colon \Omega \to \mathbb{R}^d$ satisfying $\det \operatorname{grad} \chi > 0$. We set $F = \operatorname{grad} \chi$, $E = (F^T F - I)/2$ and $M = \operatorname{grad}^2 \chi$. We subdivide $\partial\Omega$ in two disjoint parts, Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $\chi = \bar{\chi}$. The distortion of the microscopic structure is described by the plastic distortion $F_p \colon \Omega \to \mathrm{Lin}^+$. We regard F_p as the elastically relaxed distortion so we define the elastic distortion as $F_e = F F_p^{-1}$. The volumetric distortion of the microscopic structure is described by $J_p = \det F_p$ while the shear distortion by $U_p = J_p^{-1} F_p$. We impose the kinematical constrain $J_p = 1$ and to this aim we introduce the Lagrange multiplier p with the meaning of a plastic pressure. The kinematical constrain can be written also as

$$\int_\Omega (J_p - 1)\tilde{q}\, dx = 0 \quad \text{for all } \tilde{q} \colon \Omega \to \mathbb{R}.$$

8.1.2. Statics. We denote with $\tilde{v}$ the virtual rate of u.
Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi \colon \Omega \to \mathbb{R}^d$. We set $v = \dot{\chi}$ and $F = \operatorname{Grad} \chi$ and $L = \dot{F} F^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N. The plastic distortion is a map $F_p \colon \Omega \to \mathrm{Lin}^+$. We set $L_p = \dot{F}_p F_p^{-1}$ and we assume that $L_p = D_p \in \mathrm{Sym}$. We set $U_p = \operatorname{dev} D_p$ and $M_p = \operatorname{Grad} U_p \cdot F_p^{-1}$. The elastic distortion is $F_e = F F_p^{-1}$ and we set $L_e = \dot{F}_e F_e^{-1}$.

Lemma 8.1. It holds $L = L_e + F_e D_p F_e^{-1}$.

Proof. We have $L = \dot{F} F^{-1} = (\dot{F}_e F_p + F_e \dot{F}_p) F_p^{-1} F_e^{-1} = \dot{F}_e F_e^{-1} + F_e \dot{F}_p F_p^{-1} F_e^{-1} = L_e + F_e D_p F_e^{-1}$. □

We add the kinematical constrain $\operatorname{tr} D_p = 0$. To impose this constrain we consider a Lagrange multiplier $p \colon \Omega \to \mathbb{R}$. The constrain can be written in weak form as

$$-\int_\Omega \operatorname{tr} L_p \tilde{q}\, dX = 0 \quad \text{for all } \tilde{q} \colon \Omega \to \mathbb{R}.$$

8.1.3. Statics. We choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_\Omega T : \tilde{L}_e\, dX + \int_\Omega \left(T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \,\vdots\, \tilde{M}_p \right) dX.$$

Here T is the macrostress, T_p is the microstress which can be assumed deviatoric and K_p is the microhyperstress. We choose an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_\Omega f \cdot \tilde{v}\, dX + \int_{\Gamma_N} h \cdot \tilde{v}\, dA + \int_{\Gamma_D} r \cdot \tilde{v}\, dA.$$

Definition 8.2 (Principle of Virtual Powers). Equilibrium holds when $\tilde{\mathscr{W}} = \tilde{\mathscr{P}}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 8.1 and we write the internal virtual power as

$$\tilde{\mathcal{W}} = \int_\Omega T F^{-T} : \operatorname{Grad} \tilde{v}\, dX + \int_\Omega \left(-T_e : \tilde{L}_p + T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \,\vdots\, \tilde{M}_p \right) dX$$

where $T_e = F_e^T T F_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned}\tilde{\mathcal{W}} = -\int_\Omega \operatorname{Div}(TF^{-T}) : \operatorname{Grad}\tilde{v}\,dX + \int_{\partial\Omega}(TF^{-T})N\cdot\tilde{v}\,dA +\\ + \int_\Omega \left\{\left[T_p - \operatorname{dev}T_e - \operatorname{Div}\left(K_p\cdot F_p^{-T}\right)\right] : \tilde{U}_p - \left[\frac{1}{d}\operatorname{tr}T_e + p\right]\operatorname{tr}\tilde{L}_p\right\}dX +\\ + \int_{\partial\Omega}\left(K_p\cdot F_p^{-T}N\right) : \tilde{U}_p\,dA.\end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned}&\operatorname{Div}(TF^{-T}) + f = 0 \quad \text{in } \Omega\\ &(TF^{-T})N = r \quad \text{in } \Gamma_D\\ &(TF^{-T})N = h \quad \text{in } \Gamma_N\\ &T_p - \operatorname{dev}T_e - \operatorname{Div}\left(K_p\cdot F_p^{-T}\right) = 0 \quad \text{in } \Omega\\ &\frac{1}{d}\operatorname{tr}T_e + p = 0 \quad \text{in } \Omega\\ &K_p\cdot F_p^{-T}N = 0 \quad \text{in } \partial\Omega.\end{aligned}$$

Collection of useful results

APPENDIX A

Notation

A.1. Tensors

$\mathrm{Lin} = \mathbb{R}^{d\times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices
DevPla	defined as $\mathrm{DevSym} \times (\mathrm{DevSym} \otimes \mathbb{R})$
Lin^1	$d \times d$ matrices with unit determinant

APPENDIX B

Tools from mathematical analysis

In this chapter we collect some tools of linear and nonlinear analysis used in the main body of this thesis.

B.1. Some classical results

B.1.1. A general form of the Gronwall inequality.

Proposition B.1. (Gronwall inequality) Let $z \in L^\infty((0,T))$. Let $C \in \mathbb{R}$ and α, β in $L^1((0,T),[0,\infty))$ such that

$$z(t) \leq C + \int_0^t \big(\alpha(s)y(s) + \beta(s)\big)\, ds \quad \text{for a.e. } t \in (0,T).$$

Then

$$z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} ds \right) \quad \text{for a.e. } t \in (0,T).$$

Proof. Let

$$Z(t) = C + \int_0^t \big(\alpha(s)z(s) + \beta(s)\big)\, ds$$

so that $z(t) \leq Z(t)$. Clearly $Z \in \mathrm{AC}([0,T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \leq \alpha(t)Z(t) + \beta(t) \quad \text{for a.e. } t \in (0,T).$$

Multiplying by $e^{-\int_0^t \alpha(s)ds}$ one gets

$$\frac{d}{dt}\left(Z(t)e^{-\int_0^t \alpha(s)ds} \right) \leq \beta(t)e^{-\int_0^t \alpha(s)ds}$$

and integrating on $(0,t)$

$$Z(t)e^{-\int_0^t \alpha(s)ds} \leq C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} ds$$

Then

$$z(t) \leq Z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} \right) \quad \text{for a.e. } t \in (0,T).$$

This is exactly the thesis. □

The same inequality has also a discrete version that can be proved similarly.

Proposition B.2 (discrete Gronwall inequality). Let $z_n \in \mathbb{R}$, $a_n \geq 0$, $b_n \geq 0$ and $C \in \mathbb{R}$ and $\tau > 0$ such that

$$z_k \leq C + \tau \sum_{j=1}^{k-1} (a_j z_j + b_j) \quad \text{for } k \geq 0.$$

Then

$$z_k \leq \left(C + \tau \sum_{j=1}^{k-1} b_j \right) e^{\tau \sum_{j=1}^{k-1} a_j} \quad \text{for } k \geq 0.$$

B.2. Linear functional analysis

B.2.1. Other.

Proposition B.3. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz. Let $p \in [1, \infty]$ and let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \to u$ in $L^p(\Omega)$. Moreover if $p > d$ then $u_k \to u$ in $C^{0,\alpha}(\bar{\Omega})$ where $\alpha = 1 - \frac{d}{p}$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u. Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$.

The result when $p > d$ follows from Morrey's theorem and a similar argument. □

B.2.2. Bochner spaces.

Definition B.4. Let V be a Banach space. We say that $u\colon [0,T] \to V$ is absolutely continuous and we write $u \in AC([0,T], V)$ if there exists $m \in L^1((0,T),(0,\infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r)dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u\colon [0,T] \to V$ and $0 \leq a < b \leq T$ we define the total variation on $[a,b]$

$$\mathrm{Var}(u,[a,b]) = \sup\left\{ \sum_{k=1}^N \|u(t_k) - u(t_{k-1})\| \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that u has bounded variation and we write $u \in BV([0,T],V)$ if $\mathrm{Var}(u,[0,T]) < \infty$.

Proposition B.5. Let V be a Banach space. Then $W^{1,1}((0,T),V) \subset AC([0,T],V)$. If V is separable and reflexive the equality holds.

Proof. Let $u \in W^{1,1}((0,T),V)$. Then

$$\|u(t) - u(s)\| = \left\| \int_s^t \dot{u}(r)dr \right\| \leq \int_s^t \|\dot{u}(r)\|dr.$$

This proves that $u \in AC([0,T],V)$ with $m = \|\dot{u}\|$.

Now suppose that V is reflexive. We have that

$$\left\| \frac{u(t+h) - u(t)}{h} \right\| \leq \frac{1}{h} \int_t^{t+h} m(r)dr.$$

By the Lebesgue differentiation theorem the left hand side is bounded as $h \to 0$ for a.e. $t \in (0,T)$. Then for each $h_n \to 0$, up to extracting a subsequence,

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{for some } v \in V.$$

We prove that v is common to each sequence. Since V^* is separable and reflexive then there exists ξ_i dense in V^*. Now for each i the map $t \mapsto \langle \xi_i, u(t) \rangle$ is $AC([0,T])$. Then for a.e. $t \in (0,T)$

$$\lim_{h \to 0} \left\langle \xi_i, \frac{u(t+h) - u(t)}{h} \right\rangle \quad \text{exists for all } i. \tag{B.1}$$

Then if $h_n \to 0$ and $k_n \to 0$ are such that

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{and} \quad \frac{u(t+k_n) - u(t)}{k_n} \rightharpoonup w$$

from (B.1) it follows that $\langle \xi_i, v - w \rangle = 0$ for all i. This implies $v = w$. Up to now we have proved that there exists $v\colon (0,T) \to V$ such that

$$\frac{u(t+h) - u(t)}{h} \rightharpoonup v(t) \quad \text{for a.e. } t \in (0,T).$$

Now one should prove that this convergence is actually strong. This will show automatically that $u \in W^{1,1}((0,T),V)$. □

Theorem B.6 (Aubin–Lions–Simon lemma). Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is continuous. Let p and q in $[1, \infty]$. Then the space
$$\{u \in L^p((0,T), V) \mid \dot{u} \in L^q((0,T), E)\}$$
is compactly embedded in

(i) $L^p((0,T), H)$ if $p < \infty$
(ii) $C([0,T], H)$ if $p = \infty$ and $q > 1$.

Proof. See [11, Theorem II.5.16]. □

B.2.3. Generalized mixed elliptic problems. We start with the following theorem.

Theorem B.7 (Babuŝka). Let V and Q be reflexive Banach spaces. Let $b \colon Q \times V \to \mathbb{R}$ be bilinear form. Let $Z = \left\{z \in V \middle| b(q,z) = 0 \text{ for all } q \in Q\right\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that
$$b(q,v) \leq M\|q\|_Q\|v\|_V \quad \text{for all } (q,v) \in Q \times V,$$
Assume also that there exists $\beta > 0$ such that
$$\sup_{q \in Q \setminus \{0\}} \frac{b(q,w)}{\|q\|_Q} \geq \beta\|w\|_W$$
for all $w \in W$ and that if $q \in Q$ satisfies
$$b(q,v) = 0 \quad \text{for all } v \in V$$
then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that
$$b(q,w) = \langle f, q\rangle \quad \text{for all } q \in Q$$
and
$$\|w\|_W \leq \frac{1}{\beta}\|f\|_{Q'}.$$

Proof. *Step 1*. Consider the linear operator $B \colon W \to Q'$ defined by
$$\langle Bw, q\rangle = b(w,q).$$
We notice that
$$|\langle Bw, q\rangle| = \inf_{z \in Z}|b(w+z,q)| \leq M\|q\|_Q \inf_{z \in Z}\|w+z\|_V = M\|q\|_Q\|w\|_W$$
so that $B \in \mathscr{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have
$$\beta\|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = 0$$
so that $w = 0$.

Step 3. We prove that range(B) is closed. Indeed let $f_n = Bw_n$ be such that $f_n \to f$ in Q. Then f_n is a Cauchy sequence. Now
$$\beta\|w_n - w_m\| \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle B(w_n - w_m), q\rangle}{\|q\|_Q} \leq \|f_n - f_m\|_{Q'}.$$
Then also w_n is a Cauchy sequence in W and $w_n \to w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in$ range(B).

Step 4. We prove that range$(B) = Q'$. Indeed let $(B^T) = \left\{q \in Q \middle| b(q,v) = 0 \text{ for all } v \in V\right\} = \{0\}$ by hypotheses. On the other hand $(B^T) = \text{range}(B)^\perp$. Since Q is reflexive range$(B) = (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q'. Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover
$$\beta\|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = \sup_{q \in Q \setminus \{0\}} \frac{\langle f, q\rangle}{\|q\|_Q} \leq \|f\|_{Q'}.$$
This completes the proof. □

Lemma B.8. Assume that the hypotheses of Theorem B.7 holds. Then

$$\sup_{w\in W\setminus\{0\}} \frac{b(q,w)}{\|w\|_W} \geq \beta\|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q,w) = \langle f,q\rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta\|w\|_W$. Now

$$\|q\|_Q = \sup_{f\in Q'\setminus\{0\}} \frac{\langle f,q\rangle}{\|f\|_{Q'}} = \sup_{f\in Q'\setminus\{0\}} \frac{b(q,w_f)}{\|f\|_{Q'}} \leq \sup_{f\in Q'\setminus\{0\}} \frac{b(q,w_f)}{\beta\|w\|_W} \leq \sup_{w\in W\setminus\{0\}} \frac{b(q,w)}{\beta\|w\|_W}.$$

This completes the proof. □

Then we get immediately the following proposition.

Corollary B.9. Assume the hypotheses of Theorem B.7. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p,w) = \langle g,w\rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta}\|g\|_{W'}.$$

Proof. Apply first Lemma B.8 and then Theorem B.7 with the role of W and Q swapped. □

Remark B.10. Assume that $g \in V'$ is such that $\langle g,z\rangle = 0$ for all $z \in Z$. Then we can define g as an element of W'. Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g,w\rangle = \langle g,v\rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_W = \sup_{w\in W\setminus\{0\}} \frac{\langle g,w\rangle}{\|w\|_W} = \sup_{v\in V\setminus\{0\}} \frac{\langle g,v\rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary B.9 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p,v) = \langle g,v\rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta}\|g\|_{V'}.$$

The following theorem is the main result of this Section. The result is proved in [54] in the Hilbert spaces setting and we have generalized it for Banach spaces.

Theorem B.11. Let V, $\tilde{V}$, Q and $\tilde{Q}$ be reflexive Banach spaces. Let $b\colon Q\times\tilde{V} \to \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \left\{\tilde{z} \in \tilde{V} \middle| b(q,\tilde{z}) = 0 \text{ for all } q \in Q\right\}$ and define $\tilde{W} = \tilde{V}/\tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q,\tilde{v}) \leq M\|q\|_Q\|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q\in Q\setminus\{0\}} \frac{b(q,\tilde{w})}{\|q\|_Q} \geq \beta\|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q,\tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c\colon \tilde{Q}\times V \to \mathbb{R}$ be a bilinear form. Let $Z = \left\{z \in V \middle| b(\tilde{q},z) = 0 \text{ for all } \tilde{q} \in \tilde{Q}\right\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q},v) \leq N\|\tilde{q}\|_{\tilde{Q}}\|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q}\in\tilde{Q}\setminus\{0\}} \frac{c(\tilde{q},w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma\|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a\colon V \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \leq L\|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V} \setminus \{0\}} \frac{a(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geq \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$\begin{aligned} a(u, \tilde{v}) + b(p, \tilde{v}) &= \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V} \\ c(\tilde{q}, u) &= \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}. \end{aligned}$$

Moreover

$$\begin{aligned} \|u\|_V &\leq \frac{1}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'} \\ \|p\|_Q &\leq \frac{1}{\beta}\left(1 + \frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}. \end{aligned}$$

Proof. By Theorem B.7 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leq \frac{1}{\gamma}\|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w, \tilde{v}) \leq L\|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w, \cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f, \cdot \rangle - a(w, \cdot) \in \tilde{Z}'$. Then by Theorem B.7 the equation

$$a(z, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leq \frac{1}{\alpha}\left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma}\|g\|_{\tilde{Q}'}\right).$$

Now define $u = w + z \in V$ and we notice that

$$a(u, \tilde{z}) = \langle f, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \tag{B.2}$$

and

$$\|u\|_V \leq \frac{1}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}.$$

Now we notice by (B.2) we have

$$\langle f, \tilde{z} \rangle - a(u, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) - a(z, \tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark B.10 we have $\langle f, \cdot \rangle - a(u, \cdot) \in \tilde{W}'$ and the equation

$$b(p, \tilde{v}) = \langle f, \tilde{v} \rangle - a(u, \tilde{v}) \quad \text{for all } v \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leq \frac{1}{\beta}\left(\|f\|_{\tilde{V}'} + L\|u\|_V\right) \leq \frac{1}{\beta}\left(1 + \frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

This concludes the proof. □

B.3. Nonlinear functional analysis

B.3.1. Other.

Lemma B.12. Let X and Y be Banach spaces. Let U an open subset of X and $F\colon U \to Y'$. Let $x \in U$. Assume that there exists $A \in \mathscr{L}(X, Y')$ and for all $v \in X$ there exists $\omega\colon \mathbb{R} \to [0, \infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left|\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle\right| \leq \omega(\epsilon)\|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A.

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega\colon \mathbb{R} \to [0, \infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} = \sup_{y \in Y \setminus \{0\}} \frac{\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle}{\|y\|_Y} \leq \omega(\epsilon).$$

Then

$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} \to 0$$

as $\epsilon \to 0$. This is exactly the thesis. □

Lemma B.13. Let X and Y be two Banach spaces. Let A_n and A in $\mathscr{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y\rangle| \leq \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \to 0$. Then $A_n \to A$ in $\mathscr{L}(X, Y')$.

Proof. We have

$$\|A_n - A\|_{\mathscr{L}(X,Y')} = \sup_{x \in X \setminus \{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{\langle A_n x - Ax, y\rangle}{\|x\|_X \|y\|_Y} \leq \omega_n \to 0.$$

This concludes the proof. □

Proposition B.14. Let V a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then f is Lipschitz continuous on K.

Proof. By contradiction let u_n and v_n two sequences in K such that

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \geq n.$$

By compactness of K up to extracting subsequences we can assume that $u_n \to u$ and $v_n \to v$. If $u \neq v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \to \frac{|F(u) - F(v)|}{\|u - v\|} < \infty$$

which is a contradiction. If $u = v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} = \frac{|\langle DF(u), u_n - v_n\rangle + o(\|u_n - v_n\|)|}{\|u_n - v_n\|} \leq \|DF(u)\| + o(1) \to \|DF(u)\| < \infty$$

which is a contradiction. □

B.3.2. Implicit function theorem.

Theorem B.15. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \mathrm{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying

$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$

Moreover if F is of class C^r with $r \geq 2$ also χ is of class C^r.

Proof. See [55]. □

B.3.3. Fixed point theorems.

Theorem B.16 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F\colon C \to C$ continuous and with compact range. Then F admits a fixed point.

Proof. See [55]. □

B.3.4. Local existence theorem for Cauchy problems in Banach spaces.

Theorem B.17. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

Proof. Since G is of class C^1 there there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V. Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathscr{L}(X,X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L. Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathscr{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathscr{X}} = \sup_{t \in [0,T]} \|x(t)\|_X$. Since $\mathscr{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T \colon B \to C^0([0, T], B)$$

defined by

$$T(x)(t) = x_0 + \int_0^t G(x(s), s) ds.$$

We notice that $\|T(x) - x_0\|_X \leq TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T \colon B \to B$. Moreover for any x_1 and x_2 in X we have

$$\|T(x_2) - T(x_1)\|_{\mathscr{X}} \leq \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leq TL\|x_2 - x_1\|_{\mathscr{X}}.$$

Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2. □

B.3.5. Browder–Minty theorem. The results of this section are taken from [60] and [62].

Let X be a Banach space and $T \colon X \to X^*$ be a nonlinear operator and $f \in X^*$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

Definition B.18. Let X be a Banach space and let $T \colon X \to X^*$. We say that

(i) T is bounded if maps bounded sets of X into bounded sets of X^*,

(ii) T is coercive if

$$\frac{\langle T(x), x \rangle}{\|x\|} \to \infty \quad \text{as } \|x\| \to \infty$$

and that

(iii) T is monotone if

$$\langle T(x) - T(y), x - y \rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

Lemma B.19. Let $T \colon X \to X^*$ be continuous and $f \in X^*$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x \rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to te equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v \rangle \geq 0.$$

Letting $t \to 0$ we have $\langle T(x) - f, v \rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x \rangle = \langle T(y) - T(x), y - x \rangle \geq 0.$$

Then x is a solution to the variational inequality. □

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition B.20. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has at least one solution.

Proof. Without loss of generality we cn assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x)\cdot x}{|x|} = \frac{T(x)\cdot x}{|x|} - \frac{f\cdot x}{|x|} \geq \frac{T(x)\cdot x}{|x|} - |f|.$$

Now define $G\colon \mathbb{R}^n \to \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x)\cdot x = |x|^2 - T(x)\cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x)\cdot x = T(x)\cdot x > 0$ for all $x \in \mathbb{R}\setminus B(0,r)$. We define $\Pi\colon \mathbb{R}^n \to C$ with $C = \bar{B}(0,r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r\frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H\colon C \to C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \mathring{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x\cdot x = H(x)\cdot x = \frac{r}{|G(x)|}G(x)\cdot x < \frac{r}{|G(x)|}|x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x)\cdot x$ for $|x| \geq r$. This is a contradiction. □

We are now in position to prove the Browder–Minty theorem.

Theorem B.21. Let X be a reflexive and separable Banach space. Let $T\colon X \to X^*$ be bounded, continuous, coercive and monotone. Then for every $f \in X^*$ there exists a solution $x \in X$ to the equation $T(x) = f$.

Proof. As anticipated the proof relies on the Galerkin method.
Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X. Let

$$X_n = \operatorname{span}\{y_m\}_{m=1}^n.$$

We define $f_n \in X_n^*$ by $\langle f_n, x\rangle = \langle f, x\rangle$ for all $x \in X_n$. We also define $T_n\colon X_n \to X_n^*$ by $\langle T_n(x), y\rangle = \langle T(x), y\rangle$ for all x and y in X_n.
Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X_n^*} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x), y\rangle}{\|y\|_{X_n}} \leq \sup_{y\in X\setminus\{0\}} \frac{\langle T(x), y\rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows form the boundedness of T. Moreover let $x_k \to x$ in X_n. Then

$$\begin{aligned}\|T_n(x_k) - T_n(x)\|_{X_n^*} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x_k) - T_n(x), y\rangle}{\|y\|_{X_n}} &\leq \sup_{y\in X\setminus\{0\}} \frac{\langle T(x_k) - T(x), y\rangle}{\|y\|_X} = \\ &= \|T(x_k) - T(x)\|_{X^*} \to 0\end{aligned}$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x\rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x\rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T. By Proposition B.20 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.
Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n\rangle}{\|x_n\|_X} \leq \frac{\langle T_n(x_n), x_n\rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n\rangle}{\|x_n\|_X} = \frac{\langle f, x_n\rangle}{\|x_n\|_X} \leq \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X^*. Then up to a subsequence

$$\begin{aligned} x_n &\rightharpoonup x \quad \text{in } X \\ T(x_n) &\overset{*}{\rightharpoonup} g \quad \text{in } X^* \end{aligned}$$

for some $x \in X$ and $g \in X^*$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m \rangle = \lim_{n\to\infty} \langle T(x_n) - f, y_m \rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n \rangle = \langle T_n(x_n), x_n \rangle = \langle f_n, x_n \rangle = \langle f, x_n \rangle \to \langle g, x \rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \geq \langle T(y) - T(x_n), y - x_n \rangle \to \langle T(y) - g, y - x \rangle = \langle T(y) - f, y - x \rangle.$$

Applying Lemma B.19 we get the thesis. □

Actually the same existence result can be obtained with a weaken condition then monotonicity.

Definition B.22. Let X be a Banach space and let $T \colon X \to X^*$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k\to\infty} \langle T(x_k), x_k - x \rangle \leq 0$$

then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$.

If T is pseudomonotone it posses also a nice continuity property.

Proposition B.23. Let $T \colon X \to X^*$ be pseudomonotone. Then T is demicontinuous, i.e., for all $x_k \to x$ in X it holds that $T(x_k) \overset{*}{\rightharpoonup} T(x)$ in X^*.

Proof. Let $x_k \to x$ in X. Then $\{x_k\}$ is bounded in X and since T is bounded also $\{T(x_k)\}$ is bounded in X^*. Then, for all subsequence of $\{x_k\}$ there exists a further subsequence $\{x_{k_j}\}$ such that $T(x_{k_j}) \overset{*}{\rightharpoonup} g$ for some $g \in X^*$. Then

$$\liminf_{j\to\infty} \langle T(x_{k_j}), x_{k_j} - x \rangle = \langle g, x - x \rangle = 0.$$

Then, since T is pseudomonotone

$$\langle g, x - y \rangle = \limsup_{j\to\infty} \langle T(x_{k_j}), x_{k_j} - y \rangle \leq \langle T(x), x - y \rangle \quad \text{for all } y \in X.$$

By arbitrariness of y, it follows that $T(x) = g$. We conclude that the whole sequence $\{T(x_k)\}$ converges weakly-star in X^* to $T(x)$. □

The following generalization of Minty–Brower theorem holds.

Theorem B.24. Let X be a separable and reflexive Banach space. Let $T \colon X \to X^*$ coercive and pseudomonotone. Then for every $f \in X^*$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is similar to that of Theorem B.21 up to Step 4. The monotonicity is never used here and the demicontinuity (implied by pseudomonotonicity by Proposition B.23) is enough to prove that T_k are continuous and to handle the other limit passages. Then we need only to adapt Step 5.

We know that $x_k \rightharpoonup x$ in X, that $T(x_k) \overset{*}{\rightharpoonup} f$ in X^* and $\langle T(x_k), x_k \rangle \to \langle f, x \rangle$. Then in particular

$$\langle T(x_k), x_k - x \rangle = \langle T(x_k), x_k \rangle - \langle T(x_k), x \rangle \to \langle f, x \rangle - \langle f, x \rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. But

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y \rangle = \langle f, x - y \rangle.$$

Then

$$\langle f, x - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. □

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition B.25. Let X be a Banach space. Let $T \colon X \to X^*$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with
$$\limsup_{k\to\infty}\langle T(x_k), x_k - x\rangle \le 0.$$
Since T is monotone then
$$\langle T(x_k), x_k - x\rangle \ge \langle T(x), x_k - x\rangle \to 0.$$
This implies also
$$\liminf_{k\to\infty}\langle T(x_k), x_k - x\rangle \ge 0.$$
Then
$$\lim_{k\to\infty}\langle T(x_k), x_k - x\rangle = 0.$$
Now let $y \in X$ and set $y_\epsilon = (1-\epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity
$$0 \le \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon\rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x-y) + x_k - x\rangle.$$
Rearranging the terms we have
$$\epsilon\langle T(x_k), x - y\rangle \ge \langle T(y_\epsilon), x_k - x\rangle - \langle T(x_k), x_k - x\rangle + \epsilon\langle T(y_\epsilon), x - y\rangle.$$
Passing to le limit $k \to \infty$ with $\epsilon > 0$ fixed
$$\epsilon\liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge \epsilon\langle T(y_\epsilon), x - y\rangle.$$
Dividing by ϵ and passing to the limit $\epsilon \to 0^+$ we obtain using the continuity of T that
$$\liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge \langle T(x), x - y\rangle.$$
Eventually
$$\liminf_{k\to\infty}\langle T(x_k), x_k - y\rangle = \lim_{k\to\infty}\langle T(x_k), x_k - x\rangle + \liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge 0.$$
This proves that T is pseudomonotone. □

The following result is useful.

Proposition B.26. Let X be a Banach space. Let T_1 and T_2 from X into X^* be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with
$$\limsup_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - x\rangle \le 0.$$
We claim that
$$\limsup_{k\to\infty}\langle T_i(x_k), x_k - x\rangle \le 0 \tag{B.3}$$
for both $i = 1$ and $i = 2$. By contradiction
$$\limsup_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$
Up to taking a subsequence we can assume that
$$\lim_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$
Then
$$\limsup_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \le -\epsilon < 0. \tag{B.4}$$
Then since T_1 is pseudomonotone
$$\liminf_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \ge 0$$
but this contradicts (B.4). Then (B.3) holds. Since T_i is pseudomonotone then for all $y \in X$ we have
$$\liminf_{k\to\infty}\langle T_i(x_k), x_k - y\rangle \ge \langle T_i(x), x - y\rangle.$$
It follows that
$$\liminf_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - y\rangle \ge \langle T_1(x) + T_2(x), x - y\rangle.$$
This means that $T_1 + T_2$ is pseudomonotone. □

The Minty–Browder theorem can be generalized also to the case in which T is a set valued map, that is $T\colon X \rightsquigarrow X^*$. In particular we focus on the case in which $T = \partial\Phi + A$ where $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ is convex and $A\colon X \to X^*$ is pseudomonotone. In this case the problem that we want to study is that of finding $x \in X$ such that

$$\partial\Phi(x) + A(x) \ni f, \tag{B.5}$$

where $f \in X^*$. Here $\partial\Psi$ denotes the subdifferential of Ψ. The following generalization of Minty–Browder theorem holds.

Theorem B.27. Let $\Phi\colon X \to \mathbb{R}\cup\infty$ be convex, proper and lower semicontinuous. Assume that for all $\epsilon > 0$ there exist $\Phi_\epsilon\colon X \to \mathbb{R}$ convex and Gâteaux differentiable with $\Phi'_\epsilon\colon X \to X^*$ demicontinuous and bounded, such that the following properties holds:

(i) for all $x \in X$ and $\{x_\epsilon\}_{\epsilon>0} \subset X$ such that $x_\epsilon \rightharpoonup x$ in X it holds that
$$\Phi(x) \le \liminf_{\epsilon\to 0^+} \Phi_\epsilon(x_\epsilon),$$

(ii) for all $x \in X$
$$\Phi(x) \ge \limsup_{\epsilon\to 0^+} \Phi_\epsilon(x).$$

Let also $A\colon X \to X^*$ be pseudomonotone. Moreover assume that the following coercivity property holds: there exists $y \in \operatorname{dom}\Phi$ such that for all $z \in X$ the inequality

$$\frac{\Phi_\epsilon(x) + \langle A(x), x - z\rangle}{\|x\|_X} \ge \zeta(\|x\|_X) \tag{B.6}$$

holds. Here $\zeta\colon [0,\infty) \to [0,\infty)$ satisfies the condition
$$\lim_{r\to\infty} \zeta(r) = \infty.$$
Then for all $f \in X^*$ there exists $x \in X$ such that (B.5) holds.

Proof. The map Φ'_ϵ is monotone since Φ_ϵ is convex. Then by Proposition B.26 the operator $\Phi'_\epsilon + A$ is pseudomonotone. Moreover it is bounded and continuous. The coercivity property (B.6) allows to apply Theorem B.24 to conclude thet for all $\epsilon > 0$ there exists $x_\epsilon \in X$ such that $\Phi'_\epsilon(x_\epsilon) + A(x_\epsilon) \ni f$. This is equivalent to

$$\Phi_\epsilon(y) \ge \Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle \quad \text{for all } y \in X. \tag{B.7}$$

The idea is now to prove that $\{x_\epsilon\}_{\epsilon>0}$ is bounded in X in order to find a limit value by compactness. To this aim we notice that by property (ii) above, for each $\epsilon > 0$ small enough the inequality $\Phi_\epsilon(z_\epsilon) \le \Phi(z) + 1$ holds. Then by (B.6) and by (B.7) with $y = z$ we have
$$\zeta(\|x_\epsilon\|)\|x_\epsilon\|_X \le \Phi_\epsilon(x_\epsilon) + \langle A(x_\epsilon), x_\epsilon - z\rangle \le \Phi_\epsilon(z) + \langle f, z - x_\epsilon\rangle.$$
This implies that $x_\epsilon \to 0$. Then extracting a (not relabeled) subsequence we have $x_\epsilon \rightharpoonup x$. Then we take the supremum limit of (B.7), and using properties (i, ii) above we get
$$\begin{aligned}\Phi(y) &\ge \limsup_{\epsilon\to 0^+}\left[\Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle\right] \ge \liminf_{\epsilon\to 0^+}\left[\Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle\right] \ge \\ &\ge \Phi(x) + \langle f, y - x\rangle + \liminf_{\epsilon\to 0^+}\langle A(x_\epsilon), x_\epsilon - y\rangle.\end{aligned}$$
The proof of the theorem is completed if we prove that
$$\liminf_{\epsilon\to 0^+}\langle A(x_\epsilon), x_\epsilon - y\rangle \ge \langle A(x), x - y\rangle.$$
In view of the fact that A is pseudomonotone, it is enough to prove that
$$\limsup_{\epsilon\to 0^+}\langle A(x_\epsilon), x_\epsilon - x\rangle \le 0.$$
But by (B.7) it follows that
$$\begin{aligned}\limsup_{\epsilon\to 0^+}\langle A(x_\epsilon), x_\epsilon - x\rangle &\le \limsup_{\epsilon\to 0^+}\left[\Phi_\epsilon(x) - \Phi_\epsilon(x_\epsilon) + \langle f, x_\epsilon - x\rangle\right] \le \\ &\le \limsup_{\epsilon\to 0^+}\Phi_\epsilon(y) - \liminf_{\epsilon\to 0^+}\Phi_\epsilon(x_\epsilon) \le \Phi(x) - \Phi(x) = 0.\end{aligned}$$
as desired. □

Given $\Phi\colon X \to \mathbb{R} \cup \{\infty\}$ convex, proper and lower semicontinuous, a possible way to obtain its regularization Φ_ϵ for all $\epsilon > 0$ is through the Yosida approximation

$$\Phi_\epsilon(x) := \inf_{y \in X} \left(\frac{\|x-y\|^2}{2\epsilon} + \Phi(y) \right).$$

Notice that it is well defined as a function $\Phi_\epsilon\colon X \to \mathbb{R}$. The following result shows that the Yosida regularization satisfies the desired approximation properties.

Lemma B.28. Let X be a separable and reflexive Banach space and endow it with a strictly convex norm. Let $\Phi\colon X \to \mathbb{R} \cup \{\infty\}$ be convex, proper and lower semicontinuous. Then also Φ_ϵ is convex, proper and lower semicontinuous. Moreover

(i) for all $x \in X$ and $\{x_\epsilon\}_{\epsilon>0} \subset X$ such that $x_\epsilon \rightharpoonup x$ in X it holds that

$$\Phi(x) \leq \liminf_{\epsilon \to 0^+} \Phi_\epsilon(x_\epsilon),$$

(ii) for all $x \in X$

$$\Phi(x) \geq \limsup_{\epsilon \to 0^+} \Phi_\epsilon(x).$$

and Φ_ϵ is Gâteaux differentiable with $\Phi'_\epsilon\colon V \to V^*$ bounded and demicontinuous.

Proof. See [62, Lemma 5.17]. □

Corollary B.29. Let X be a reflexive and separable Banach space. Let $\Phi\colon X \to \mathbb{R} \cup \{\infty\}$ be convex, proper and lower semicontinuous. Let $A\colon V \to V^*$ pseudomonotone. Suppose then one of the following two conditions holds:

(i) there exists $\alpha > 1$ such that $\Phi(x) \geq \|x\|^\alpha$ for all $x \in X$ and there exists $z \in \operatorname{dom}\Phi$ such that

$$x \mapsto \frac{\langle A(x), x-y \rangle}{\|x\|}$$

is bounded from below,

(ii) Φ is bounded from below and A is coercive in the sense that there exists $z \in \operatorname{dom}\Phi$ such that

$$\frac{\langle A(x), x-z \rangle}{\|x\|} \to \infty \quad \text{as } \|x\| \to \infty.$$

Then for each $f \in X^*$ there exists at least one $x \in X$ such that (B.5) holds.

Proof. For all $\epsilon > 0$ let Φ_ϵ be the Yosida approximation of Φ. In view of Lemma B.28 and Theorem B.27 get the thesis if we prove that (B.6) holds.

In case that (ii) is assumed the result follows immediately. Indeed since $\Phi \geq C$ for some $C \in \mathbb{R}$ also $\Phi_\epsilon \geq C$ for all $\epsilon > 0$.

When (i) is assumed, we fix $\delta > 0$ and for all $\epsilon \leq \delta$ we estimate

$$\Phi_\epsilon(x) \geq \Phi_\delta(x) \geq \inf_{y \in X} \left(\frac{\|x-y\|^2}{2\delta} + \Phi(y) \right) \geq \inf_{y \in X} \left(\frac{(\|x\| - \|y\|)^2}{2\delta} + \|y\|^\alpha \right) = |\cdot|^\alpha_\delta(\|x\|),$$

where $|\cdot|^\alpha_\delta$ is the Yosida approximation of $|\cdot|^\alpha$. Now let $r \in \mathbb{R}$ be the optimal value such that

$$|\cdot|^\alpha_\delta(\|x\|) = \frac{(r - \|x\|)^2}{2\epsilon} + |r|^\alpha.$$

so that $|\cdot|^\alpha_\delta(\|x\|) \geq r^\alpha$. The optimality of r implies that $\|x\| = r + \epsilon\alpha r^{\alpha-1}$ which in turn implies that

$$\|x\|^{\min\{1, \frac{1}{\alpha-1}\}} \leq Cr$$

Collecting the results obtained, we get $\Phi_\epsilon(x) \geq \|x\|^{\min\{\alpha,\alpha^*\}}$. This implies the desired coercivity. □

B.3.6. Lower semicontinuity of integral functionals.

Theorem B.30 (Lower semicontinuity of integral functionals). Let $\Omega \subset \mathbb{R}^d$ open bounded. Let $f\colon \Omega \times \mathbb{R}^n \times \mathbb{R}^m \to (-\infty, \infty]$ such that

(i) $f(\cdot, z, p)$ is measurable for all $(z,p) \in \mathbb{R}^n \times \mathbb{R}^m$
(ii) $f(x, \cdot, \cdot)$ is continuous for a.e. $x \in \Omega$
(iii) $f(x, z, \cdot)$ is convex for all $x \in \Omega$ and $z \in \mathbb{R}^n$
(iv) there exists $\phi \in L^1(\Omega)$ such that $f(x,z,p) \geq \phi(x)$ for a.e. $x \in \Omega$ and all $(z,p) \in \mathbb{R}^n \times \mathbb{R}^m$.

Then for all measurable $y\colon \Omega \to \mathbb{R}^n$ and $u\colon \Omega \to \mathbb{R}^m$ is well defined

$$I(y,u) = \int_\Omega f(x,y,u)dx.$$

Moreover if $y_k, y\colon \Omega \to \mathbb{R}^n$ measurable satisfy $y_k \to y$ in measure and $u_k, u \in L^1(\Omega, \mathbb{R}^m)$ satisfy $u_k \to u$ in $L^1(\Omega, \mathbb{R}^n)$, then

$$I(y,u) \le \liminf_{k\to\infty} I(y_k, u_k).$$

B.4. Convex analysis

Example B.31. Let $f(v) = \frac{1}{2}\|v\|^2$. Then $f^*(\xi) = \frac{1}{2}\|\xi\|^2$. Indeed

$$\langle \xi, v\rangle - \frac{1}{2}\|v\|^2 \le \frac{2\|\xi\|\|v\| - \|v\|^2}{2} \le \frac{\|\xi\|^2}{2}.$$

This implies $f^*(\xi) \le \frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon > 0$ and let $v \in V$ such that $\|v\| = \|\xi\|$ and $\langle \xi, v\rangle \ge \|\xi\|^2 - \epsilon$. Then

$$f^*(\xi) \ge \|\xi\|^2 - \epsilon - \frac{1}{2}\|\xi\|^2 \ge \frac{1}{2}\|\xi\|^2 - \epsilon.$$

By arbitrariness of ϵ we conclude.

Let f and g from V to $(-\infty, \infty]$. Their inf-convolution $f\Box g\colon V \to (-\infty,\infty]$ is

$$f\Box g(v) = \inf_{w\in V}(f(w) + g(v-w)).$$

Proposition B.32. (inf-convolution duality formula) Let f and g from V to $(-\infty,\infty]$. Then

(i) $(f\Box g)^* = f^* + g^*$
(ii) $(f+g)^* \le f^*\Box g^*$
(iii) if f and g are convex and proper and there exists $v \in \operatorname{dom} f \cap \operatorname{dom} g$ and f is continuous at v then $(f+g)^* = f^*\Box g^*$.

Proof. *Step 1.* It holds that

$$\begin{aligned}
(f\Box g)^* &= \sup_{v\in V}\left(\langle \xi, v\rangle - \inf_{w\in V}\big(f(w) + g(v-w)\big)\right) = \\
&= \sup_{v\in V}\sup_{w\in V}\big(\langle \xi, w\rangle - f(w) + \langle \xi, v-w\rangle - g(v-w)\big) = \\
&= \sup_{w\in V}\sup_{v\in V}\big(\langle \xi, w\rangle - f(w) + \langle \xi, v-w\rangle - g(v-w)\big) = \\
&= \sup_{w\in V}\big(\langle \xi, w\rangle - f(w) + g^*(\xi)\big) = f^*(\xi) + g^*(\xi).
\end{aligned}$$

This establish (i).

Step 2. By the Young–Fenchel inequality

$$f^*(\zeta) + g^*(\xi - \zeta) \ge \langle \xi, v\rangle - f(v) - g(v).$$

Taking the supremum over $v \in V$

$$f^*(\zeta) + g^*(\xi-\zeta) \ge (f+g)^*$$

and taking the infimum over $\zeta \in V$

$$f^*\Box g^* \ge (f+g)^*.$$

This proves (ii).

Step 3. Now we prove (iii). If $(f+g)^*(\xi) = \infty$ then (iii) is satisfied at ξ. Then suppose that $(f+g)^*(\xi) \in [-\infty,\infty)$. Since $\operatorname{dom} f \cap \operatorname{dom} g \ne \emptyset$ then actually $(f+g)^*(\xi) = c \in (-\infty,\infty)$. Let

$$A = \{(a,v) \in \mathbb{R}\times V \mid a \le \langle \xi, v\rangle - g(v) - c\}.$$

Clearly A is nonempty and convex. We show that A and $\operatorname{int}(\operatorname{epi} f)$ are disjoint. Indeed if by contradiction $(a,v) \in A \cap \operatorname{int}(\operatorname{epi} f)$ then

$$f(v) < a \le \langle \xi, v\rangle - g(v) - c$$

and

$$c < \langle \xi, v\rangle - f(v) - g(v) \le (f+g)^*(\xi) = c.$$

Then by Hahn–Banach theorem there exists $(\gamma,\zeta) \in \mathbb{R}\times V^*$ such that

$$\gamma a + \langle \zeta, v\rangle \le \gamma b + \langle \zeta, w\rangle \tag{B.8}$$

for all $(a,v) \in \operatorname{int}(\operatorname{epi} f)$ and $(b,w) \in A$. If we fix $v = w \in \operatorname{dom} f \cap \operatorname{dom} g$, $b \in \mathbb{R}$ sufficiently small such that $(b,v) \in A$ and we let $a \to \infty$, we discover that $\gamma \leq 0$ necessarily. If $\gamma = 0$ then inequality (B.8) and Hahn–Banach theorem imply that $\operatorname{dom} f$ and $\operatorname{dom} g$ are separated by a closed hyperplane. This in turn implies that $\operatorname{int}(\operatorname{dom} f)$ and $\operatorname{dom} g$ are disjoint, but this contradicts the hypotheses so that $\beta < 0$. In this case we set $\theta = |\beta|^{-1} f \zeta$ and from inequality (B.8) we get

$$\begin{aligned} f^*(\theta) &= \sup\left\{\langle \theta, v\rangle - f(v) \mid v \in V\right\} = \\ &= \sup\left\{\langle \theta, v\rangle - a \mid (a,v) \in \operatorname{epi} f\right\} \leq \\ &\leq \inf\left\{\langle \theta, v\rangle - a \mid (a,v) \in A\right\} = \\ &= c + \inf\{\langle \theta - \xi, v\rangle + g(v) \mid v \in V\} = c - g^*(\xi - \theta). \end{aligned}$$

Then

$$f^* \square g^*(\xi) \leq f^*(\theta) + g^*(\xi - \theta) \leq c = (f+g)^*(\xi).$$

This concludes the proof. □

Proposition B.33 (properties of positively 1-homogeneous convex functions). Let $f\colon V \to (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0) = 0$ and f is subadditive
(ii) for each $x \in \operatorname{dom} f$ it holds $\partial f(x) = \{\xi \in \partial f(0) \mid f(x) = \langle \xi, x\rangle\}$
(iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

Proof. *Step 1.* Let $v \in \operatorname{dom} f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n\to\infty} f\left(\frac{1}{n} v\right) = \liminf_{n\to\infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2} f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).

Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1} f^*(\xi)$ and this implies $f^*(\xi) = 0$.

Step 3. Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0\rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^{**} = \iota_K^* = \sigma_K$. This establishes (iii). □

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